

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\130-C1& C2-Final 23rd March.pvdb

Date of Analysis : Sep 9, 2026 9:30am

PV Elite 28, January 2026



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 FileName : 130-C1& C2-Final 23rd March Page 2 of 317
 Vessel Design Summary: Step: 48 9:30am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1200.000 x 1800.000 x 1200.000 mm ID
 Vessel Design Length, Tangent to Tangent 16036.35 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 65 °C
 Channel Side Design Temperature 100 °C
 Shell Side Design Pressure 1550.000 KPa.
 Channel Side Design Pressure 4550.000 KPa.
 Shop Shell Side Test Pressure 2548.008 KPa.
 Field Shell Side Test Pressure 2015.000 KPa.
 Shop Channel Side Test Pressure 6495.683 KPa.
 Field Channel Side Test Pressure 5915.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Cone	SA-516 70	...	...	K02700	
Flange	SA-765 II	...	...	K03047	
Nozzle	SA-765 II	...	...	K03047	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-765 II	...	...	K03047	
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-765 II	...	...	K03047	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flg	4565.683	103.42	4804.000	3.0000	N/A	No
LHS Ch Flange	4561.911	103.42	5570.000	3.0000	N/A	No
LHS Ch Shell	4561.940	103.42	5570.000	3.0000	N/A	No
LHS Cone Shell	1557.512	0.00	3818.000	3.0000	N/A	No
LHS Cone Shell	1567.673	0.00	2235.000	3.0000	N/A	No
Main Shell	1561.258	0.00	1958.000	3.0000	N/A	No
RHS Cone shell	1561.258	0.00	2241.000	3.0000	N/A	No
RHS Cone shell	1557.512	0.00	3818.000	3.0000	N/A	No
Rhs Ch Shell	4561.940	103.42	5570.000	3.0000	N/A	No
RHS Ch Flg	4561.911	103.42	5570.000	3.0000	N/A	No
RHS Blind Flang	4565.487	103.42	4804.000	3.0000	N/A	No

Liquid Level: 1800.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



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Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	159.2	156.0	164.0	156.0	151.9	...	1.00	1.00
Body Flg	418.2	259.0	211.0	203.0	198.0	...	1.00	1.00
Cylinder	1918.2	1500.0	28.0	28.0	23.3	...	1.00	1.00
Cylinder	2236.2	200.0	20.0	20.0	9.9	...	1.00	1.00
Conical	3326.2	1090.0	20.0	20.0	14.8	...	1.00	1.00
Cylinder	12710.2	9384.0	16.0	16.0	13.3	...	1.00	1.00
Conical	13800.2	1090.0	20.0	20.0	14.7	...	1.00	1.00
Cylinder	14000.2	200.0	20.0	20.0	9.9	...	1.00	0.80
Cylinder	15618.2	1500.0	28.0	28.0	23.3	...	1.00	1.00
Body Flg	15880.3	259.0	211.0	203.0	198.0	...	1.00	1.00
Body Flg	16036.3	156.0	164.0	156.0	151.9	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	156	108.585	103.419	No Calc
20	30	203	197.993	103.419	No Calc
30	Tsht	28	7.3409	103.419	2992
Tsht	40	28	No Calc	103.419	No Calc
40	50	20	No Calc	...	422
50	60	20	No Calc	...	198
60	70	16	No Calc	...	103
70	80	20	No Calc	...	198
80	Tsht	20	No Calc	...	422
Tsht	90	20	No Calc	...	No Calc
90	100	28	7.34096	103.419	2992
100	110	203	197.993	103.419	No Calc
110	120	156	108.585	103.419	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1499.97	99166	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	11964	No Calc	No Calc	No Calc
50	60	11964	No Calc	No Calc	No Calc
60	70	11964	No Calc	No Calc	No Calc
70	80	11964	No Calc	No Calc	No Calc
80	Tsht	11964	No Calc	No Calc	No Calc
Tsht	90	No Calc	No Calc	No Calc	No Calc
90	100	1500.03	99168.6	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
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Noz N6	WRC-297	0.949	Shell	Passed
Npz N2 250	WRC-297	1.090	Shell	Failed
Noz N3 80	WRC-107/537	0.792	n/a	Passed
Noz N1 150	WRC-297	0.860	Shell	Passed
Noz N4 100	WRC-297	0.783	Shell	Passed
Noz N5 200	WRC-297	1.319	Shell	Failed
Noz N7	WRC-297	0.949	Shell	Passed

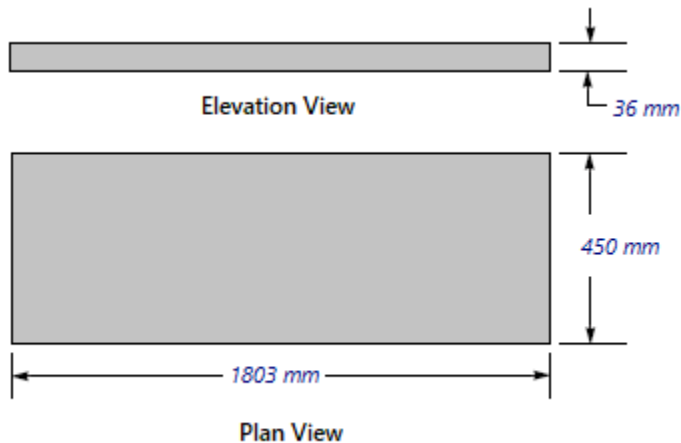
Foundation/Support Summary

Saddle Parameters:

Saddle Width	425.000	mm
Saddle Bearing Angle	130.000	deg.
Centerline Dimension	1575.000	mm
Wear Pad Width	625.000	mm
Wear Pad Thickness	16.000	mm
Wear Pad Bearing Angle	142.000	deg.
Distance from Saddle to Tangent	1110.000	mm

Baseplate Length	1803.000	mm
Baseplate Thickness	36.000	mm
Baseplate Width	450.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	24.000	mm
Web Thickness	24.000	mm
Height of Center Web	675.000	mm
Number of Bolts in Baseplate	4	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	43560.95	Kgf
Maximum Transverse Saddle Shear Load	2269.95	Kgf
Maximum Longitudinal Saddle Shear Load	4156.43	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	40617.23	Kgf
Maximum Transverse Saddle Shear Load	377.28	Kgf
Maximum Longitudinal Saddle Shear Load	188.97	Kgf

Weights:

Fabricated - Bare w/o removable internals	47701.9	kg.
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Vessel Design Summary:

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Shop Test	- Fabricated + water (full)	77895.8 kg.
Shipping	- Fab. + rem. intls.+ shipping app.	47701.9 kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	47701.9 kg.
Empty	- Fab. + intls. + details + wghts.	49927.0 kg.
Operating	- Empty + operating liquid (no ca)	72990.4 kg.
Field Test	- Empty weight + test medium (full)	78315.0 kg.

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 FileName : 130-C1& C2-Final 23rd March
 Nozzle Summary: Step: 44 9:30am Page 6 of 317
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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Noz N6	4629	OK	...	OK	12.92	OK	Passed
Npz N2 250	1579	...	...	OK	11.10	OK	Passed
Noz N3 80	1958	...	...	OK	8.73	OK	Passed
Noz N1 150	1958	...	...	OK	10.16	OK	Passed
Noz N1 150	1744	...	...	OK	10.16	OK	Passed
Noz N4 100	1958	...	...	OK	9.22	OK	Passed
Noz N4 100	1904	...	...	OK	9.22	OK	Passed
Noz N5 200	1755	...	...	OK	10.16	OK	Passed
D1 nozzle 50	1958	...	...	OK	7.80	OK	No Calc[*]
K1 50	1958	...	...	OK	7.80	OK	No Calc[*]
K2 50	1958	...	...	OK	7.80	OK	No Calc[*]
K3 50	1958	...	...	OK	7.80	OK	No Calc[*]
K4 50	1958	...	...	OK	7.80	OK	No Calc[*]
K5 50	1958	...	...	OK	7.80	OK	No Calc[*]
K6 50	1958	...	...	OK	7.80	OK	No Calc[*]
K7 50	1958	...	...	OK	7.80	OK	No Calc[*]
K8 50	1958	...	...	OK	7.80	OK	No Calc[*]
Noz N7	4617	OK	...	OK	12.95	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1579 Nozzle : Npz N2 250 [Shellside]
 Minimum MAWP Nozzles : 4617 Nozzle : Noz N7 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	Noz N6	1068.175	0.000	1161.280
60	Npz N2 250	5326.175	0.000	513.650
60	Noz N3 80	7326.174	180.000	164.400
60	Noz N1 150	7326.174	56.917	316.800
60	Noz N4 100	9326.175	56.917	215.200
60	Noz N5 200	9326.175	0.000	399.350
60	D1 nozzle 50	6826.175	180.000	113.600
60	K1 50	6826.175	109.075	113.600
60	K2 50	6826.175	70.925	113.600
60	K3 50	5826.174	70.925	113.600
60	K4 50	5826.174	109.075	113.600
60	K5 50	8626.175	109.075	113.600
60	K6 50	8626.175	70.925	113.600
60	K7 50	9726.175	70.925	113.600
60	K8 50	9626.175	109.075	113.600
90	Noz N7	14968.175	180.000	1161.280

The nozzle spacing is computed by the following:

$$= \sqrt{I^2 + I_c^2} \text{ where}$$

I - Arc length along the inside vessel surface in the long. direction.



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Nozzle Summary:

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Ic - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 43 9:30am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1 nozzle 50	2.000 in	Actual	LWN	3.320	16.764	...	...	223.36	300
K1 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K2 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K3 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K4 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K5 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K6 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K7 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
K8 50	2.000 in	Actual	LWN	3.320	16.764	...	...	433.06	300
Noz N3 80	3.000 in	Actual	LWN	4.620	20.574	...	...	217.94	300
Noz N4 100	4.000 in	Actual	LWN	5.760	22.352	...	...	346.65	300
Noz N1 150	6.000 in	Actual	LWN	8.120	26.924	...	...	367.77	300
Noz N5 200	8.000 in	80	LWN	8.625	12.700	...	...	289.23	300
Npz N2 250	10.000 in	60	WNF	10.750	12.700	...	...	276.42	150
Noz N6	24.000 in	Actual	WNF	24.000	17.480	...	...	511.19	300
Noz N7	24.000 in	Actual	WNF	24.000	17.480	...	...	511.19	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1 nozzle 5	SA-765 II	16.000	10.000	...	...	...
K1 50	SA-765 II	16.000	10.000	...	...	...
K2 50	SA-765 II	16.000	10.000	...	...	...
K3 50	SA-765 II	16.000	10.000	...	...	...
K4 50	SA-765 II	16.000	10.000	...	...	...
K5 50	SA-765 II	16.000	10.000	...	...	...
K6 50	SA-765 II	16.000	10.000	...	...	...
K7 50	SA-765 II	16.000	10.000	...	...	...
K8 50	SA-765 II	16.000	10.000	...	...	...
Noz N3 80	SA-765 II	16.000	10.000	...	...	...
Noz N4 100	SA-765 II	16.000	10.000	...	...	...
Noz N1 150	SA-765 II	16.000	10.000	...	...	...
Noz N5 200	SA-765 II	16.000	10.000	...	...	...
Npz N2 250	SA-765 II	16.000	10.000	...	...	...
Noz N6	SA-765 II	14.000	10.000	...	...	...
Noz N7	SA-765 II	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
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Nozzle Schedule:

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D1 nozzle 50	6826.175	180.0	206.38	0.00	Main Shell
K1 50	6826.175	109.1	400.00	0.00	Main Shell
K2 50	6826.175	70.9	400.00	0.00	Main Shell
K3 50	5826.174	70.9	400.00	0.00	Main Shell
K4 50	5826.174	109.1	400.00	0.00	Main Shell
K5 50	8626.175	109.1	400.00	0.00	Main Shell
K6 50	8626.175	70.9	400.00	0.00	Main Shell
K7 50	9726.175	70.9	400.00	0.00	Main Shell
K8 50	9626.175	109.1	400.00	0.00	Main Shell
Noz N3 80	7326.175	180.0	200.02	0.00	Main Shell
Noz N4 100	9326.175	56.9	273.05	0.00	Main Shell
Noz N1 150	7326.175	56.9	268.29	0.00	Main Shell
Noz N5 200	9326.175	0.0	263.52	0.00	Main Shell
Npz N2 250	5326.175	0.0	250.00	0.00	Main Shell
Noz N6	1068.175	0.0	400.00	0.00	LHS Ch Shell
Noz N7	14968.174	180.0	400.00	0.00	Rhs Ch Shell

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 Bill of Material: Step: 42 9:30am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 164.0mm THK X 1580.0mm OD	SA-765 II
2	WELD NECK FLANGE: 211.0mm THK X 1580.0mm OD	SA-765 II
2	INSULATION: 156mm X 70mm THK	Cold Insulation
2	INSULATION: 259mm X 70mm THK	Cold Insulation
2	INSULATION: 1500mm X 70mm THK	Cold Insulation
4	BAFFLES: 1158.0mm DIA X 19.0mm THK	SA-516 70
10	TIE RODS: 19mm DIA X 6100mm	SA-193 B7M
2	TUBESHEET: 1256.0mm X 118.0mm THK	SA-765 II
1151	TUBES: 12200.0mm X 25.4mm DIA X 2.1mm THK	SA-334 1
2	INSULATION: 200mm X 70mm THK	Cold Insulation
2	INSULATION: 1090mm X 70mm THK	Cold Insulation
2	SADDLE: 425.0mm X 130 DEG	SA-516 70
1	INSULATION: 9384mm X 70mm THK	Cold Insulation
2	GASKET: 1264.0mm OD X 1232.0mm ID	...
56	BODY FLANGE BOLTS: 57.1mm DIA	SA-193 B7
112	NUTS FOR BODY FLANGE BOLTS: 57.1mm DIA	...
1	CYLINDRICAL SEGMENT 28.0mm THK X 3048.0mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 721.9mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 932.0mm X 200.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 2837.9mm X 200.0mm	SA-516 70
1	ECC. CONICAL SEGMENT 20.0mm THK X 4570.9mm X 1090.0mm X 3047.3mm	SA-516 70
1	ECC. CONICAL SEGMENT 20.0mm THK X 1083.8mm X 1090.0mm X 722.6mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3047.3mm X 9384.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 2607.5mm X 9384.0mm	SA-516 70
1	ECC. CONICAL SEGMENT 20.0mm THK X 3047.3mm X 1090.0mm X 2031.5mm	SA-516 70
1	ECC. CONICAL SEGMENT 20.0mm THK X 2607.5mm X 1090.0mm X 1738.3mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1738.3mm X 200.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 2031.5mm X 200.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 932.0mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 2837.9mm X 1500.0mm	SA-516 70



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Bill of Material:

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NAMEPLATE

...

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 FileName : 130-C1& C2-Final 23rd March Page 12 of 317
 MDMT Summary: Step: 47 9:30am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Cone Shell[8]	B	-24	-48	-29	0.409	20.000	1.00	No
LHS Cone Shell[8]	B	-24	-41	-29	0.699	20.000	1.00	No
Main Shell [8]	B	-31	-42		0.800	16.000	1.00	No
RHS Cone shell[8]	B	-24	-41	-29	0.699	20.000	1.00	No
RHS Cone shell[8]	B	-24	-48	-29	0.409	20.000	1.00	No
Left Saddle [24]	B	-31	-42		0.800	16.000	1.00	No
Sdl 2 Fr50 [24]	B	-31	-42		0.800	16.000	1.00	No
Npz N2 250 [1]	B	-31	-43		0.786	16.000	1.00	No
Nozzle Flg [5]	A	-18	-29					
Noz N3 80 [1]	B	-31	-42		0.791	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Noz N1 150 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Noz N4 100 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Noz N5 200 [1]	B	-31	-43		0.786	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
D1 nozzle 50 [1]	B	-31	-42		0.791	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K1 50 [1]	B	-31	-42		0.789	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K2 50 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K3 50 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K4 50 [1]	B	-31	-42		0.789	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K5 50 [1]	B	-31	-42		0.789	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K6 50 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K7 50 [1]	B	-31	-43		0.787	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K8 50 [1]	B	-31	-42		0.789	16.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: SS[13]	B	-13	-104		0.285	29.500	1.00	No

Warmest MDMT: -13 -29

LHS Blind Flg[11]	B	11	7		0.924	39.000	1.00	No
LHS Ch Flange[11]	B	2	-12		0.739	28.000	1.00	No
RHS Ch Flg [11]	B	-15	-29		0.739	28.000	1.00	No
RHS Blind Fla[11]	B	-6	-10		0.924	39.000	1.00	No
LHS Ch Shell [8]	B	2	-8		0.826	28.000	1.00	No
Rhs Ch Shell [8]	B	-15	-24		0.826	28.000	1.00	No
Noz N6 [1]	B	2	-8		0.811	28.000	1.00	No
Nozzle Flg [5]	A	-18	-24					
Noz N7 [1]	B	-15	-25		0.814	28.000	1.00	No
Nozzle Flg [5]	A	-18	-24					
Tubesheet: CS[14]	B	4	-6		0.826	29.500	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						

Warmest MDMT: 11 7

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
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 MDMT Summary: Step: 47 9:30am Sep 9,2026

Shell	-29.0	-33.0	Fail
Channel/Tube	7.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1550	KPa.
Channel Side Design Pressure	4550	KPa.
Shell Side Design Temperature	65.0	°C
Channel Side Design Temperature	100.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-33.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/11	
Element From Node	10	
Element To Node	20	
Element Type	Flange	
Description	LHS Blind Flg	
Distance "FROM" to "TO"	156	mm
Flange Outside Diameter	1580	mm
Element Thickness	156	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	164	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	100	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-765 II	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	227	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03047	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1580	mm
Liquid Density	0.001011	kg./cm ³
Element From Node	10	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm



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Height/Length of Insulation	156	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element#	2/11
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flange
Distance "FROM" to "TO"	259 mm
Flange Inside Diameter	1200 mm
Element Thickness	203 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	211 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.2
Material Name	SA-765 II
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1200 mm
Liquid Density	0.001011 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	259 mm
Thickness of Insulation	70 mm
Density	0.000352 kg./cm ³

Element#	3/11
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1200 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm



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Yield Stress, Operating	239	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1200	mm
Liquid Density	0.001011	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1500	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Noz N6	
Dist. from "FROM" Node / Offset dist	650	mm
Nozzle Diameter	24	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	482.05	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element#	4/11	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	LHS Cone Shell	
Distance "FROM" to "TO"	200	mm
Inside Diameter	1200	mm
Element Thickness	20	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1200	mm



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Liquid Density	0.000636	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	200	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element#	5/11	
Element From Node	50	
Element To Node	60	
Element Type	Conical	
Description	LHS Cone Shell	
Distance "FROM" to "TO"	1090	mm
Inside Diameter	1200	mm
Element Thickness	20	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	1800	mm
Design Length of Cone	1090	mm
Half Apex Angle of Cone	28.830969	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1800	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1090	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element#	6/11	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	9384	mm
Inside Diameter	1800	mm
Element Thickness	16	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm



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Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	Yes	

Element From Node	60	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	1110	mm
Width of Saddle	425	mm
Height of Saddle at Bottom	1575	mm
Saddle Contact Angle	130.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	625	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	142.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	60	
Detail Type	Saddle	
Detail ID	Sdl 2 Fr50	
Dist. from "FROM" Node / Offset dist	8110	mm
Width of Saddle	425	mm
Height of Saddle at Bottom	1575	mm
Saddle Contact Angle	130.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	625	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	142.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1800	mm
Liquid Density	0.000636	kg./cm³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	9384	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm³

Element From Node	60	
Detail Type	Nozzle	
Detail ID	Npz N2 250	
Dist. from "FROM" Node / Offset dist	2000	mm



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Nozzle Diameter	10	in.
Nozzle Schedule	60	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	69.496	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765	II

Element From Node	60	
Detail Type	Nozzle	
Detail ID	Noz N3 80	
Dist. from "FROM" Node / Offset dist	4000	mm
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	18.725	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765	II

Element From Node	60	
Detail Type	Nozzle	
Detail ID	Noz N1 150	
Dist. from "FROM" Node / Offset dist	4000	mm
Nozzle Diameter	6	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	56.917099	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	55.12	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765	II

Element From Node	60	
Detail Type	Nozzle	
Detail ID	Noz N4 100	
Dist. from "FROM" Node / Offset dist	6000	mm
Nozzle Diameter	4	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	56.917099	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	32.345	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765	II

Element From Node	60	
Detail Type	Nozzle	
Detail ID	Noz N5 200	
Dist. from "FROM" Node / Offset dist	6000	mm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	57.504	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765	II

Element From Node	60	
Detail Type	Nozzle	
Detail ID	D1 nozzle 50	
Dist. from "FROM" Node / Offset dist	3500	mm



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Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.467	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K1 50	
Dist. from "FROM" Node / Offset dist	3500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	109.075	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.807	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K2 50	
Dist. from "FROM" Node / Offset dist	3500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	70.925499	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.807	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K3 50	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	70.925499	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.807	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K4 50	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	109.075	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K5 50	
Dist. from "FROM" Node / Offset dist	5300	mm



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Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	109.075	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.807	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K6 50	
Dist. from "FROM" Node / Offset dist	5300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	70.925499	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	15.807	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K7 50	
Dist. from "FROM" Node / Offset dist	6400	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	70.925499	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Nozzle	
Detail ID	K8 50	
Dist. from "FROM" Node / Offset dist	6300	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	109.075	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-765 II	

Element From Node	60	
Detail Type	Weight	
Detail ID	INTERNAL T/S	
Dist. from "FROM" Node / Offset dist	4692	mm
Miscellaneous Weight	2400	Kgf
Offset from Element Centerline	0	mm

Element#	7/11	
Element From Node	70	
Element To Node	80	
Element Type	Conical	
Description	RHS Cone shell	
Distance "FROM" to "TO"	1090	mm
Inside Diameter	1800	mm
Element Thickness	20	mm



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Internal Corrosion Allowance	3	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1550	KPa.
Design Temperature Internal Pressure	65	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	65	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	1200	mm
Design Length of Cone	1090	mm
Half Apex Angle of Cone	28.830969	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1800 mm
Liquid Density	0.000636 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1090 mm
Thickness of Insulation	70 mm
Density	0.000352 kg./cm ³

Element#	8/11
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	RHS Cone shell
Distance "FROM" to "TO"	200 mm
Inside Diameter	1200 mm
Element Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1550 KPa.
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	65 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	0.8
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1200 mm
Liquid Density	0.000636 kg./cm ³

Element From Node	80
Detail Type	Insulation
Detail ID	Insulation



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Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	200	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element#	9/11
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Rhs Ch Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1200 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4550 KPa.
Design Temperature Internal Pressure	100 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	100 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	Yes

Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1200 mm
Liquid Density	0.001011 kg./cm ³

Element From Node	90
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	70 mm
Density	0.000352 kg./cm ³

Element From Node	90
Detail Type	Nozzle
Detail ID	Noz N7
Dist. from "FROM" Node / Offset dist	850 mm
Nozzle Diameter	24 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	482.05 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-765 II

Element#	10/11
Element From Node	100
Element To Node	110
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	259 mm
Flange Inside Diameter	1200 mm



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Element Thickness	203	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	211	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	100	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-765 II	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	227	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03047	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 100	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1200	mm
Liquid Density	0.001011	kg./cm ³

Element From Node	100	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	259	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

Element#	11/11	
Element From Node	110	
Element To Node	120	
Element Type	Flange	
Description	RHS Blind Flang	
Distance "FROM" to "TO"	156	mm
Flange Outside Diameter	1580	mm
Element Thickness	156	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	164	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	4550	KPa.
Design Temperature Internal Pressure	100	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	100	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-765 II	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	Yes	

Element From Node	110	
Detail Type	Liquid	
Detail ID	Liquid: 110	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1580	mm



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Liquid Density	0.0009984	kg./cm ³
Element From Node	110	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	156	mm
Thickness of Insulation	70	mm
Density	0.000352	kg./cm ³

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XY Coordinate Calculations: Step: 2 9:30am Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flg		159.175	...	156	...
LHS Ch Flange		418.175	...	259	...
LHS Ch Shell		1918.17	...	1500	...
LHS Cone Shell		2236.17	...	200	...
LHS Cone Shell		3326.18	...	1090	...
Main Shell		12710.2	...	9384	...
RHS Cone shell		13800.2	...	1090	...
RHS Cone shell		14000.2	...	200	...
Rhs Ch Shell		15618.2	...	1500	...
RHS Ch Flg		15880.3	...	259	...
RHS Blind Flang		16036.3	...	156	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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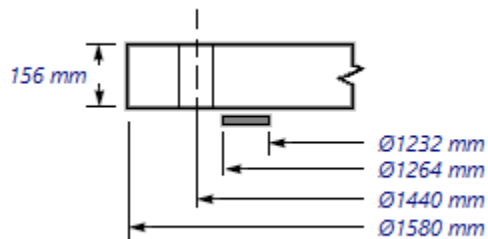
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 Flg Calc [Int P]: CH CO FL Flng: 9 9:30am Sep 9,2026

Flange Input Data Values

Description: CH CO FL :

LHS Blind Flg

Flange Type		Blind	
Design Pressure	P	4565.68	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1580.000	mm
Flange Thickness	t	156.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4523.894	mm
Diameter of Bolt Circle	C	1440.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	1275.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1264.000	mm
Gasket Inside Diameter	Gi	1232.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1249.745 mm

Basic Flange and Bolt Loads:



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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1250^2 \cdot 4566$$

$$= 571130.062 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1250 \cdot 3 \cdot 4566$$

$$= 78172.953 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(571130 + 78173 + 0, 0)$$

$$= 649303.062 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1250 + 0 \cdot 0 \cdot 0$$

$$= 196747.234 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((649303 + 0 + 4 \cdot 0 / 1250) / 172, 196747 / 172)$$

$$= 370.216 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	370.216	618.347	
Radial Distance between Bolts and Edge:	57.150	70.000	
Circ. Spacing between the Bolts:	120.650	161.229	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 618 \cdot 172 / (68.9 \cdot \pi (1264 + 1232))$$

$$= 19.671 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 618$$

$$= 1084486.88 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 649303.06 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1440 - 1250) / 2$$

$$= 95.1274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 649303 \cdot 95.1 \cdot 1 / (153^2 \cdot 1250) + 0.3 \cdot 1 \cdot 4566 \cdot 1250^2 / 153^2$$

$$= 130.73 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1084486 \cdot 95.1 \cdot 1 / (153^2 \cdot 1250)$$



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 Flg Calc [Int P]: CH CO FL Flng: 9 9:30am Sep 9,2026

= 65.71 N./mm²

Bolt Stress, Operating [BSO]:

= Wm1 / Ab

= 649303/618

= 102.98 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)

= (196747/618)

= 31.20 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	130.73	138.00	65.71	138.00
Bolting	102.98	172.00	31.20	172.00

Reqd. Blind Flange Thickness at Center 151.918 mm

Reqd. Blind Flange Thickness at Gasket 108.575 mm

Estimated M.A.W.P. (Operating) 4819 KPa.

Estimated Finished Weight of Flange at given Thk. 2370.6 kg.

Estimated Unfinished Weight of Forging at given Thk 2370.6 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.92, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 11 °C

Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	125.75	138.00	63.20	138.00
Bolting	102.98	172.00	31.20	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 148.920 mm

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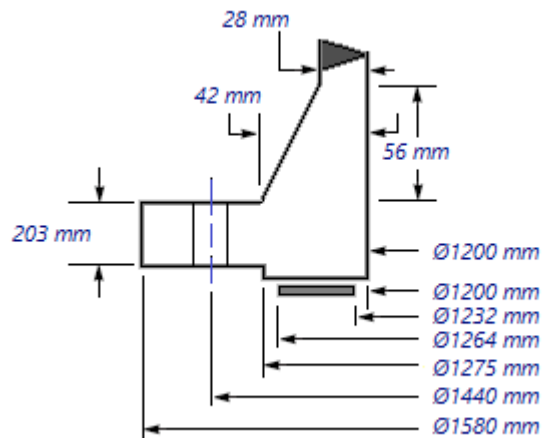
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 Flg Calc [Int P]: LHS Ch Flang Flng: 10 9:30am Sep 9,2026

Flange Input Data Values

Description: LHS Ch Flang :

LHS Ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	4561.91	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1200.000	mm
Flange Outside Diameter	A	1580.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	28.0000	mm
Thickness of Hub at Large End	gl	42.0000	mm
Length of Hub	h	56.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1440.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		28	
Flange Face Outside Diameter	Fod	1275.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1264.000	mm
Gasket Inside Diameter	Gi	1232.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4562 \cdot (1200/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 4562) + Ca$$

$$= 23.3385 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 25) / (603 + 0.6 \cdot 25)$$

$$= 5582.100 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 28) / (600 + 0.6 \cdot 28)$$

$$= 6264.115 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	200.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1206.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	39.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	25.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	78.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $b_o = N/2$	8.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1249.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1250^2 \cdot 4562$$

$$= 570658.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1250 \cdot 3 \cdot 4562$$

$$= 78108.375 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1206^2 \cdot 4562$$

$$= 531407.688 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 570658 - 531408$$

$$= 39250.574 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(570658 + 78108 + 0, 0)$$

$$= 648766.688 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1250 + 0 \cdot 0 \cdot 0$$

$$= 196747.234 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((648767 + 0 + 4 \cdot 0 / 1250) / 172, 196747 / 172)$$

$$= 369.910 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsm_{max}]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 57.1 + 6 \cdot 200/(3 + 0.5)$
 $= 457.157 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 1440 \cdot \sin(\pi/28)$
 $= 161.229 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{Bs/(2a + t)}, 1)$
 $= \max(\sqrt{161/(2 \cdot 57.1 + 200)}, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	369.910	618.347	
Radial Distance between Hub and Bolts:	69.850	78.000	
Radial Distance between Bolts and Edge:	57.150	70.000	
Circ. Spacing between the Bolts:	120.650	161.229	457.157

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [N_{min}]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 618 \cdot 172/(68.9 \cdot \pi(1264 + 1232))$
 $= 19.671 \text{ mm}$ [Note: Exceeds actual gasket width, 16.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or W_{gs}]:
 $= Sa \cdot Ab$
 $= 172 \cdot 618$
 $= 1084486.88 \text{ Kg}$

Gasket Load for the Operating Condition [HG]:
 $= Wo - H$
 $= 648767 - 570658$
 $= 78108.39 \text{ Kg}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (1440 - 1250) / 2$
 $= 95.1274 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5((C - B)/2 + hg)$
 $= 0.5((1440 - 1206)/2 + 95.1)$
 $= 106.0637 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (1440 - 1206 - 39) / 2$
 $= 97.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	531408.	97.5000	1.0000	51813.
Face Pressure, Mt	39251.	106.0637	1.0000	4163.
Gasket Load, Mg	78108.	95.1274	1.0000	7430.
Gasket Seating, Matm	1084487.	95.1274	1.0000	103166.

Total Moment for Operation, Mo 63407. Kg-m.
Total Moment for Gasket seating, Mg 103166. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 173.638 mm
Hub Ratio, h/h0 = HL / H0 0.323
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.560

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.875
Factor V per 4.16.5 0.351
Factor f per 4.16.5 1.194

Stress factor from Table 4.16.4 [K]:

= A / B
= 1580/1206
= 1.310

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.31² • log10(1.31))) / (1.31² - 1))/(1.31 - 1)
= 7.337

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
= (1.31²(1 + 8.55246 • log10(1.31)) - 1)/((1.0472 + 1.9448 • 1.31²) • (1.31 - 1))
= 1.793

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
= (1.31²(1 + 8.55246 • log10(1.31)) - 1)/(1.36136(1.31² - 1) • (1.31 - 1))
= 8.062

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
= (1.31² + 1)/(1.31² - 1)
= 3.792

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 8.06 • 25² • 174/0.35
= 2492336.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.88/174
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (200 • 0.005 + 1)/1.79 + 200³/2492336
= 4.330

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.19•63407/1206)/(4.33•39²)
= 93.52 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.19 \cdot 103166 / 1206) / (4.33 \cdot 39^2)$$

$$= 152.16 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 63407 / 1206) / (4.33 \cdot 200^2)$$

$$= 6.98 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 103166 / 1206) / (4.33 \cdot 200^2)$$

$$= 11.35 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (7.34 \cdot 63407 / (200^2 \cdot 1206)) - 3.79 \cdot 6.98$$

$$= 68.11 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (7.34 \cdot 103166 / (200^2 \cdot 1206)) - 3.79 \cdot 11.4$$

$$= 110.82 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (93.5 + \max(6.98, 68.1)) / 2$$

$$= 80.81 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (152 + \max(11.4, 111)) / 2$$

$$= 131.49 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= \text{Wm1} / \text{Ab}$$

$$= 648767 / 618$$

$$= 102.89 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (196747 / 618)$$

$$= 31.20 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	93.52	207.00	152.16	207.00
Radial Flange	6.98	138.00	11.35	138.00
Tangential Flange	68.11	138.00	110.82	138.00
Maximum Average	80.81	138.00	131.49	138.00
Bolting	102.89	172.00	31.20	172.00

Minimum Required Flange Thickness 197.993 mm

Estimated M.A.W.P. (Operating) 5582 KPa.
 Estimated Finished Weight of Flange at given Thk. 1364.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 1665.5 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	477.71	499.99	0.955	477.71	499.99	0.955
Tangential Flange	St	348.91	374.99	0.930	348.91	374.99	0.930
Tangential Flange	Sto	143.82	250.00	0.575	143.82	250.00	0.575
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	384.66	749.99	0.513	384.66	749.99	0.513
Combined	Sh + Sto	621.54	749.99	0.829	621.54	749.99	0.829
Radial Flange	Sr	35.75	499.99	0.071	35.75	499.99	0.071
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.53/1.28)

= 1.1911

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.44/1.23) if h/h0 < 0.35, else max (0.25; 0.28/0.76)

= 0.3586

In this case, the hub ratio h/h0, 0.3225 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	541.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	227.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.465 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1264^2 - 1232^2)$$

$$= 627.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207 \cdot 627 / (28 \cdot 22.1)$$

$$= 210.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(210; 508)$$

$$= 210.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(210; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 542)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (627 / (28 \cdot 22.1))$$

$$:: 290.0 \geq 142.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 627 + \pi/4 \cdot 4562 \cdot 1232^2) / (0.7 \cdot 22.1 \cdot 28)$$

$$:: 290.0 \geq 266.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 290 \leq 276 (627 / (28 \cdot 22.1))$$

$$:: 290.0 \leq 280.0 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 290 \leq 542 (1/0.47)$$



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:: 290.0 ≤ 1163.5 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.2 \cdot 22.1 \cdot 57.1$$

$$= 746.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 114 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 103166 / 1 \cdot 9807 \cdot 0.35 / (4.08 \cdot 201334 \cdot 25^2 \cdot 174 \cdot 0.3)$$

$$= 0.692 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 63407 / 1 \cdot 9807 \cdot 0.35 / (4.08 \cdot 196784 \cdot 25^2 \cdot 174 \cdot 0.3)$$

$$= 0.435 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 103166 / 1 \cdot 9807 \cdot 0.35 / (4.33 \cdot 201334 \cdot 25^2 \cdot 174 \cdot 0.3)$$

$$= 0.652 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 63407 / 1 \cdot 9807 \cdot 0.35 / (4.33 \cdot 196784 \cdot 25^2 \cdot 174 \cdot 0.3)$$

$$= 0.410 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.74, Temperature Reduction per Fig. UCS 66.1 = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	90.29	207.00	144.85	207.00
Radial Flange	7.70	138.00	12.35	138.00
Tangential Flange	63.24	138.00	101.46	138.00
Maximum Average	76.77	138.00	123.15	138.00
Bolting	102.89	172.00	31.20	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 190.627 mm

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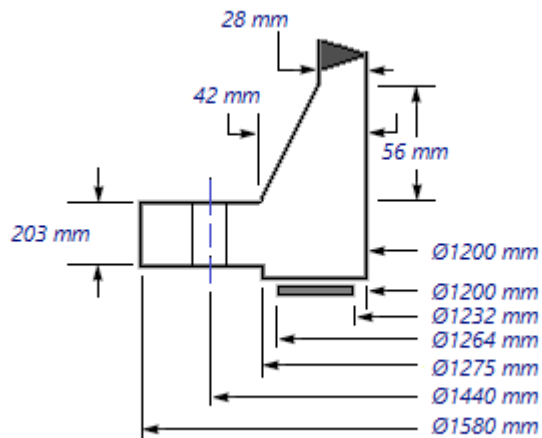
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Flange Input Data Values

Description: New Flange :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	4561.91	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1200.000	mm
Flange Outside Diameter	A	1580.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	28.0000	mm
Thickness of Hub at Large End	gl	42.0000	mm
Length of Hub	h	56.0000	mm
Flange Material	SA-765 II		
Flange Material UNS number	K03047		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1440.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	28		
Flange Face Outside Diameter	Fod	1275.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1264.000	mm
Gasket Inside Diameter	Gi	1232.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4562 \cdot (1200/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 4562) + Ca$$

$$= 23.3385 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 25) / (603 + 0.6 \cdot 25)$$

$$= 5582.100 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 28) / (600 + 0.6 \cdot 28)$$

$$= 6264.115 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	200.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1206.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	39.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	25.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	78.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	16.000 mm
Basic Gasket Width, $bo = N/2$	8.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1249.745 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1250^2 \cdot 4562$$

$$= 570658.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1250 \cdot 3 \cdot 4562$$

$$= 78108.375 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1206^2 \cdot 4562$$

$$= 531407.688 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 570658 - 531408$$

$$= 39250.574 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(570658 + 78108 + 0, 0)$$

$$= 648766.688 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1250 + 0 \cdot 0$$

$$= 196747.234 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((648767 + 0 + 4 \cdot 0/1250)/172, 196747/172)$$

$$= 369.910 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 57.1 + 6 \cdot 200/(3 + 0.5)$$

$$= 457.157 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1440 \cdot \sin(\pi / 28)$$

$$= 161.229 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{161/(2 \cdot 57.1 + 200)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	369.910	618.347	
Radial Distance between Hub and Bolts:	69.850	78.000	
Radial Distance between Bolts and Edge:	57.150	70.000	
Circ. Spacing between the Bolts:	120.650	161.229	457.157

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 618 \cdot 172 / (68.9 \cdot \pi (1264 + 1232))$$

$$= 19.671 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 618$$

$$= 1084486.88 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 648767 - 570658$$

$$= 78108.39 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1440 - 1250) / 2$$

$$= 95.1274 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (1440 - 1206) / 2 + 95.1$$

$$= 106.0637 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1440 - 1206 - 39) / 2$$

$$= 97.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	531408.	97.5000	1.0000	51813.
Face Pressure, Mt	39251.	106.0637	1.0000	4163.
Gasket Load, Mg	78108.	95.1274	1.0000	7430.
Gasket Seating, Matm	1084487.	95.1274	1.0000	103166.

Total Moment for Operation, Mo 63407. Kg-m.
Total Moment for Gasket seating, Mg 103166. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 173.638 mm
Hub Ratio, h/h0 = HL / H0 0.323
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.560

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.875
Factor V per 4.16.5 0.351
Factor f per 4.16.5 1.194

Stress factor from Table 4.16.4 [K]:

= A / B
= 1580/1206
= 1.310

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K² * log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.31² * log10(1.31))) / (1.31² - 1)) / (1.31 - 1)
= 7.337

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.31² * (1 + 8.55246 * log10(1.31)) - 1) / ((1.0472 + 1.9448 * 1.31²) * (1.31 - 1))
= 1.793

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.31² * (1 + 8.55246 * log10(1.31)) - 1) / (1.36136 * (1.31² - 1) * (1.31 - 1))
= 8.062

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.31² + 1) / (1.31² - 1)
= 3.792

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 8.06 * 25² * 174 / 0.35
= 2492336.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.88/174
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / (T + t³/d)
= (200 * 0.005 + 1) / (1.79 + 200³/2492336)
= 4.330

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.19 * 63407 / 1206) / (4.33 * 39²)
= 93.52 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.19 \cdot 103166 / 1206) / (4.33 \cdot 39^2)$$

$$= 152.16 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 63407 / 1206) / (4.33 \cdot 200^2)$$

$$= 6.98 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 103166 / 1206) / (4.33 \cdot 200^2)$$

$$= 11.35 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (7.34 \cdot 63407 / (200^2 \cdot 1206)) - 3.79 \cdot 6.98$$

$$= 68.11 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (7.34 \cdot 103166 / (200^2 \cdot 1206)) - 3.79 \cdot 11.4$$

$$= 110.82 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (93.5 + \max(6.98, 68.1)) / 2$$

$$= 80.81 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (152 + \max(11.4, 111)) / 2$$

$$= 131.49 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= \text{Wm1} / \text{Ab}$$

$$= 648767 / 618$$

$$= 102.89 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (196747 / 618)$$

$$= 31.20 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	93.52	207.00	152.16	207.00
Radial Flange	6.98	138.00	11.35	138.00
Tangential Flange	68.11	138.00	110.82	138.00
Maximum Average	80.81	138.00	131.49	138.00
Bolting	102.89	172.00	31.20	172.00

Minimum Required Flange Thickness

197.993 mm

Estimated M.A.W.P. (Operating)

5582 KPa.

Estimated Finished Weight of Flange at given Thk.

1364.3 kg.

Estimated Unfinished Weight of Forging at given Thk

1665.5 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	477.71	499.99	0.955	477.71	499.99	0.955
Tangential Flange	St	348.91	374.99	0.930	348.91	374.99	0.930
Tangential Flange	Sto	143.82	250.00	0.575	143.82	250.00	0.575
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	384.66	749.99	0.513	384.66	749.99	0.513
Combined	Sh + Sto	621.54	749.99	0.829	621.54	749.99	0.829
Radial Flange	Sr	35.75	499.99	0.071	35.75	499.99	0.071
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:
 = max(1; F1num / F1den)
 = max(1; 1.53/1.28)
 = 1.1911

Hub Stress Correction Factor [f2]:
 = max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.44/1.23) if h/ho < 0.35, else max (0.25; 0.28/0.76)
 = 0.3586

In this case, the hub ratio h/ho, 0.3225 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	541.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	22.0838 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	227.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.465 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1264^2 - 1232^2)$$

$$= 627.3 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207 \cdot 627 / (28 \cdot 22.1)$$

$$= 210.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(210; 508)$$

$$= 210.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(210; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 542)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (627 / (28 \cdot 22.1))$$

$$\therefore 290.0 \geq 142.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 627 + \pi/4 \cdot 4562 \cdot 1232^2) / (0.7 \cdot 22.1 \cdot 28)$$

$$\therefore 290.0 \geq 266.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 290 \leq 276 (627 / (28 \cdot 22.1))$$

$$\therefore 290.0 \leq 280.0 \text{ N./mm}^2 \text{ Check Fails}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 290 \leq 542 (1/0.47)$$



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:: 290.0 ≤ 1163.5 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.2 \cdot 22.1 \cdot 57.1$$

$$= 746.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 114 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 103166 / 1 \cdot 9807 \cdot 0.35 / (4.08 \cdot 201334 \cdot 25^2 \cdot 174 \cdot 0.3)$
 $= 0.692$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 63407 / 1 \cdot 9807 \cdot 0.35 / (4.08 \cdot 196784 \cdot 25^2 \cdot 174 \cdot 0.3)$
 $= 0.435$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 103166 / 1 \cdot 9807 \cdot 0.35 / (4.33 \cdot 201334 \cdot 25^2 \cdot 174 \cdot 0.3)$
 $= 0.652$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 63407 / 1 \cdot 9807 \cdot 0.35 / (4.33 \cdot 196784 \cdot 25^2 \cdot 174 \cdot 0.3)$
 $= 0.410$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.74, Temperature Reduction per Fig. UCS 66.1 = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -15 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	90.29	207.00	144.85	207.00
Radial Flange	7.70	138.00	12.35	138.00
Tangential Flange	63.24	138.00	101.46	138.00
Maximum Average	76.77	138.00	123.15	138.00
Bolting	102.89	172.00	31.20	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 190.627 mm

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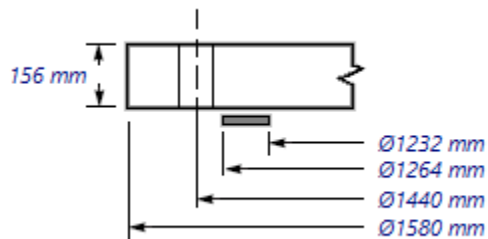
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Flange Input Data Values

Description: New Flange :

RHS Blind Flang

Flange Type		Blind	
Design Pressure	P	4565.49	KPa.
Design Temperature		100	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1580.000	mm
Flange Thickness	t	156.0000	mm
Flange Material		SA-765 II	
Flange Material UNS number		K03047	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4523.894	mm
Diameter of Bolt Circle	C	1440.000	mm
Nominal Bolt Diameter	a	57.1500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	1275.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1264.000	mm
Gasket Inside Diameter	Gi	1232.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	16.000 mm
Basic Gasket Width,	$bo = N/2$	8.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.127 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1249.745 mm

Basic Flange and Bolt Loads:



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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1250^2 \cdot 4565$$

$$= 571105.625 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1250 \cdot 3 \cdot 4565$$

$$= 78169.609 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(571106 + 78170 + 0, 0)$$

$$= 649275.250 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 7.13 \cdot \pi \cdot 1250 + 0 \cdot 0 \cdot 0$$

$$= 196747.234 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((649275 + 0 + 4 \cdot 0 / 1250) / 172, 196747 / 172)$$

$$= 370.200 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	370.200	618.347	
Radial Distance between Bolts and Edge:	57.150	70.000	
Circ. Spacing between the Bolts:	120.650	161.229	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 618 \cdot 172 / (68.9 \cdot \pi (1264 + 1232))$$

$$= 19.671 \text{ mm [Note: Exceeds actual gasket width, 16.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 618$$

$$= 1084486.88 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 649275.25 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1440 - 1250) / 2$$

$$= 95.1274 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 649275 \cdot 95.1 / (153^2 \cdot 1250) + 0.3 \cdot 1 \cdot 4565 \cdot 1250^2 / 153^2$$

$$= 130.73 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1084486 \cdot 95.1 / (153^2 \cdot 1250)$$



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= 65.71 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b
 = 649275/618
 = 102.97 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)
 = (196747/618)
 = 31.20 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	130.73	138.00	65.71	138.00
Bolting	102.97	172.00	31.20	172.00

Reqd. Blind Flange Thickness at Center 151.915 mm
 Reqd. Blind Flange Thickness at Gasket 108.575 mm

Estimated M.A.W.P. (Operating) 4819 KPa.
 Estimated Finished Weight of Flange at given Thk. 2370.6 kg.
 Estimated Unfinished Weight of Forging at given Thk 2370.6 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.92, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 11 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -10 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	125.75	138.00	63.20	138.00
Bolting	102.97	172.00	31.20	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 148.920 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flg		4565.7	164	3	1580	138	1
LHS Ch Flange		4561.9	211	3	1200	138	2
LHS Ch Shell		4561.9	28	3	1200	138	3
LHS Cone Shell		1557.5	20	3	1200	138	4
LHS Cone Shell		1567.7	20	3	1800	138	5
Main Shell		1561.3	16	3	1800	138	6
RHS Cone shell		1561.3	20	3	1800	138	7
RHS Cone shell		1557.5	20	3	1200	138	8
Rhs Ch Shell		4561.9	28	3	1200	138	9
RHS Ch Flg		4561.9	211	3	1200	138	10
RHS Blind Flang		4565.5	164	3	1580	138	11

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flg		4550	4804	5010	156	151.918	1
LHS Ch Flange		4550	5570	6264	203	197.993	2
LHS Ch Shell		4550	5570	6264	28	23.3387	3
LHS Cone Shell		1550	3818	4509	20	9.85257	4
LHS Cone Shell		1550	2235	2655	20	14.7963	5
Main Shell		1550	1958	2427	16	13.2867	6
RHS Cone shell		1550	2241	2655	20	14.7477	7
RHS Cone shell		1550	3818	4509	20	9.85257	8
Rhs Ch Shell		4550	5570	6264	28	23.3387	9
RHS Ch Flg		4550	5570	6264	203	197.993	10
RHS Blind Flang		4550	4804	5010	156	151.915	11

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1579 KPa.
 Shell Side MAPnc = 1960 KPa.
 Channel Side MAWP = 4629 KPa.
 Channel Side MAPnc = 5010 KPa.

Elements Suitable for Design Internal Pressure.

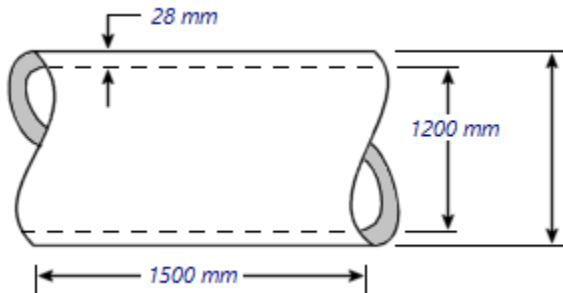
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 FileName : 130-C1& C2-Final 23rd March Page 53 of 317
 Element: LHS Ch Shell Step: 8 9:30am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	4561.940 KPa.	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	100 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1200.000 mm	Actual Stress	112.8 N./mm ²
Spec. Min. Thk t,e, or sa	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	59187.6 cm ²	SG of Contents	1.013
Element Volume	1696763.9 cm ³	Weight of Contents	1717.1 Kg
Empty Weight	1255.8 Kg	Operating Weight	2972.9 Kg
		Comp. All. Ope. Stress	143.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4562 \cdot 603) / (138 \cdot 1 - 0.6 \cdot 4562) \\
 &= 20.3387 + 3.0000 = 23.3387 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 11.940 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 25) / (603 + 0.6 \cdot 25) \\
 &= 5582 - 11.9 = 5570 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 28) / (600 + 0.6 \cdot 28) \\
 &= 6264 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4562 \cdot (603 + 0.6 \cdot 25)) / (1 \cdot 25) \\
 &= 112.778 \text{ N./mm}^2
 \end{aligned}$$



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 Element: LHS Ch Shell Step: 8 9:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.280 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 28$, $t_r = 20.6$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -8 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4628.898 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.826	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-8 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1256.00	1499.97	50.24	1.1942	0.0031466	112.74

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 113) / (3 \cdot 50.2) = 2992 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.341	1256.00	1499.97	289.34	1.1942	0.0002245	22.44

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.4) / (3 \cdot 289) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1256.00	99165.99	50.24	50.0000	0.0004363	43.62

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 43.6) / (3 \cdot 50.2) = 1158 \text{ KPa.}$$

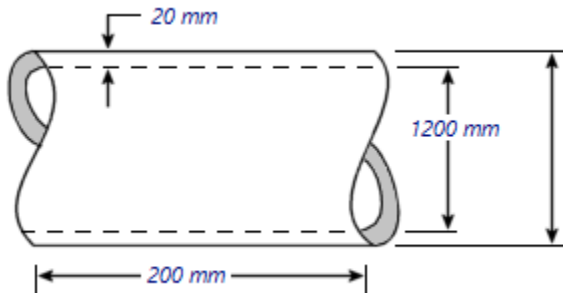
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 Element: LHS Cone Shell Step: 9 9:30am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Cone Shell

Cylindrical Shell Design Information:			
Design Pressure P	1557.512 KPa.	Int. Design Temperature	65 °C
Material	SA-516 70	Ext. Design Temperature	65 °C
Length	200.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1200.000 mm	Actual Stress	56.2 N./mm ²
Spec. Min. Thk t,e, or sa	20.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	7791.2 cm ²	SG of Contents	0.637
Element Volume	109570.2 cm ³	Weight of Contents	69.8 Kg
Empty Weight	118.8 Kg	Operating Weight	188.6 Kg
		Comp. All. Ope. Stress	137.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1558 \cdot 603) / (138 \cdot 1 - 0.6 \cdot 1558) \\
 &= 6.8526 + 3.0000 = 9.8526 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.512 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 17) / (603 + 0.6 \cdot 17) \\
 &= 3826 - 7.51 = 3818 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 20) / (600 + 0.6 \cdot 20) \\
 &= 4509 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1558 \cdot (603 + 0.6 \cdot 17)) / (1 \cdot 17) \\
 &= 56.184 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Cone Shell Step: 9 9:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 6.95$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

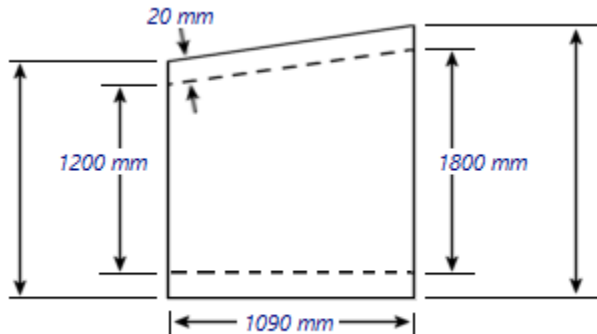
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1578.586 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.409	Temperature Difference	24 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Cone Shell

Conical Head/Section Design Information:					
Design Pressure	P	1567.673 KPa.	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		1090.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1800.000 mm	Actual Stress	96.0 N./mm ²	
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		54896.7 cm ²	SG of Contents	1.000	
Element Volume		1316397.2 cm ³	Weight of Contents	1315.6 Kg	
Empty Weight		922.7 Kg	Operating Weight	2238.3 Kg	
			Comp. All. Ope. Stress	139.3 N./mm ²	
			To End Diameter	1200.000 mm	
Line of Support	Neither End				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$$
 per Appendix 1-4 (e)

$$= (1568 \cdot 1807) / (2 \cdot 0.88 \cdot (138 \cdot 1 - 0.6 \cdot 1568))$$

$$= 11.7963 + 3.0000 = 14.7963 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 17.673 KPa.

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138 \cdot 1 \cdot 17 \cdot 0.88) / (1807 + 1.2 \cdot 17 \cdot 0.88)$$

$$= 2252 - 17.7 = 2235 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138 \cdot 1 \cdot 20 \cdot 0.88) / (1800 + 1.2 \cdot 20 \cdot 0.88)$$

$$= 2655 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:



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$$= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (1568 \cdot (1807 + 1.2 \cdot 17 \cdot 0.88)) / (2 \cdot 1 \cdot 17 \cdot 0.88)$$

$$= 96.044 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.636 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 11.9$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.7$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

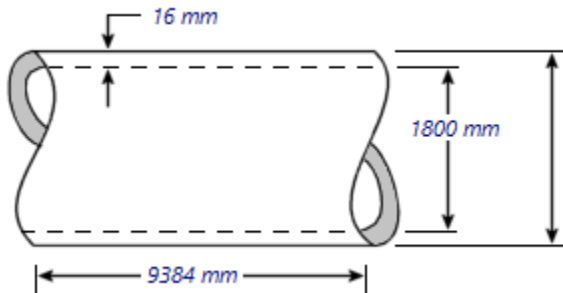
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1578.586 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.699	Temperature Difference	17 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	1561.258 KPa.	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		9384.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1800.000 mm	Actual Stress	109.4 N./mm ²	
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		540086.6 cm ²	SG of Contents	0.637	
Element Volume		18409732.0 cm ³	Weight of Contents	11720.2 Kgf	
Empty Weight		6639.0 Kgf	Operating Weight	18359.2 Kgf	
			Comp. All. Ope. Stress	121.2 N./mm ²	
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency	100 %		Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1561 \cdot 903) / (138 \cdot 1 - 0.6 \cdot 1561) \\
 &= 10.2867 + 3.0000 = 13.2867 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 11.258 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 13) / (903 + 0.6 \cdot 13) \\
 &= 1970 - 11.3 = 1958 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 16) / (900 + 0.6 \cdot 16) \\
 &= 2427 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1561 \cdot (903 + 0.6 \cdot 13)) / (1 \cdot 13) \\
 &= 109.390 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell Step: 11 9:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.881 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 16$, $t_r = 10.4$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

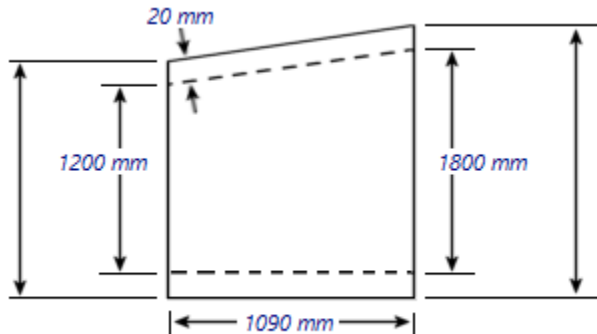
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1578.586 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	16.000 mm	Basic MDMT	-31 °C
Stress/Thk. Ratio	0.800	Temperature Difference	11 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-42 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Cone shell

Conical Head/Section Design Information:					
Design Pressure	P	1561.258 KPa.	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	65 °C	
Length		1090.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1800.000 mm	Actual Stress	95.7 N./mm ²	
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.	20.000 mm	
Surface Area		54896.7 cm ²	SG of Contents	0.637	
Element Volume		1316397.4 cm ³	Weight of Contents	838.1 Kg	
Empty Weight		922.7 Kg	Operating Weight	1760.8 Kg	
			Comp. All. Ope. Stress	121.2 N./mm ²	
			To End Diameter	1200.000 mm	
Line of Support	Neither End				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D) / (2 \cdot \cos(a) \cdot (S \cdot E - 0.6 \cdot P))$$
 per Appendix 1-4 (e)

$$= (1561 \cdot 1807) / (2 \cdot 0.88 \cdot (138 \cdot 1 - 0.6 \cdot 1561))$$

$$= 11.7477 + 3.0000 = 14.7477 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 11.258 KPa.

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138 \cdot 1 \cdot 17 \cdot 0.88) / (1807 + 1.2 \cdot 17 \cdot 0.88)$$

$$= 2252 - 11.3 = 2241 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D + 1.2 \cdot t \cdot \cos(a))$$
 per App 1-4(e)

$$= (2 \cdot 138 \cdot 1 \cdot 20 \cdot 0.88) / (1800 + 1.2 \cdot 20 \cdot 0.88)$$

$$= 2655 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:



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Element: RHS Cone shell Step: 12 9:30am Sep 9,2026

$$\begin{aligned} &= (P \cdot (D + 1.2 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a)) \\ &= (1561 \cdot (1807 + 1.2 \cdot 17 \cdot 0.88)) / (2 \cdot 1 \cdot 17 \cdot 0.88) \\ &= 95.651 \text{ N./mm}^2 \end{aligned}$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.636 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 11.9$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.7$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

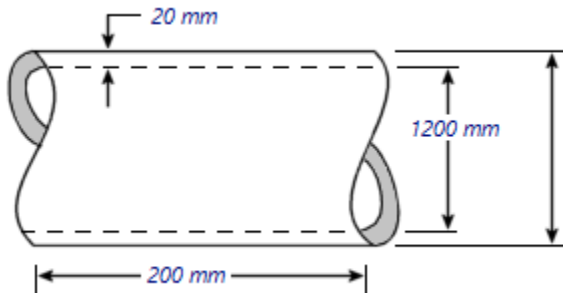
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1578.586 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.699	Temperature Difference	17 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS Cone shell

Cylindrical Shell Design Information:					
Design Pressure	P	1557.512 KPa.	Int. Design Temperature	65	°C
Material		SA-516 70	Ext. Design Temperature	65	°C
Length		200.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1200.000 mm	Actual Stress	56.2	N./mm ²
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.	20.000	mm
Surface Area		7791.2 cm ²	SG of Contents	0.637	
Element Volume		109569.9 cm ³	Weight of Contents	69.8	Kgf
Empty Weight		118.8 Kg	Operating Weight	188.6	Kgf
			Comp. All. Ope. Stress	131.4	N./mm ²
Radiography Information:					
Spot Radiography per UW-11(b) Type 2			User Defined		
Circ. Joint Efficiency		80 %	Long. Joint Eff.		E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1558 \cdot 603) / (138 \cdot 1 - 0.6 \cdot 1558) \\
 &= 6.8526 + 3.0000 = 9.8526 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 7.512 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 17) / (603 + 0.6 \cdot 17) \\
 &= 3826 - 7.51 = 3818 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 20) / (600 + 0.6 \cdot 20) \\
 &= 4509 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1558 \cdot (603 + 0.6 \cdot 17)) / (1 \cdot 17) \\
 &= 56.184 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 130-C1& C2-Final 23rd March Page 64 of 317
Element: RHS Cone shell Step: 13 9:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.639 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 6.95$, $c = 3$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.41$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1578.586 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-33 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.409	Temperature Difference	24 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

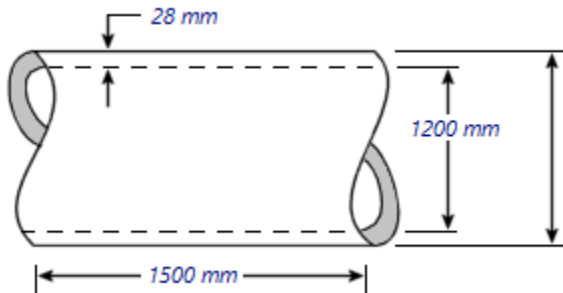
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 Element: Rhs Ch Shell Step: 14 9:30am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Rhs Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	4561.940 KPa.	Int. Design Temperature	100 °C
Material	SA-516 70	Ext. Design Temperature	100 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1200.000 mm	Actual Stress	112.8 N./mm ²
Spec. Min. Thk t,e, or sa	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	59187.6 cm ²	SG of Contents	1.013
Element Volume	1696763.9 cm ³	Weight of Contents	1717.1 Kg
Empty Weight	1255.8 Kg	Operating Weight	2972.9 Kg
		Comp. All. Ope. Stress	137.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4562 \cdot 603) / (138 \cdot 1 - 0.6 \cdot 4562) \\
 &= 20.3387 + 3.0000 = 23.3387 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 11.940 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 25) / (603 + 0.6 \cdot 25) \\
 &= 5582 - 11.9 = 5570 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 28) / (600 + 0.6 \cdot 28) \\
 &= 6264 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4562 \cdot (603 + 0.6 \cdot 25)) / (1 \cdot 25) \\
 &= 112.778 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Rhs Ch Shell Step: 14 9:30am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.280 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 28$, $t_r = 20.6$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -15 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	4628.898 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	-15 °C
Stress/Thk. Ratio	0.826	Temperature Difference	10 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-24 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Rhs Ch Shell

Cylindrical Shell From 90 to 100 Ext. Chart: CS-2 at 100 °C

Elastic Modulus from Chart: CS-2 at 100 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1256.00	1500.03	50.24	1.1943	0.0031465	112.74

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 113) / (3 \cdot 50.2) = 2992 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.341	1256.00	1500.03	289.34	1.1943	0.0002245	22.44

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 22.4) / (3 \cdot 289) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	1256.00	99168.58	50.24	50.0000	0.0004363	43.62

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 43.6) / (3 \cdot 50.2) = 1158 \text{ KPa.}$$



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 Shop Pressure Test Results: Step: 15 9:30am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2052.161 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2015.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1736.444 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2216.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2216.500 KPa.
App 27-4	= M.A.W.P.	1578.586 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6017.568 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6495.683 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5091.789 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4628.898 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	160.9	234.0	0.688	6507.45
LHS Cone Shell	92.3	234.0	0.395	2559.77
LHS Cone Shell	157.2	234.0	0.672	2565.65
Main Shell	179.8	234.0	0.768	2565.65
RHS Cone shell	157.2	234.0	0.672	2565.65
RHS Cone shell	92.3	234.0	0.395	2559.77
Rhs Ch Shell	160.9	234.0	0.688	6507.45

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Noz N6	Nozzle	138.00	138.00	1.000
Npz N2 250	Nozzle	138.00	138.00	1.000
Noz N3 80	Nozzle	138.00	138.00	1.000
Noz N1 150	Nozzle	138.00	138.00	1.000
Noz N4 100	Nozzle	138.00	138.00	1.000
Noz N5 200	Nozzle	138.00	138.00	1.000
D1 nozzle 50	Nozzle	138.00	138.00	1.000
K1 50	Nozzle	138.00	138.00	1.000
K2 50	Nozzle	138.00	138.00	1.000
K3 50	Nozzle	138.00	138.00	1.000
K4 50	Nozzle	138.00	138.00	1.000
K5 50	Nozzle	138.00	138.00	1.000
K6 50	Nozzle	138.00	138.00	1.000
K7 50	Nozzle	138.00	138.00	1.000
K8 50	Nozzle	138.00	138.00	1.000
Noz N7	Nozzle	138.00	138.00	1.000

Minimum

1.000



Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flange	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Cone Shell	138.00	138.00	1.000
LHS Cone Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Cone shell	138.00	138.00	1.000
RHS Cone shell	138.00	138.00	1.000
Rhs Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Noz N6	134.38	179.40	0.749
Npz N2 250	42.15	179.40	0.235
Noz N3 80	7.54	179.40	0.042
Noz N1 150	10.03	179.40	0.056
Noz N4 100	8.67	179.40	0.048
Noz N5 200	27.95	179.40	0.156
D1 nozzle 50	6.83	179.40	0.038
K1 50	6.83	179.40	0.038
K2 50	6.83	179.40	0.038
K3 50	6.83	179.40	0.038
K4 50	6.83	179.40	0.038
K5 50	6.83	179.40	0.038
K6 50	6.83	179.40	0.038
K7 50	6.83	179.40	0.038
K8 50	6.83	179.40	0.038
Noz N7	134.38	179.40	0.749

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2052.161 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2015.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2548.008 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1736.444 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2216.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2216.500 KPa.
App 27-4	= M.A.W.P.	1578.586 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6017.568 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	5915.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6495.683 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5091.789 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	6506.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	6506.500 KPa.
App 27-4	= M.A.W.P.	4628.898 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1550 • 1

= 2015.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 4550 • 1

= 5915.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	146.5	234.0	0.626	5926.76
LHS Cone Shell	73.1	234.0	0.312	2026.76
LHS Cone Shell	124.5	234.0	0.532	2032.64
Main Shell	142.4	234.0	0.609	2032.64
RHS Cone shell	124.5	234.0	0.532	2032.64
RHS Cone shell	73.1	234.0	0.312	2026.76
Rhs Ch Shell	146.5	234.0	0.626	5926.76

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Noz N6	Nozzle	138.00	138.00	1.000
Npz N2 250	Nozzle	138.00	138.00	1.000
Noz N3 80	Nozzle	138.00	138.00	1.000
Noz N1 150	Nozzle	138.00	138.00	1.000
Noz N4 100	Nozzle	138.00	138.00	1.000
Noz N5 200	Nozzle	138.00	138.00	1.000
D1 nozzle 50	Nozzle	138.00	138.00	1.000
K1 50	Nozzle	138.00	138.00	1.000
K2 50	Nozzle	138.00	138.00	1.000



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K3 50	Nozzle	138.00	138.00	1.000
K4 50	Nozzle	138.00	138.00	1.000
K5 50	Nozzle	138.00	138.00	1.000
K6 50	Nozzle	138.00	138.00	1.000
K7 50	Nozzle	138.00	138.00	1.000
K8 50	Nozzle	138.00	138.00	1.000
Noz N7	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flange	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Cone Shell	138.00	138.00	1.000
LHS Cone Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Cone shell	138.00	138.00	1.000
RHS Cone shell	138.00	138.00	1.000
Rhs Ch Shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS Blind Flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	108.00	108.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Noz N6	122.39	179.40	0.682
Npz N2 250	33.40	179.40	0.186
Noz N3 80	5.97	179.40	0.033
Noz N1 150	7.95	179.40	0.044
Noz N4 100	6.87	179.40	0.038
Noz N5 200	22.14	179.40	0.123
D1 nozzle 50	5.41	179.40	0.030
K1 50	5.41	179.40	0.030
K2 50	5.41	179.40	0.030
K3 50	5.41	179.40	0.030
K4 50	5.41	179.40	0.030
K5 50	5.41	179.40	0.030
K6 50	5.41	179.40	0.030
K7 50	5.41	179.40	0.030
K8 50	5.41	179.40	0.030
Noz N7	122.39	179.40	0.682

Elements Suitable for Test Pressure.

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 ASME TS Calc: Pressure Test

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 Case: 1 9:30a Sep 9,2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Test Pressure		2015.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		5915.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	21.11 °C
Channel Operating Temperature	Tcop	21.11 °C
Tube Operating Temperature	Ttop	21.11 °C
Tubesheet Operating Temperature	Tshop	21.11 °C
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm
Circumferential Joint Efficiency	Eecc,w	1.00
Length	Lecc	1090.000 mm

Large Diameter Main Shell Cylinder Properties:



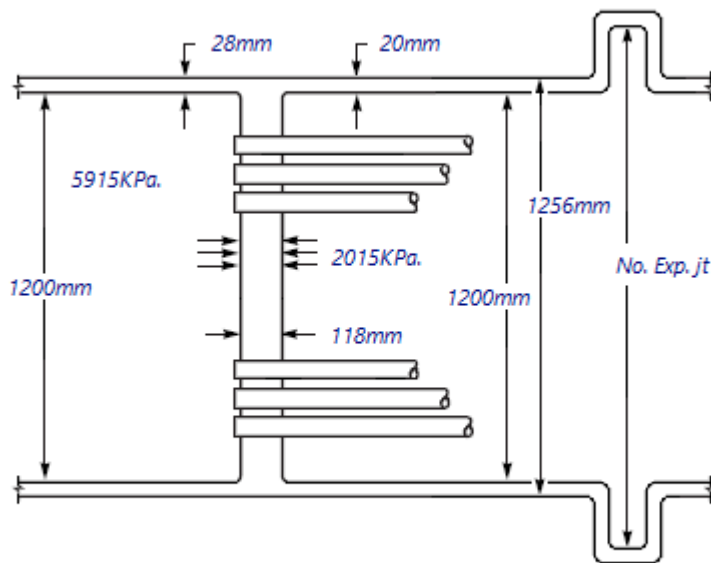
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Material	SA-516 70
Material UNS Number	K02700
Allowable Stress at Temperature	138.00 N./mm ²
Allowable Stress at Operating Temperature	138.00 N./mm ²
Allowable Stress at Ambient	138.00 N./mm ²
Thickness	ts,L 16.0000 mm
Internal Corrosion Allowance	3.0000 mm
External Corrosion Allowance	0.0000 mm
Inside Diameter	DS,L 1800.000 mm
Circumferential Joint Efficiency	Es,L,w 1.00
Length	Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%



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Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (5915 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 5915)$$

$$= 0.6807 + 0.0000 = 0.6807 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 122.73) / (3 \cdot 12) = 13593 \text{ KPa.}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6084	25.40	11964.00	41.75	50.0000	0.0006311	63.10
$EMAWP = (4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 63.1) / (3 \cdot 41.7) = 2015 \text{ KPa.}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6807 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	5915.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.



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Required External Design Pressure 2015.00 KPa.
 Required Thickness due to Shell Side pressure 0.6084 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/118$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(25.4 - 2 \cdot 2.11 (202701232/202701232) \cdot (108/138) \cdot (0); 25.4 - 2 \cdot 2.11)$
 $= 25.4000 \text{ mm}$

Equivalent outer tube limit circle diameter $[Do]$:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 564 + 25.4$
 $= 1154.200 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:
 $= (p - dt) / p$
 $= (31.8 - 25.4) / 31.8$
 $= 0.200$

Equivalent outer tube limit radius $[ao]$:
 $= Do / 2$
 $= 1154/2$
 $= 577.1000 \text{ mm}$

Effective tube pitch $[p^*]$:
 $= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$
 $= 31.8 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1154 \cdot 31.8) / (\pi \cdot 1154^2) }$
 $= 31.7500 \text{ mm}$

Effective ligament efficiency for bending $[\mu^*]$:
 $= (p^* - d^*) / p^*$
 $= (31.8 - 25.4) / 31.8$
 $= 0.2000$

Ratio $[ps]$:
 $= as / ao$
 $= 600/577$
 $= 1.0397$

Ratio $[pc]$:
 $= ac / ao$
 $= 600/577$
 $= 1.0397$

Compute parameter $[xt]$:
 $= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$
 $= 1 - 1151 ((25.4 - 2 \cdot 2.11) / (2 \cdot 577))^2$
 $= 0.6124$



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Determine parameter [xs]:

$$= 1 - Nt / (2 \cdot a_o)^2$$

$$= 1 - 1151 (25.4 / (2 \cdot 577))^2$$

$$= 0.4426$$

Determine the value [h'g]:

$$= \max(h_g - CATC; 0) \text{ (For pressure only cases)}$$

$$= \max(0 - 0; 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Stiffness of the Conical Section [Kecc]:

$$= 0.8 \cdot \pi \cdot tecc (Decc, S + tecc) Eecc / Lecc$$

$$= 0.8 \cdot \pi \cdot 20 (1200 + 20) 202701232 / 1090$$

$$= 1162936.8750 \text{ Kg/mm}$$

Axial Stiffness of the Large Diameter Section [Ks,L]:

$$= \pi \cdot ts, L (Ds, L + ts, L) Es, L / Ls, L$$

$$= \pi \cdot 16 (1800 + 16) 202701232 / 9384$$

$$= 201071.5312 \text{ Kg/mm}$$

Axial Shell Stiffness of the Small Diameter Section [Ks]:

$$= \pi \cdot ts (Ds + ts) Es / Ls$$

$$= 3.14 \cdot 20 (1200 + 20) 202701232 / 200$$

$$= 7922507.0000 \text{ Kg/mm}$$

Overall Effective Shell Stiffness for Kettle Exchangers [Ks*]:

$$= Ks \cdot Ks, L \cdot Kecc / (2Kecc \cdot Ks, L + 2Ks, L \cdot Ks + Ks \cdot Kecc)$$

$$= 7922507 \cdot 201072 \cdot 1162936 /$$

$$(2 \cdot 1162936 \cdot 201072 + 2 \cdot 201072 \cdot 7922507 + 7922507 \cdot 1162936)$$

$$= 143976.3594 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 202701232 / 11964$$

$$= 266.7340 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 143976 / (1151 \cdot 267)$$

$$= 0.46896$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 143976 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1200 + 20)20)$$

$$= 0.0116 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.012 \cdot 202701232 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 3456324096.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1200 + 20) \cdot 3.46E+09 / 118^3 \cdot (1 + 118 \cdot 0.012 + (118 \cdot 0.012)^2 / 2)$$

$$= 51063192.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - \nu_s / (2(Ds + ts)))$$



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$$= 1200^2 / (4 \cdot 202701232 \cdot 20) \cdot (1 - 1200 \cdot 0.3 / (2(1200 + 20)))$$

$$= 0.75695E-01 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1200 + 28)28}$$

$$= 0.0098 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0098 \cdot 202701232 \cdot 28^3 / (6(1 - 0.3^2))$$

$$= 7989422592.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1200 + 28 / \cos(0)) \cdot 7.99E+09 / 118^3 \cdot (1 + 118 \cdot 0.0098 \cdot (1 + 1) / 2 + (118 \cdot 0.0098)^2 / 2 \cdot 1)$$

$$= 101245488.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1200^2 / (4 \cdot 202701232 \cdot 28) \cdot (1 - 1200 \cdot 0.3 / (2(1200 + 28)))$$

$$= 0.54128E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zeta α]:

$$zeta\alpha = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 3.716536 ; \mu^* = 0.200000 ; E^* = 31489462 \text{ KPa}$$

$$E^*/E = 0.155349 ; \nu^* = 0.458572$$

Note: As h/p (3.717) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2)1151 \cdot 202701232 \cdot 2.11(25.4 - 2.11)577^2) / (31489462 \cdot 11964 \cdot 118^3))^{1/4}$$

$$= 3.2883$$

Values from Table 4.18.3

$$Z_d = 0.041677 ; Z_v = 0.087307 ; Z_m = 0.436628$$

$$Z_a = 0.310011E+01 ; Z_w = 0.087307$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1256 / 1154$$

$$= 1.0882$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31489462 \cdot (51063192 + 101245488 + 202701232 \cdot \ln(1.09))$$

$$= 2.9134$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)^F$$

$$= (1 + 0.46)^{2.91}$$

$$= 4.2494$$



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Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.04 - 1 - 4.25 \cdot 0.087) / (1 + 4.25 \cdot 0.44)$$

$$= -0.116032$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.042 + -0.12 \cdot 0.087) / 2 \cdot 3.29^4$$

$$= 1.8442$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.087 + -0.12 \cdot 0.44) / 2 \cdot 3.29^4$$

$$= 2.1421$$

Compute parameter [U]:

$$= (Zw + (ps - 1)Zm)Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.087 + (1.04 - 1)0.44)3.29^4 / (1 + 4.25 \cdot 0.44)$$

$$= 4.2843$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [Y]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= ps \cdot ks \cdot \beta s \cdot \delta S (1 + h \cdot \beta s)$$

$$= 1.04 \cdot 3.46E+09 \cdot 0.012 \cdot 0.075694516 (1 + 118 \cdot 0.012)$$

$$= 7512.8462 \text{ mm}^2$$

Calculate parameter [ω*s]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega s$$

$$= 577^2 (1.04^2 - 1) \cdot (1.04 - 1) / 4 - 7513$$

$$= -7245.4404 \text{ mm}^2$$

Parameter [wc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.04 \cdot 7.99E+09 [0.054128472 (1 + 0.0098 \cdot 1 \cdot 118) -$$

$$0.000 (1/118 + 1 \cdot 0.0098/2)]$$

$$= 9507.2402 \text{ mm}^2$$

Parameter [ω*c]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega c$$

$$= 577^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.04 - 1) / 2] - 9507$$

$$= -9239.8330 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Small End Coefficient [As]:

$$= Ds(Ds + ts)$$

$$= 1200 (1200 + 20)$$

$$= 14640.0010 \text{ cm}^2$$

Large End Coefficient [As,L]:



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$$= Ds, L(Ds, L + ts, L)$$

$$= 1800 (1800 + 16)$$

$$= 32688.0020 \text{ cm}^2$$

Diameter Difference [Decc]:

$$= Decc, L - Decc, S$$

$$= 1800 - 1200$$

$$= 600.0000 \text{ mm}$$

Equivalent Poisson's ratio for Kettle Exchangers [vs*]:

$$= 2 \cdot Ks \cdot As [As \cdot vs / Ks + (vecc - (Lecc^2 - 2 \cdot vecc \cdot Lecc^2 - 3 \cdot \Delta ecc^2) / (8 \cdot Lecc^3) \cdot (Decc^2 + Lecc^2) 0.5) (Decc, L + Decc, S) (Decc, S + tecc) / (5 \cdot Kecc) + As, L \cdot vs, L / (2 \cdot Ks, L)]$$

$$= 2 \cdot 143976 / 14640 [14640 \cdot 0.3 / 7922507 + [0.3 \cdot (1090^2 - 2 \cdot 0.3 \cdot 1090^2 - 3 \cdot 0.3^2) / (8 \cdot 1090^3) \cdot \sqrt{(600^2 + 1090^2)}] (1800 + 1200) (1200 + 20) / (5 \cdot 1162936) + 32688 \cdot 0.3 / (2 \cdot 201072)]$$

$$= 0.5367$$

Note: Use vs* in place of vs when determining P's

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \}$$

$$= 2015 \{ 0.44 + 2(1 - 0.44) 0.3 + [2/0.47 \cdot (1200/1154)^2] 0.54 - [(1.04^2 - 1) / (1 \cdot 0.47)] - [(1 - 1) / (2 \cdot 1 \cdot 0.47)] [(0^2 - (1200)^2) / 1154^2] \}$$

$$= 6203.0732 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.61 + 2(1 - 0.61) 0.3 + 1/(1 \cdot 0.47)] 5915$$

$$= 17610.9434 \text{ KPa.}$$

Pressure [Py]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 1151 \cdot 267 \cdot 0 / (3.14 \cdot 577^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 4.28 \cdot 0 / (2 \cdot \pi \cdot 577^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(4.28/577^2) \cdot (-7245 \cdot 2015 - -9240 \cdot 5915)$$

$$= -515.2565 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega_s \cdot P*s - \omega_c \cdot P*c)$$

$$= (4.28/577^2) \cdot (7513 \cdot 0 - 9507 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot (P's - P't + Py + Pw + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.47 / (1 + 1 \cdot 0.47 (1.84 + (1.04 - 1) 2.14)) \cdot (6203 - 17611 + 0 + 0 + -0 + -515)$$

$$= -2935.6008 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega_s \cdot P*s - \omega_c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$



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$$= [((-7245 \cdot 2015 - -9240 \cdot 5915) - (7513 \cdot 0 - 9507 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.25 \cdot 0.44)$$

$$= 1430.4642 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.12 + 2 \cdot 1430 / (-2936 \cdot 577^2)$$

$$= -0.14473$$

Fm Value from Table 4.18.3 = 0.072364

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.072 / 0.2) \cdot (2 \cdot 577 / (118 - 0))^2 \cdot -2936$$

$$= -152.4413 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.057 / 0.2) \cdot (2 \cdot 577 / (62.9 - 0))^2 \cdot -2479$$

$$= -356.2377 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 62.9 + 0 + 0$$

$$= 62.9121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.2 \cdot 118 / 577) \cdot 190$$

$$= 15.5395 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp)$$

$$= (9690 \cdot -1064) / (0.2 \cdot 7.7 \cdot 3629)$$

$$= -184.4543 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + Cats + Catc$$

$$= 7.7 + 0 + 0$$

$$= 7.7033 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(HreqB; HreqS)$$

$$= \max(62.9; 7.7)$$

$$= 62.9121 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.056, F_{tmax} = 1.7$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs)$$

$$= ((2015 \cdot 0.44 - 5915 \cdot 0.61) - (-2936) \cdot 0.056) / (0.61 - 0.44)$$

$$= -15.1089 \text{ N./mm}^2$$



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Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((2015 \cdot 0.44 - 5915 \cdot 0.61) - (-2936) \cdot 1.7) / (0.61 - 0.44)$$

$$= 13.2680 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 15.1 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.042 + -0.14 \cdot 0.087)3.29^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (d_t^2 + (d_t - 2 \cdot t_t)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 202701232 / 205) }$$

$$= 138.9286$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 1500 / 8.27$$

$$= 181.4227$$

Buckling Allowable Stress [Stb]:

$$= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) (C_t \leq F_t)$$

$$= (1 / 2) \cdot (\pi^2 \cdot 202701232 / 181^2)$$

$$= 194.746 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 15.1 \cdot 1.54$$

$$= 237.85 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 195$$

$$= 3065.793 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 195$$

$$= 3000.1914 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$



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= 195/195
 = 1.000

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:
 = $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 195$
 = 1941.3005 Kgf

Minimum Length of Weld [ar]:
 = $2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$
 = $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$
 = 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm
 Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 3066 is >= tube strength 3066 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Small Shell Cylinder Axial Membrane Allowable Stress:
 = $Ss \cdot Es, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Small Cylinder [σ_s, m]:
 = $ao^2 / (ts(Ds + ts)) [Pe + (Ds^2 / (4 \cdot ao^2) - 1)(Ps - Pt)] + Ds^2 \cdot Pt / (4 \cdot ts(Ds + ts))$
 = $577^2 / (20 (1200 + 20)) [-2936 + (1200^2 / (4 \cdot 577^2) - 1)(2015 - 5915)] + 1200^2 \cdot 5915 / (4 \cdot 20 (1200 + 20))$
 = 42.8954 N./mm²

Large Shell Cylinder Axial Membrane Allowable Stress:
 = $Ss, L \cdot Es, L, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Large Cylinder [σ_s, L, m]:
 = $ao^2 / (ts, L(Ds, L + ts, L)) [Pe + (Ds, L^2 / (4 \cdot ao^2) - 1)(Ps - Pt)] + Ds, L^2 \cdot Pt / (4 \cdot ts, L(Ds, L + ts, L))$
 = $577^2 / (20 (1800 + 16)) [-2936 + (1800^2 / (4 \cdot 577^2) - 1)(2015 - 5915)] + 1800^2 \cdot 5915 / (4 \cdot 16 (1200 + 20))$
 = 67.2306 N./mm²

Cone Axial Membrane Allowable Stress:
 = $Secc \cdot Eecc, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Cone Large End [$\sigma_{ecc, L, m}$]:
 = $ao^2 / (tecc(Decc, L + tecc) \cos(\text{angle})) [Pe + (Decc, L^2 / (4 \cdot ao^2) - 1)(Ps - Pt)] + Decc, L^2 \cdot Pt / (4 \cdot tecc(Decc, L + tecc) \cos(\text{angle}))$
 = $577^2 / (20 (1800 + 20) \cos(28.8)) [-2936 + (1800^2 / (4 \cdot 577^2) - 1)(2015 - 5915)] + 1800^2 \cdot 5915 / (4 \cdot 20 (1800 + 20) \cos(28.8))$
 = 61.2596 N./mm²

Circumferential Load, Cone Large End (used in Appendix 1-5) [f1']:
 = $\sigma_{ecc, L, m} \cdot tecc \cdot \cos(\text{theta ecc}) - Ps \cdot Decc, L / 4$
 = $61.3 \cdot 20 \cdot \cos(28.8) - 2015 \cdot 1800 / 4$
 = 16.9803 Kgf/mm

Circumferential Load, Cone Large End, (used in Appendix 1-8) [f1']:
 = $-\sigma_{ecc, L, m} \cdot tecc \cdot \cos(\text{theta ecc}) + Ps \cdot Decc, L / 4$
 = $-61.3 \cdot 20 \cdot \cos(28.8) + 2015 \cdot 1800 / 4$
 = -16.9803 Kgf/mm



Axial Membrane Stress in Cone Small End [$\sigma_{ecc,S,m}$]:
 $= a o^2 / (tecc(Decc, S + tecc) \cos(\text{angle})) [Pe + (Decc, S^2 / (4 \cdot a o^2) - 1)(Ps - Pt)] + Decc, S^2 \cdot Pt / (4 \cdot tecc(Decc, S + tecc) \cos(\text{angle}))$
 $= 577^2 / (20(1200 + 20) \cos(28.8)) [-2936 + (1200^2 / (4 \cdot 577^2) - 1)(2015 - 5915)] + 1200^2 \cdot 5915 / (4 \cdot 20(1200 + 20) \cos(28.8))$
 $= 48.9648 \text{ N./mm}^2$

Circumferential Load, Cone Small End (used in Appendix 1-5) [f_2']:
 $= \sigma_{ecc,S,m} \cdot tecc \cdot \cos(\text{theta ecc}) - Ps \cdot Decc, S / 4$
 $= 49 \cdot 20 \cdot \cos(28.8) - 2015 \cdot 1200 / 4$
 $= 25.8364 \text{ Kg/mm}$

Circumferential Load, Cone Small End, (used in Appendix 1-8) [f_2']:
 $= -\sigma_{ecc,S,m} \cdot tecc \cdot \cos(\text{theta ecc}) + Ps \cdot Decc, S / 4$
 $= -49 \cdot 20 \cdot \cos(28.8) + 2015 \cdot 1200 / 4$
 $= -25.8364 \text{ Kg/mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 20 mm, the minimum Shell length adjacent to the tubesheet should be 278.855 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	152.4 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	33.2 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	15.1 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-15.1 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	237.9 <=	3065.8 Kg	Pass
Axial Membrane Stress in Shell	42.9 <=	138.0 N./mm ²	Pass
Large Cyl. Axial Membrane Stress	67.2 <=	138.0 N./mm ²	Pass
Large End Cone Axial Membrane Stress	61.3 <=	138.0 N./mm ²	Pass
Small End Cone Axial Membrane Stress	49.0 <=	138.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	62.912	118.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	86.503 ...	229	356	51 190	Ps+Pt-Th	D1 Ok
D2uc	36.600 ...	77	356	18 190	Ps+Pt-Th	D2 Ok
D3uc	62.912 ...	152	356	33 190	Ps+Pt-Th	D3 Ok
D1c	88.587 ...	235	356	53 190	Ps+Pt-Th-c	D1 Ok
D2c	38.964 ...	80	356	19 190	Ps+Pt-Th-c	D2 Ok
D3c	64.419 ...	155	356	34 190	Ps+Pt-Th-c	D3 Ok
Max:	88.5873 ... mm	0.659	0.277	(Str. Ratio)		

Load Case Definitions:



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[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					---- Shell Band Stress ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	19	137	...	...	:	...	...	...	...	Ok
D2uc	24	137	...	...	:	...	...	...	...	Ok
D3uc	43	137	...	...	:	...	...	...	...	Ok
D1c	17	137	...	...	:	...	...	...	...	Ok
D2c	25	137	...	...	:	...	...	...	...	Ok
D3c	42	137	...	...	:	...	...	...	...	Ok
Max RATIO	0.311		...	...	:	...	...	...	...	...

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	16	137	...	...	Ok
D2uc	51	137	...	...	Ok
D3uc	67	137	...	...	Ok
D1c	18	137	...	...	Ok
D2c	64	137	...	...	Ok
D3c	82	137	...	...	Ok
Max RATIO	0.596		...	...	...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---				Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	22	137	...	...	:	966	9078	...	...	Ok
D2uc	27	137	...	...	:	3064	9078	...	...	Ok
D3uc	49	137	...	...	:	4030	9078	...	...	Ok
D1c	23	137	...	...	:	1023	9078	...	...	Ok
D2c	34	137	...	...	:	3691	9078	...	...	Ok
D3c	57	137	...	...	:	4714	9078	...	...	Ok
Max RATIO	0.412		...	...	:	0.519		...	...	...

Tube, Shell and Channel Stress Summary:



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Case#	Ten	---- Tube Stresses			Tube Loads		Shell Stress		Channel Stress		Pass
		Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	24	194	-20	-194	381	3065	...	370	...	370	Ok
D2uc	11	194	-11	-194	172	3065	...	370	...	370	Ok
D3uc	15	194	-15	-194	238	3065	...	370	...	370	Ok
D1c	26	194	-20	-194	404	3065	...	370	...	370	Ok
D2c	12	194	-12	-194	185	3065	...	370	...	370	Ok
D3c	15	194	-15	-194	238	3065	...	370	...	370	Ok
Max RATIO		0.132		0.104		0.132		...		...	

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	88.587	118.000	Ok
Tube thickness :	0.681	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.219 mm
 Due to Pressure + Thermal Shell Extends by : +.219 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2187	0.2187	...	...	...	...
2	-0.1756	-0.1756	...	...	...	...
3	0.0431	0.0431	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	153	No Calc	No Calc
20	30	No Calc	...	200	No Calc	No Calc
30	Tsht	1499.97	1256	25	0.0031466	112.736
Tsht	40	No Calc	...	25	No Calc	No Calc
40	50	11964	1240	17	0.0002309	23.0853
50	60	11964	1840	17	0.00016066	16.0619
60	70	11964	1832	13	0.00010856	10.8536
70	80	11964	1840	17	0.00016066	16.0619
80	Tsht	11964	1240	17	0.0002309	23.0853
Tsht	90	No Calc	...	17	No Calc	No Calc
90	100	1500.03	1256	25	0.0031465	112.736
100	110	No Calc	...	200	No Calc	No Calc
110	120	No Calc	...	153	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	156	108.585	103.419	No Calc
20	30	203	197.993	103.419	No Calc
30	Tsht	28	7.3409	103.419	2992
Tsht	40	28	No Calc	103.419	No Calc
40	50	20	No Calc	...	422
50	60	20	No Calc	...	198
60	70	16	No Calc	...	103
70	80	20	No Calc	...	198
80	Tsht	20	No Calc	...	422
Tsht	90	20	No Calc	...	No Calc
90	100	28	7.34096	103.419	2992
100	110	203	197.993	103.419	No Calc
110	120	156	108.585	103.419	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1499.97	99166	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	11964	No Calc	No Calc	No Calc
50	60	11964	No Calc	No Calc	No Calc
60	70	11964	No Calc	No Calc	No Calc
70	80	11964	No Calc	No Calc	No Calc
80	Tsht	11964	No Calc	No Calc	No Calc
Tsht	90	No Calc	No Calc	No Calc	No Calc
90	100	1500.03	99168.6	No Calc	No Calc



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External Pressure Calculations: Step: 17 9:30am Sep 9,2026

100	110	No Calc	No Calc	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	2492.17	...	2492.17	...	249.217
20	30	1415.79	304417	1392.32	305052	141.579
30	40	1255.82	1696763	1124.01	1713774	125.582
40	50	118.823	109570	101.247	111838	11.8823
50	60	922.701	1316397	786.059	1334030	92.2701
60	70	6639.01	18409732	5403.11	18569222	663.901
70	80	922.701	1316397	786.059	1334030	92.2701
80	90	118.823	109570	101.247	111838	11.8823
90	100	1255.82	1696763	1124.01	1713774	125.582
100	110	1415.79	304417	1392.32	305052	141.579
110	120	2492.17	...	2492.17	...	249.217
Total		19049	25264030.00	17194	25498612.00	1904



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	78	...	...	Liquid: 10
10	Insl	63.2141	78	...	...	Insulation
20	Liqd	308.064	129.5	...	...	Liquid: 20
20	Insl	29.1695	129.5	...	...	Insulation
30	Liqd	1717.1	750	...	...	Liquid: 30
30	Insl	154.146	750	...	...	Insulation
30	Nozl	530.252	650	904.8	...	Noz N6
40	Liqd	69.7558	100	...	...	Liquid: 40
40	Insl	20.3049	100	...	...	Insulation
50	Liqd	1315.59	367.498	-150	...	Liquid: 50
50	Insl	154.103	545	...	...	Insulation
60	Sadl	1151.41	1110	1221.5	...	Left Saddle
60	Sadl	1151.41	8110	1221.5	...	Sdl 2 Fr50
60	Liqd	11720.2	4692	...	...	Liquid: 60
60	Insl	1383.24	4692	...	...	Insulation
60	Nozl	76.4456	2000	1036.53	...	Npz N2 250
60	Nozl	20.5978	4000	-938.1	...	Noz N3 80
60	Nozl	60.6317	4000	532.861	817.94	Noz N1 150
60	Nozl	35.579	6000	518.996	796.658	Noz N4 100
60	Nozl	63.2547	6000	996.837	...	Noz N5 200
60	Nozl	11.5135	3500	-925.4	...	D1 nozzle 50
60	Nozl	17.3874	3500	-302.426	874.588	K1 50
60	Nozl	17.3874	3500	302.418	874.59	K2 50
60	Nozl	17.3874	2500	302.418	874.59	K3 50
60	Nozl	17.3267	2500	-302.426	874.588	K4 50
60	Nozl	17.3874	5300	-302.426	874.588	K5 50
60	Nozl	17.3874	5300	302.418	874.59	K6 50
60	Nozl	17.3267	6400	302.418	874.59	K7 50
60	Nozl	17.3267	6300	-302.426	874.588	K8 50
60	Wght	2400	4692	...	...	INTERNAL T/S
70	Liqd	838.058	367.498	-150	...	Liquid: 70
70	Insl	154.103	545	...	...	Insulation
80	Liqd	69.7555	100	...	...	Liquid: 80



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80	Ins1	20.3049	100	...	...	Insulation
90	Liqd	1717.1	750	...	...	Liquid: 90
90	Ins1	154.146	750	...	...	Insulation
90	Noz1	530.252	850	-904.8	...	Noz N7
100	Liqd	308.064	129.5	...	...	Liquid: 100
100	Ins1	29.1695	129.5	...	...	Insulation
110	Liqd	...	78	...	...	Liquid: 110
110	Ins1	63.2141	78	...	...	Insulation
30	FTsh	659.718	1559	...	...	
30	Tube	18482.5	7600	...	...	Tubes
30	RTsh	659.718	13641	...	...	Tubesheet
40	Bafl	193.664	1500	...	...	Baffles
40	Bafl	193.664	3000	...	...	Baffles
40	Bafl	193.664	4500	...	...	Baffles
40	Bafl	194.194	6000	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2302.8
Liquid	18063.7
Insulation	2225.1
Nozzles	1467.4
Weights	2400.0
Exchanger Components	20577.1
Liquid in Tubes	4999.6

Sum of the Detail Weights 52035.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	20954.6	20954.6	20954.6	20954.6	20954.6	20954.6
Saddles	2302.8	2302.8	2302.8	2302.8	2302.8	2302.8
Nozzles	1467.4	1467.4	1467.4	1467.4	1467.4	1467.4
Wld Weights	2400.0	2400.0	2400.0	2400.0	2400.0	2400.0
Exchanger	20577.1	20577.1	20577.1	20577.1	20577.1	20577.1
Insulation	...	...	...	...	2225.1	2225.1 *
Ope. Liquid	...	...	...	...	...	18063.7
Tube Ope Lqd	...	...	...	...	...	4999.6
Test Liquid	...	25248.6	...	...	...	...
Tube Tst Lqd	...	4945.3	...	...	...	...
Totals	47701.9	77895.8	47701.9	47701.9	49927.0	72990.4

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	47701.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	77895.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	47701.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	47701.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	49927.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	72990.4 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	70950.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	78315.0 kg.



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Element and Detail Weights: Step: 18 9:30am Sep 9,2026

Exchanger Tube DataVolume of Exchanger tubes : 4948286.0 cm³

Weight of Ope Liq in tubes : 4999.6 kg.

Weight of Water in tubes : 4945.3 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	27747.2	...
20	30	21029.3	10065.7
30	40	59187.6	56548.7
40	50	7791.15	7539.82
50	60	54896.7	53275
60	70	540087	530653
70	80	54896.7	53275
80	90	7791.15	7539.82
90	100	59187.6	56548.7
100	110	21029.3	10065.7
110	120	27747.2	...

Total 881390.5 785511.1 cm²Total 948.7 845.5 ft²

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 Nozzle Flange MAWP: Step: 19 9:30am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - - UG-44	Max Pressure 50%	DNV
Noz N6	4670.1	5100.0	100	300	GR 1.1	...	...	...	...
Npz N2 250	1868.0	1960.0	65	150	GR 1.1	2278.5	1831.2	1868.0	523.6
Noz N3 80	4887.1	5100.0	65	300	GR 1.1	5340.7	1989.9	4887.1	1989.9
Noz N1 150	4887.1	5100.0	65	300	GR 1.1	3380.5	3950.0	4887.1	3950.1
Noz N4 100	4887.1	5100.0	65	300	GR 1.1	4320.9	3009.7	4887.1	3009.7
Noz N5 200	4887.1	5100.0	65	300	GR 1.1	4122.7	3207.9	4887.1	3207.9
D1 nozzle 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K1 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K2 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K3 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K4 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K5 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K6 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K7 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
K8 50	4887.1	5100.0	65	300	GR 1.1	...	...	...	...
Noz N7	4670.1	5100.0	100	300	GR 1.1	1985.2	4670.1	4670.1	4670.1

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1831.225 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 4670.067 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5100.005 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
Npz N2 250	302.387	1.20	1261	0	1578.6	1831.2	Pass
Noz N3 80	111.125	0.50	130	612	1578.6	1989.9	Pass
Noz N1 150	196.088	0.50	454	1147	1578.6	3950.0	Pass
Noz N4 100	138.176	0.50	202	765	1578.6	3009.7	Pass
Noz N5 200	248.539	0.50	1153	1836	1578.6	3207.9	Pass
Noz N7	657.225	0.50	10380	5507	4628.9	4670.1	Pass

Evaluating Flange on Nozzle [Npz N2 250]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1261) + 4 \cdot \text{abs}(0) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot ((1868.034 - 1578.586) + 1.200 \cdot 1868.034)$$

$$= 0.20183E+05 \leq 0.22421E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Npz N2 250 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1868.034, 1.200 \cdot 1868.034 + 1868.034 - 20182.9 / (\pi \cdot 302.387^3))$$

$$= \min(1868.034, 1831.225)$$



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 Nozzle Flange MAWP: Step: 19 9:30am Sep 9, 2026

= 1831.225 KPa.

Evaluating Flange on Nozzle [Noz N3 80]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(130) + 4 \cdot \text{abs}(612) 111.125 \leq \pi \cdot 111.125^3 [(4887.066 - 1578.586) + .500 \cdot 4887.066]$$

$$= 0.23479E+04 \leq 0.25288E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N3 80 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 2347.9 / (\pi \cdot 111.125^3))$$

$$= \min(4887.066, 1989.911)$$

$$= 1989.911 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz N1 150]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(454) + 4 \cdot \text{abs}(1147) 196.088 \leq \pi \cdot 196.088^3 [(4887.066 - 1578.586) + .500 \cdot 4887.066]$$

$$= 0.81657E+04 \leq 0.13894E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N1 150 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 8165.7 / (\pi \cdot 196.088^3))$$

$$= \min(4887.066, 3950.049)$$

$$= 3950.049 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz N4 100]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(202) + 4 \cdot \text{abs}(765) 138.176 \leq \pi \cdot 138.176^3 [(4887.066 - 1578.586) + .500 \cdot 4887.066]$$

$$= 0.36520E+04 \leq 0.48615E+04 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N4 100 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 3652.0 / (\pi \cdot 138.176^3))$$

$$= \min(4887.066, 3009.664)$$

$$= 3009.664 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz N5 200]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(1153) + 4 \cdot \text{abs}(1836) 248.539 \leq \pi \cdot 248.539^3 [(4887.066 - 1578.586) + .500 \cdot 4887.066]$$

$$= 0.20278E+05 \leq 0.28292E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N5 200 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4887.066, 0.500 \cdot 4887.066 + 4887.066 - 20277.9 / (\pi \cdot 248.539^3))$$

$$= \min(4887.066, 3207.884)$$

$$= 3207.884 \text{ KPa.}$$

Evaluating Flange on Nozzle [Noz N7]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(10380) + 4 \cdot \text{abs}(5507) 657.225 \leq \pi \cdot 657.225^3 [(4670.067 - 4628.898) + .500 \cdot 4670.067]$$

$$= 0.18055E+06 \leq 0.21611E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N7 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4670.067, 0.500 \cdot 4670.067 + 4670.067 - 180553.8 / (\pi \cdot 657.225^3))$$

$$= \min(4670.067, 5019.877)$$

$$= 4670.067 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 FileName : 130-C1& C2-Final 23rd March
 Wind Load Calculation: Step: 20 9:30am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.22 \cdot 0.94) / (1 + 1.7 \cdot 3.4 \cdot 0.22)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.612
Structure Height to Diameter ratio	9.139

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1575 + 9500 = 11075 \text{ mm}$
 $= 36.3 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (36.3 \text{ ft.}) > 15 \text{ ft.}$
 $= 2.01(z / Zg)^{2/\alpha}$
 $= 2.01(36.3/900)^{2/9.5}$
 $= 1.02$

Type of Hill: No Hill



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 Wind Load Calculation: Step: 20 9:30am Page 94 of 317
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Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = (1 + K1 • K2 • K3)²
 = (1 + 0 • 0 • 0)²
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
 = max(16, 0.00256 • 1.02 • 1 • 1 • 1 • 60.4²)
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = qz • G • Cf • WindArea
 = 16 • 0.85 • 0.61 • 3.47
 = 28.8 lbs. [13.1] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flg	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
LHS Ch Flange	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
LHS Ch Shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
LHS Cone Shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
LHS Cone Shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
Main Shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
RHS Cone shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
RHS Cone shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
Rhs Ch Shell	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
RHS Ch Flg	11075.0	0.000	0.000	0.000	1.023	1.000	78.120
RHS Blind Flang	11075.0	0.000	0.000	0.000	1.023	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
INTERNAL T/S	0.00	11075.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	11075	2064	3219.84	78.12	14.3906
20	30	11075	1608	4164.72	78.12	18.6136
30	40	11075	1675.2	25128	78.12	112.306
40	50	11075	1656	3312	78.12	14.8025
50	60	11075	2016	21974.4	78.12	98.2113
60	70	11075	2366.4	222063	78.12	992.477
70	80	11075	2016	21974.4	78.12	98.2113
80	90	11075	1656	3312	78.12	14.8025
90	100	11075	1675.2	25128	78.12	112.306
100	110	11075	1608	4164.72	78.12	18.6136
110	120	11075	2064	3219.84	78.12	14.3906

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



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Wind Load Calculation: Step: 20 9:30am Sep 9, 2026

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.120.75)$
 = 0.096 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.096$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 70688 \cdot 1.25$
 = 4156.428 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 70688$
 = 2216.8 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	790	5437.5	350.062



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Earthquake Load Calculation:

Step: 21

9:30am

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20	30	600	5437.5	265.87
30	40	600	5437.5	265.87
40	50	600	5437.5	265.87
50	60	600	5437.5	265.87
60	Sadl	900	5437.5	398.804
Sadl	70	900	5437.5	398.804
60	70	900	5437.5	398.804
70	80	900	5437.5	398.804
80	90	600	5437.5	265.87
90	100	600	5437.5	265.87
100	110	600	5437.5	265.87
110	120	790	5437.5	350.062

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 22 9:30am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	7936.175 mm
Center of Gravity of the Liquid	7858.619 mm
Center of Gravity of the Insulation	8018.217 mm
Center of Gravity of the Nozzles	7891.833 mm
Center of Gravity of the Added Weights (Operating)	8018.175 mm
Center of Gravity of the Added Weights (Empty)	8018.175 mm
Center of Gravity of the Tubesheet(s)	8018.175 mm
Center of Gravity of the Tubes	8018.175 mm
Center of Gravity of the Baffles	5787.715 mm
Center of Gravity of Bare Shell New and Cold	8018.827 mm
Center of Gravity of Bare Shell Corroded	8018.892 mm
Vessel CG in the Operating Condition	7948.101 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	7974.369 mm
Vessel CG in the Test Condition	7991.362 mm

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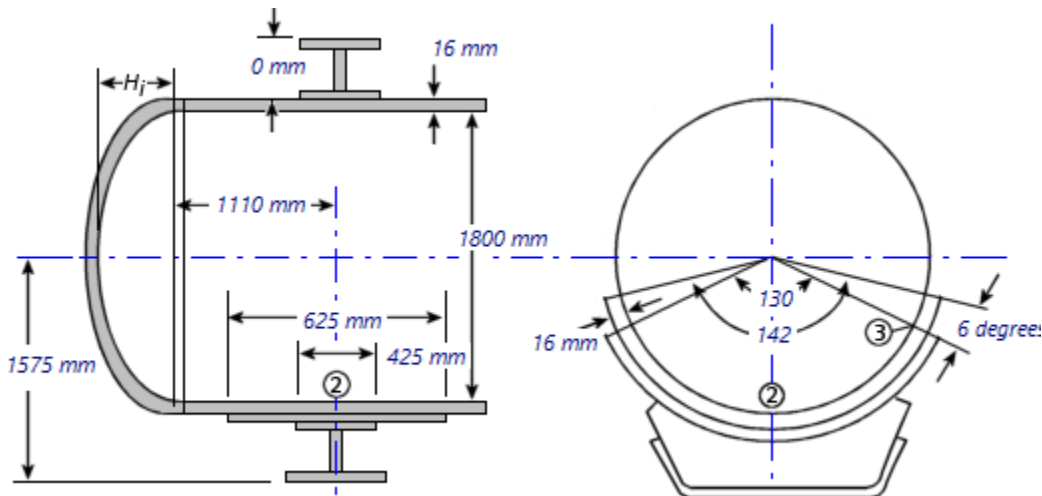
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 Saddle Calcs: Road Transport: Step: 23 9:30am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	908.00	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.30	

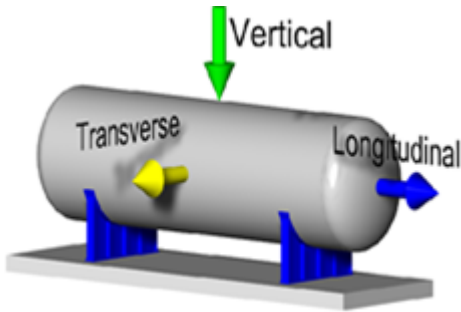


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		48116.79	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	0.0	Km/hr



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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-21.81	-234.00
Long. Stress at Top of Midspan	21.81	138.00
Long. Stress at Bottom of Midspan	21.81	138.00
Long. Stress at Top of Saddles	6.34	138.00
Long. Stress at Bottom of Saddles	-3.55	-234.00
Long. Stress at Bottom of Saddles	3.55	138.00
Tangential Shear in Shell	27.04	110.40
Circ. Stress at Horn of Saddle	37.06	207.00
Circ. Compressive Stress in Shell	1.75	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 363 • 0.00256 • 0² • 1
 = 0.0 lbf
 = 0.000 Kgf

Longitudinal Force due to Wind during Transport [F_{lw}]:
 = π • D²/4 • 0.00256 • wv² • wlsf • Wind Dia. Mult
 = π • 6.47²/4 • 0.00256 • 0² • 1 • 1.2
 = 0.0 lbf
 = 0.0 Kgf

Transverse Force due to Lateral Acceleration [F_t]:
 = g_z • Shipping Weight • g_w + F_{tw}
 = 0.1 • 45399 • 1 + 0
 = 4539.9 Kgf

Longitudinal Force due to Axial Acceleration [F_l]:
 = g_x • Shipping Weight • g_w + F_{lw}
 = 0.1 • 45399 • 1 + 0
 = 4539.9 Kgf

Saddle Reaction Force due to Transverse Load [F_{wt}]:
 = F_t/Num of Saddles • B / E
 = 4540/2 • 1575/1689
 = 2116.3 Kgf

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:
 = F_l • B / L_s
 = 4540 • 1575/7000
 = 1021.5 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:
 = Saddle Load • g_y
 = 22439 • 1.05



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= 23561.227 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 22439 + 23561 + max(1021, 2116)
 = 48116.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	49268.20	Kgf
Transverse Shear Load	2269.95	Kgf
Longitudinal Shear Load	4539.91	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(1800/2 + 16 + 16)sin(130/2), 2•908)
 = 1689.358 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -48117•1110[1-(1-1110/11964+(908²-0²)/
 (2•1110•11964))/(1+(4•0)/(3•11964))]
 = -3297.4 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 48117•11964/4((1+2(908²-0²)/(11964²))/(1+(4•0)/
 (3•11964))-4•1110/11964)
 = 92167.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 908/(2•16) - 92167/(π•908²•16)
 = -21.81 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 908/(2 • 16) + 92167/(π • 908² • 16)
 = 21.81 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•908/(2•16)--3297/(0.12•π•908²•16)
 = 6.34 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•908/(2•16)+-3297/(0.22•π•908²•16)
 = -3.55 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 48117 (11964 - 2 • 1110)/(11964 + (4 • 0/3))
 = 39188.4 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.02 \cdot 39188 / (908 \cdot 16)$$

$$= 27.04 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(908 \cdot 16)}$$

$$= 94.015 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(908 \cdot 16)}, 2 \cdot 1110)$$

$$= 613.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.73 \cdot 48117 \cdot 0.1 / (613(16 + 1 \cdot 16))$$

$$= -1.75 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -48117 / (4(16 + 1 \cdot 16)613) -$$

$$3 \cdot 0.045 \cdot 48117 / (2(16 + 1 \cdot 16)^2)$$

$$= -37.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	675.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(900 \cdot 16)} + 625)16^3) / 12$$

$$= 27.723 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 27.7 + 130(295 - 8)^2$$

$$= 107272.969 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	812.2	16.0	8.0	130.0	103961.6	0.107E+06
Wearplate	625.0	16.0	24.0	100.0	240000.0	0.736E+05
Web	24.0	607.0	335.5	145.7	4887564.0	0.471E+05
BasePlate	450.0	36.0	657.0	162.0	10643399.0	0.212E+06
Totals	...	...	...	537.6	15874925.0	0.440E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 15874925/538$$

$$= 295.275 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 130/2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115) \cos(115))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.22 \cdot 48117$$

$$= 10673.2666 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 2.5675 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1575 - 900 \cdot \sin(130/2) / 1.13$$

$$= 856.002 \text{ mm}$$

Bending Moment, M = Fh * d = 9136.5205 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 6.0112 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(48117 + 1151)450) / (4 \cdot 1803 \cdot 173)}$$

$$= 22.843 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130/2) (900 + 16 + 16)$$

$$= 1689.358 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1689 / (5 - 1)$$

$$= 422.339 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 422 \cdot 450/2$$

$$= 950.264 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 49268 / (1803 \cdot 450)$$

$$= 6.072 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 950 \cdot 6.07$$

$$= 5770.355 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 96.2 + 50.7$$

$$= 146.921 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 5770/147$$

$$= 3.852 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 425 - 24$$

$$= 401.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15353/147)}$$

$$= 102.225 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1575 - \cos(130/2) (900 + 16 + 16) - 36$$

$$= 1145.120 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1145/102$$

$$= 11.202$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 4540 / (2 \cdot 1803) \cdot 422 \cdot 1145/2$$

$$= 304.448 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.85/152 + (304 \cdot 213/153531280) / 173$$



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= 0.049

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/198
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 425/2
 = 212.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1575 - cos(932/2)(36 + 0 + 0) - 0
 = 607.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(15399/198)}$
 = 88.278 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 607/88.3
 = 6.876

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 4540/(2 • 1803)
 = 1.259 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 1.26 • 422 • 607
 = 322.760 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 323/724660
 = 4.368 N./mm²

Compressive Allowable, KL/r < Cc (6.88 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (6.88)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (6.88) / (8 \cdot 123) - (6.88^3) / (8 \cdot 123^3))$
 = 153.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.73/154 + (323 • 213/15399)/173
 = 0.062

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	908.00	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		49183.93	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	1.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	0.0	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-22.29	-234.00	
Long. Stress at Top of Midspan	22.29	138.00	
Long. Stress at Bottom of Midspan	22.29	138.00	
Long. Stress at Top of Saddles	6.48	138.00	
Long. Stress at Bottom of Saddles	-3.63	-234.00	
Long. Stress at Bottom of Saddles	3.63	138.00	
Tangential Shear in Shell	27.64	110.40	
Circ. Stress at Horn of Saddle	37.89	207.00	
Circ. Compressive Stress in Shell	1.78	138.00	

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 363 \cdot 0.00256 \cdot 0^2 \cdot 1 \\
 &= 0.0 \text{ lbf} \\
 &= 0.000 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 6.47^2/4 \cdot 0.00256 \cdot 0^2 \cdot 1 \cdot 1.2 \\
 &= 0.0 \text{ lbf} \\
 &= 0.0 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 45399 \cdot 1 + 0
 \end{aligned}$$



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= 4539.9 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$
 = $0.1 \cdot 45399 \cdot 1 + 0$
 = 4539.9 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$
 = $4540/2 \cdot 1575/1689$
 = 2116.3 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$
 = $4540 \cdot 1575/7000$
 = 1021.5 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$
 = $22960 \cdot 1.05$
 = 24107.814 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = $22960 + 24108 + \max(1021, 2116)$
 = 49183.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	50335.34	Kgf
Transverse Shear Load	2269.95	Kgf
Longitudinal Shear Load	4539.91	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(1800/2 + 16 + 16) \sin(130/2), 2 \cdot 908)$
 = 1689.358 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 = $-49184 \cdot 1110 [1 - (1 - 1110/11964 + (908^2 - 0^2) / (2 \cdot 1110 \cdot 11964)) / (1 + (4 \cdot 0) / (3 \cdot 11964))]$
 = -3370.5 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 = $49184 \cdot 11964 / 4 ((1 + 2(908^2 - 0^2)/(11964^2)) / (1 + (4 \cdot 0) / (3 \cdot 11964)) - 4 \cdot 1110/11964)$
 = 94211.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 = $0 \cdot 908 / (2 \cdot 16) - 94212 / (\pi \cdot 908^2 \cdot 16)$
 = -22.29 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 908 / (2 \cdot 16) + 94212 / (\pi \cdot 908^2 \cdot 16)$$

$$= 22.29 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 908 / (2 \cdot 16) - 3371 / (0.12 \cdot \pi \cdot 908^2 \cdot 16)$$

$$= 6.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 908 / (2 \cdot 16) + 3371 / (0.22 \cdot \pi \cdot 908^2 \cdot 16)$$

$$= -3.63 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 49184 \cdot (11964 - 2 \cdot 1110) / (11964 + (4 \cdot 0 / 3))$$

$$= 40057.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.02 \cdot 40058 / (908 \cdot 16)$$

$$= 27.64 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{908 \cdot 16}$$

$$= 94.015 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{908 \cdot 16}, 2 \cdot 1110)$$

$$= 613.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot r))$$

$$= -0.73 \cdot 49184 \cdot 0.1 / (613(16 + 1 \cdot 16))$$

$$= -1.78 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot r)^2)$$

$$= -49184 / (4(16 + 1 \cdot 16) 613) -$$

$$3 \cdot 0.045 \cdot 49184 / (2(16 + 1 \cdot 16)^2)$$

$$= -37.89 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	675.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(900 \cdot 16)} + 625)16^3)/12$
 $= 27.723 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 27.7 + 130(295 - 8)^2$
 $= 107272.969 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	812.2	16.0	8.0	130.0	103961.6	0.107E+06
Wearplate	625.0	16.0	24.0	100.0	240000.0	0.736E+05
Web	24.0	607.0	335.5	145.7	4887564.0	0.471E+05
BasePlate	450.0	36.0	657.0	162.0	10643399.0	0.212E+06
Totals	...	...	...	537.6	15874925.0	0.440E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15874925/538$
 $= 295.275 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 130/2$
 $= 115.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115)\cos(115))$
 $= 0.2218$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.22 \cdot 49184$
 $= 10909.9805 \text{ Kgf}$

Tension Stress, $St = (Fh/As) = 2.6244 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1575 - 900 \cdot \sin(130/2) / 1.13$
 $= 856.002 \text{ mm}$

Bending Moment, $M = Fh \cdot d = 9339.1514 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 6.1445 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(49184 + 1151)450 / (4 \cdot 1803 \cdot 173))}$
 $= 23.089 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 130/2) (900 + 16 + 16)$
 $= 1689.358 \text{ mm}$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 1689 / (5 - 1) \\
 &= 422.339 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 422 \cdot 450 / 2 \\
 &= 950.264 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 50335 / (1803 \cdot 450) \\
 &= 6.204 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 950 \cdot 6.2 \\
 &= 5895.340 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 96.2 + 50.7 \\
 &= 146.921 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 5895 / 147 \\
 &= 3.935 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 425 - 24 \\
 &= 401.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 147 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 425 / 2 \\
 &= 212.500 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 15353 / 147 } \\
 &= 102.225 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1575 - \cos(130 / 2) (900 + 16 + 16) - 36 \\
 &= 1145.120 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199943008 / 260 }
 \end{aligned}$$



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$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1145/102$$

$$= 11.202$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 4540 / (2 \cdot 1803) \cdot 422 \cdot 1145/2$$

$$= 304.448 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.94/152 + (304 \cdot 213/153531280) / 173$$

$$= 0.050$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/198$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1575 - \cos(932/2) (36 + 0 + 0) - 0$$

$$= 607.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (15399/198) }$$

$$= 88.278 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 607/88.3$$

$$= 6.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4540 / (2 \cdot 1803)$$

$$= 1.259 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.26 \cdot 422 \cdot 607$$



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= 322.760 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 323/724660

= 4.368 N./mm²

Compressive Allowable, $KL/r < C_c$ (6.88 < 123) per AISC E2-1 [Sca]:

= (1-(Kl/r)²/(2• C_c ²)) F_y /(5/3+3•(Kl/r)/(8• C_c)-(Kl/r ³)/(8• C_c ³))

= (1-(6.88)²/(2 • 123²))260/

(5/3+3•(6.88)/(8• 123)-(6.88³)/(8•123³))

= 153.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 5.85/154 + (323 • 213/15399)/173

= 0.063

Review notes about nozzle loadings and supports in the Warnings report.

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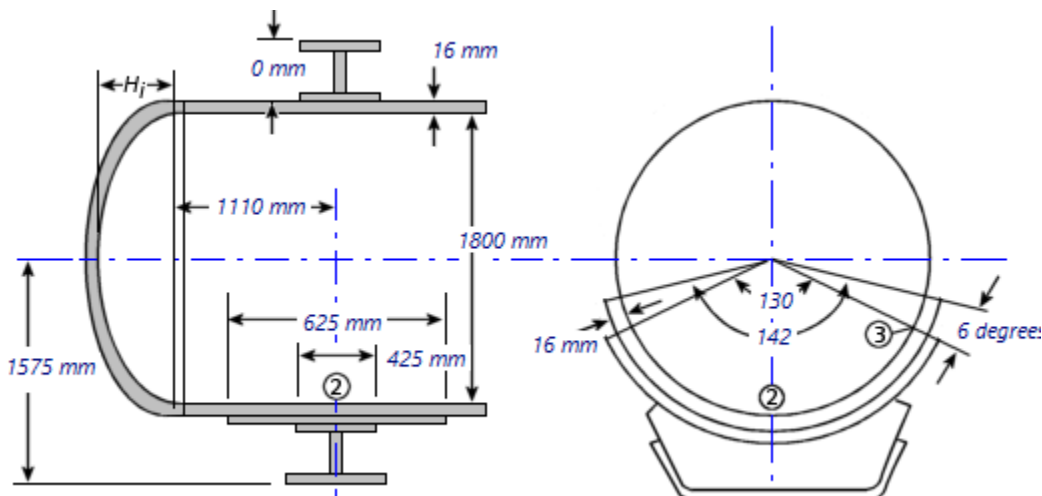
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	909.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	42119.96	Kgf
Pressure used in Analysis	P	1555.620 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	31.00	138.00
Long. Stress at Bottom of Midspan	77.84	138.00
Long. Stress at Top of Saddles	61.21	138.00
Long. Stress at Bottom of Saddles	50.61	138.00
Tangential Shear in Shell	29.09	110.40
Circ. Stress at Horn of Saddle	39.08	172.50
Circ. Compressive Stress in Shell	1.74	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1509/2 + 0) \cdot 1575/1689$$

$$= 2110.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(378, 0) \cdot 1575/7000$$

$$= 85.0 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(4156, 0) \cdot 1575/7000$$

$$= 935.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (4156/2 + 0) \cdot 1575/1689$$

$$= 5812.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 36307 + \max(85, 2110, 935, 5813)$$

$$= 42120.0 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (1.97 \cdot 1.2)^2)$$

$$= 3706.263 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	43271.37	Kgf
Transverse Shear Load	2078.21	Kgf
Longitudinal Shear Load	4156.43	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1800/2 + 16 + 16)\sin(130/2), 2 \cdot 910)$$

$$= 1689.358 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -42120 \cdot 1110 [1 - (1 - 1110/11964 + (910^2 - 0^2)/(2 \cdot 1110 \cdot 11964))/(1 + (4 \cdot 0)/(3 \cdot 11964))]$$

$$= -2881.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 42120 \cdot 11964/4 ((1 + 2(910^2 - 0^2)/(11964^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 11964) - 4 \cdot 1110 / 11964 \\ = 80685.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1556 \cdot 910 / (2 \cdot 13) - 80685 / (\pi \cdot 910^2 \cdot 13) \\ = 31.00 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1556 \cdot 910 / (2 \cdot 13) + 80685 / (\pi \cdot 910^2 \cdot 13) \\ = 77.84 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1556 \cdot 910 / (2 \cdot 13) - 2882 / (0.12 \cdot \pi \cdot 910^2 \cdot 13) \\ = 61.21 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 1556 \cdot 910 / (2 \cdot 13) + 2882 / (0.22 \cdot \pi \cdot 910^2 \cdot 13) \\ = 50.61 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 42120 (11964 - 2 \cdot 1110) / (11964 + (4 \cdot 0 / 3)) \\ = 34304.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.02 \cdot 34304 / (910 \cdot 13) \\ = 29.09 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{910 \cdot 13} \\ = 84.814 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(425 + 1.56 \cdot \sqrt{910 \cdot 13}, 2 \cdot 1110) \\ = 594.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.73 \cdot 42120 \cdot 0.1 / (595(13 + 1 \cdot 16)) \\ = -1.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -42120 / (4(13 + 1 \cdot 16)595) - 3 \cdot 0.045 \cdot 42120 / (2(13 + 1 \cdot 16)^2) \\ = -39.08 \text{ N./mm}^2$$

Distance between Saddle Supports [L_s]:

$$= 7000.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.11875 \text{E-04} \cdot 7000 (65 - 21.1)$$



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= 3.648 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	239.0	N./mm ²
Height of Web at Center	Hw,c	675.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(903 \cdot 13)} + 625)13^3)/12$
 $= 14.537 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 14.5 + 103(308 - 6.5)^2$
 $= 93602.125 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	794.0	13.0	6.5	103.2	67094.8	0.936E+05
Wearplate	625.0	16.0	21.0	100.0	210000.0	0.822E+05
Web	24.0	607.0	332.5	145.7	4843859.5	0.456E+05
BasePlate	450.0	36.0	654.0	162.0	10594800.0	0.195E+06
Totals	...	...	...	510.9	15715755.0	0.416E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 15715755/511$
 $= 307.608 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 130/2$
 $= 115.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115)\cos(115))$
 $= 0.2218$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.22 \cdot 42120$
 $= 9343.0518 \text{ Kg}$

Tension Stress, St = (Fh/As) = 2.2475 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 143.3985 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1575 - 903 \cdot \sin(130/2) / 1.13$
 $= 853.606 \text{ mm}$

Bending Moment, M = Fh * d = 7975.4395 Kg-m.



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Bending Stress, $S_b = (M * C_1 / I) = 5.7840 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 159.3317 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(42120 + 1151)450 / (4 * 1803 * 159))}$
 $= 22.328 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 130/2) (900 + 16 + 16)$
 $= 1689.358 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1689 / (5 - 1)$
 $= 422.339 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 422 * 450 / 2$
 $= 950.264 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 43271 / (1803 * 450)$
 $= 5.333 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 950 * 5.33$
 $= 5067.999 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 96.2 + 50.7$
 $= 146.921 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5068 / 147$
 $= 3.383 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 425 - 24$
 $= 401.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 147$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 425/2
 = 212.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{15353/147}$
 = 102.225 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1575 - cos(130/2)(900 + 16 + 16) - 36
 = 1145.120 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/239}$
 = 128.509

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1145/102
 = 11.202

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 4156 / (2 \cdot 1803) \cdot 422 \cdot 1145/2$
 = 278.731 Kg-m.

Compressive Allowable, $KL/r < Cc$ (11.2 < 129) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (11.2)^2 / (2 \cdot 129^2)) 239 / (5/3 + 3 \cdot (11.2) / (8 \cdot 129) - (11.2)^3 / (8 \cdot 129^3))$
 = 140.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 3.38/140 + (279 \cdot 213/153531280) / 159$
 = 0.048

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/198$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 425/2
 = 212.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1575 - cos(932/2)(36 + 0 + 0) - 0
 = 607.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{15399/198}$$

$$= 88.278 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 607/88.3$$

$$= 6.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 4156 / (2 \cdot 1803)$$

$$= 1.153 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.15 \cdot 422 \cdot 607$$

$$= 295.497 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 295/724660$$

$$= 3.999 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.88 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.88)^2 / (2 \cdot 129^2)) 239 / (5/3 + 3 \cdot (6.88) / (8 \cdot 129) - (6.88)^3 / (8 \cdot 129^3))$$

$$= 141.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 5.03/141 + (295 \cdot 213/15399) / 159$$

$$= 0.061$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	37458.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	935.2 Kg
Maximum Transverse Force	Ft	2078.2 Kg
Maximum Longitudinal Force	F1	4156.4 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4156 / (4.87 \cdot 4)$$

$$= 20.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2078 / (4.87 \cdot 4)$$

$$= 10.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $4156 / (82.8 \cdot 4)$
 = 1.2307 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1575 \cdot 2078 + 0$
 = 3273.25 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $3273 / 37459$
 = 87.38 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1.23, 0]$
 = 1.2307 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, tr = 10.4, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	909.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 42409.54 Kgf
 Pressure used in Analysis P 1555.620 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	30.84	138.00
Long. Stress at Bottom of Midspan	78.00	138.00
Long. Stress at Top of Saddles	61.26	138.00
Long. Stress at Bottom of Saddles	50.58	138.00
Tangential Shear in Shell	29.29	110.40
Circ. Stress at Horn of Saddle	39.35	172.50
Circ. Compressive Stress in Shell	1.75	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (1509/2 + 0) \cdot 1575/1689$
 $= 2110.5\ Kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(378, 0) \cdot 1575/7000$
 $= 85.0\ Kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(4156, 0) \cdot 1575/7000$
 $= 935.2\ Kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (4156/2 + 0) \cdot 1575/1689$
 $= 5812.6\ Kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 36597 + \max(85, 2110, 935, 5813)$
 $= 42409.5\ Kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(1.97 \cdot 1.2)^2)$
 $= 3706.263\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	43560.95	Kgf
Transverse Shear Load	2078.21	Kgf
Longitudinal Shear Load	4156.43	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(1800/2 + 16 + 16) \sin(130/2), 2 \cdot 910)$
 $= 1689.358\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			



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The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / (1 + (4hm)/3L) \right]$$

$$= -42410 \cdot 1110 \left[1 - \left(1 - \frac{1110}{11964} + \frac{(910^2 - 0^2)}{(2 \cdot 1110 \cdot 11964)} \right) / (1 + (4 \cdot 0)/(3 \cdot 11964)) \right]$$

$$= -2901.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h m^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 42410 \cdot 11964 / 4 \left(\frac{(1 + 2(910^2 - 0^2)/(11964^2))}{(1 + (4 \cdot 0)/(3 \cdot 11964))} - \frac{4 \cdot 1110}{11964} \right)$$

$$= 81240.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 1556 \cdot 910 / (2 \cdot 13) - 81240 / (\pi \cdot 910^2 \cdot 13)$$

$$= 30.84 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1556 \cdot 910 / (2 \cdot 13) + 81240 / (\pi \cdot 910^2 \cdot 13)$$

$$= 78.00 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 1556 \cdot 910 / (2 \cdot 13) - 2901 / (0.12 \cdot \pi \cdot 910^2 \cdot 13)$$

$$= 61.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1556 \cdot 910 / (2 \cdot 13) + 2901 / (0.22 \cdot \pi \cdot 910^2 \cdot 13)$$

$$= 50.58 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 42410 (11964 - 2 \cdot 1110) / (11964 + (4 \cdot 0/3))$$

$$= 34540.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.02 \cdot 34540 / (910 \cdot 13)$$

$$= 29.29 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(910 \cdot 13)}$$

$$= 84.814 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(910 \cdot 13)}, 2 \cdot 1110)$$

$$= 594.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.73 \cdot 42410 \cdot 0.1 / (595(13 + 1 \cdot 16))$$



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= -1.75 N./mm²

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ_{7,r}]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -42410 / (4(13 + 1 \cdot 16) 595) -$$

$$3 \cdot 0.045 \cdot 42410 / (2(13 + 1 \cdot 16)^2)$$

$$= -39.35 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	239.0	N./mm ²
Height of Web at Center	Hw,c	675.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(903 \cdot 13)} + 625) 13^3) / 12$$

$$= 14.537 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 14.5 + 103 (308 - 6.5)^2$$

$$= 93602.125 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	794.0	13.0	6.5	103.2	67094.8	0.936E+05
Wearplate	625.0	16.0	21.0	100.0	210000.0	0.822E+05
Web	24.0	607.0	332.5	145.7	4843859.5	0.456E+05
BasePlate	450.0	36.0	654.0	162.0	10594800.0	0.195E+06
Totals	...	...	...	510.9	15715755.0	0.416E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15715755 / 511$$

$$= 307.608 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 130 / 2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115) \cos(115))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.22 \cdot 42410$$

$$= 9407.2842 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.2629 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 143.3985 \text{ N./mm}^2$$



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Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1575 - 903 \cdot \sin(130/2) / 1.13$
 $= 853.606 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 8030.2705 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 5.8237 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 159.3317 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$
 $= \sqrt{3(42410 + 1151) \cdot 450 / (4 \cdot 1803 \cdot 159)}$
 $= 22.403 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 130/2) (900 + 16 + 16)$
 $= 1689.358 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1689 / (5 - 1)$
 $= 422.339 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 422 \cdot 450 / 2$
 $= 950.264 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 43561 / (1803 \cdot 450)$
 $= 5.369 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 950 \cdot 5.37$
 $= 5101.915 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 96.2 + 50.7$
 $= 146.921 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5102 / 147$
 $= 3.405 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 425 - 24$
 $= 401.000 \text{ mm}$



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15353/147)}$$

$$= 102.225 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1575 - \cos(130/2) (900 + 16 + 16) - 36$$

$$= 1145.120 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/239)}$$

$$= 128.509$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1145/102$$

$$= 11.202$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 4156 / (2 \cdot 1803) \cdot 422 \cdot 1145/2$$

$$= 278.731 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 129$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 129^2)) 239 / (5/3 + 3 \cdot (11.2) / (8 \cdot 129) - (11.2)^3 / (8 \cdot 129^3))$$

$$= 140.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.41/140 + (279 \cdot 213/153531280) / 159$$

$$= 0.048$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/198$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$



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Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1575 - cos(932/2)(36 + 0 + 0) - 0
 = 607.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(15399/198)}$
 = 88.278 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 607/88.3
 = 6.876

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = 4156/(2 • 1803)
 = 1.153 Kgf/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 1.15 • 422 • 607
 = 295.497 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 295/724660
 = 3.999 N./mm²

Compressive Allowable, KL/r < Cc (6.88 < 129) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (6.88)^2 / (2 \cdot 129^2)) 239 / (5/3 + 3 \cdot (6.88) / (8 \cdot 129) - (6.88)^3 / (8 \cdot 129^3))$
 = 141.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.07/141 + (295 • 213/15399)/159
 = 0.061

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	37748.3 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	935.2 Kgf
Maximum Transverse Force	Ft	2078.2 Kgf
Maximum Longitudinal Force	F1	4156.4 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 11237.6 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 = max(Ff, F1) / (Bolt Area • Number of Bolts)



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= $11238 / (4.87 \cdot 4)$
 = 56.6 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $2078 / (4.87 \cdot 4)$
 = 10.5 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $4156 / (82.8 \cdot 4)$
 = 1.2307 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1575 \cdot 2078 + 0$
 = 3273.25 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $3273 / 37748$
 = 86.71 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1.23, 0]$
 = 1.2307 cm²

MDMT Calculations for: Sdl 2 Fr50, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, tr = 10.4, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Review notes about nozzle loadings and supports in the Warnings report.

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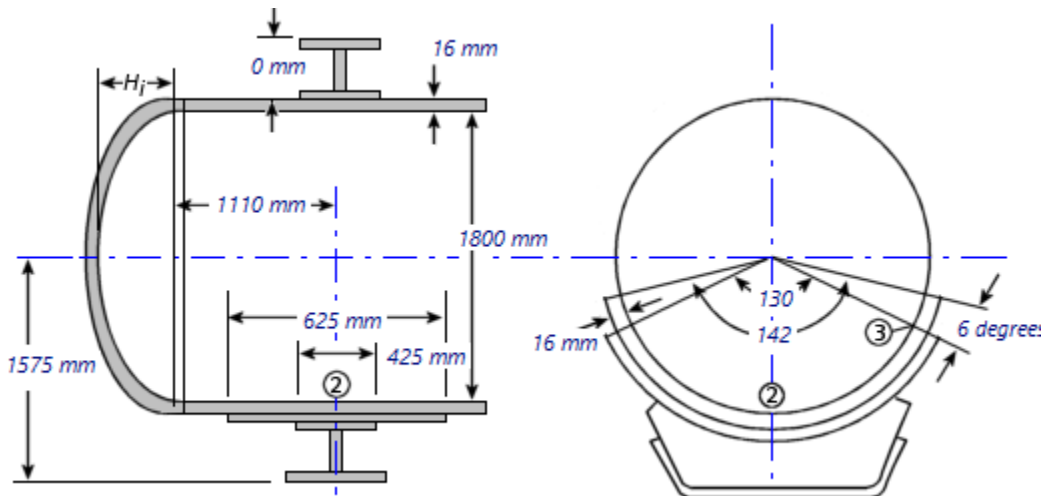
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	909.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		38237.59	Kgf
Pressure used in Analysis	P	2023.915	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	49.54	234.00
Long. Stress at Bottom of Midspan	92.06	234.00
Long. Stress at Top of Saddles	76.97	234.00
Long. Stress at Bottom of Saddles	67.34	234.00
Tangential Shear in Shell	26.40	138.58
Circ. Stress at Horn of Saddle	35.48	350.99
Circ. Compressive Stress in Shell	1.58	234.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (755/2 + 0) \cdot 1575/1689$
 $= 1055.2 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(189, 0) \cdot 1575/7000$
 $= 42.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 37182 + \max(42.5, 1055, 0, 0)$
 $= 38237.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(1.97 \cdot 1.2)^2)$
 $= 3706.263 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	39389.00	Kgf
Transverse Shear Load	377.28	Kgf
Longitudinal Shear Load	188.97	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1800/2 + 16 + 16)\sin(130/2), 2 \cdot 910)$
 $= 1689.358 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -38238 \cdot 1110 [1 - (1 - 1110/11964 + (910^2 - 0^2)/(2 \cdot 1110 \cdot 11964))/(1 + (4 \cdot 0)/(3 \cdot 11964))]$
 $= -2616.0 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 38238 \cdot 11964/4((1 + 2(910^2 - 0^2)/(11964^2))/(1 + (4 \cdot 0)/(3 \cdot 11964)) - 4 \cdot 1110/11964)$
 $= 73248.2 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 $= 2024 \cdot 910/(2 \cdot 13) - 73248/(\pi \cdot 910^2 \cdot 13)$
 $= 49.54 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$



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$$= 2024 \cdot 910 / (2 \cdot 13) + 73248 / (\pi \cdot 910^2 \cdot 13)$$

$$= 92.06 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 2024 \cdot 910 / (2 \cdot 13) - 2616 / (0.12 \cdot \pi \cdot 910^2 \cdot 13)$$

$$= 76.97 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2024 \cdot 910 / (2 \cdot 13) + 2616 / (0.22 \cdot \pi \cdot 910^2 \cdot 13)$$

$$= 67.34 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 38238 (11964 - 2 \cdot 1110) / (11964 + (4 \cdot 0 / 3))$$

$$= 31142.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.02 \cdot 31142 / (910 \cdot 13)$$

$$= 26.40 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(910 \cdot 13)}$$

$$= 84.814 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(910 \cdot 13)}, 2 \cdot 1110)$$

$$= 594.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.73 \cdot 38238 \cdot 0.1 / (595(13 + 1 \cdot 16))$$

$$= -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -38238 / (4(13 + 1 \cdot 16)595) -$$

$$3 \cdot 0.045 \cdot 38238 / (2(13 + 1 \cdot 16)^2)$$

$$= -35.48 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	675.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(903 \cdot 13)} + 625)^3) / 12$$

$$= 14.537 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.5 + 103 (308 - 6.5)^2$$

$$= 93602.125 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	794.0	13.0	6.5	103.2	67094.8	0.936E+05
Wearplate	625.0	16.0	21.0	100.0	21000.0	0.822E+05
Web	24.0	607.0	332.5	145.7	4843859.5	0.456E+05
BasePlate	450.0	36.0	654.0	162.0	10594800.0	0.195E+06
Totals	...	...	...	510.9	15715755.0	0.416E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 15715755/511$$

$$= 307.608 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 130/2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115) \cos(115))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.22 \cdot 38238$$

$$= 8481.8623 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.0403 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1575 - 903 \cdot \sin(130/2) / 1.13$$

$$= 853.606 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 7240.3086 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.2508 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt[3]{(38238 + 1151) \cdot 450 / (4 \cdot 1803 \cdot 173)}$$

$$= 20.425 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130/2) (900 + 16 + 16)$$

$$= 1689.358 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1689 / (5 - 1)$$

$$= 422.339 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 422 \cdot 450 / 2$$

$$= 950.264 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 39389 / (1803 \cdot 450)$$

$$= 4.855 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 950 \cdot 4.85$$

$$= 4613.291 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 96.2 + 50.7$$

$$= 146.921 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 4613 / 147$$

$$= 3.079 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 425 - 24$$

$$= 401.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 147$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 425 / 2$$

$$= 212.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (15353 / 147) }$$

$$= 102.225 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1575 - \cos(130 / 2) (900 + 16 + 16) - 36$$

$$= 1145.120 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1145/102$$

$$= 11.202$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 189 / (2 \cdot 1803) \cdot 422 \cdot 1145/2$$

$$= 12.672 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.2 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.2)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$$

$$= 152.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 3.08/152 + (12.7 \cdot 213/153531280) / 173$$

$$= 0.021$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/198$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1575 - \cos(932/2) (36 + 0 + 0) - 0$$

$$= 607.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (15399/198) }$$

$$= 88.278 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 607/88.3$$

$$= 6.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 189 / (2 \cdot 1803)$$

$$= 0.052 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.052 \cdot 422 \cdot 607$$

$$= 13.435 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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= 13.4/724660
 = 0.182 N./mm²

Compressive Allowable, $KL/r < C_c$ (6.88 < 123) per AISC E2-1 [Sca]:
 $= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (6.88)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.88) / (8 \cdot 123) - (6.88^3) / (8 \cdot 123^3))$
 = 153.8 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
 = 4.58/154 + (13.4 • 213/15399)/173
 = 0.031

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load QO (Weight)	QO	38333.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	42.5 Kg
Maximum Transverse Force	Ft	377.3 Kg
Maximum Longitudinal Force	F1	189.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 189/(4.87 • 4)
 = 1.0 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 377/(4.87 • 4)
 = 1.9 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = 189/(29 • 4)
 = 0.1599 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = 1575 • 377 + 0
 = 594.23 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = 594/38334
 = 15.50 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.16, 0]
 = 0.1599 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	909.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	11964.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1110.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	130.00	degrees
Wear Plate Width	b2 or b1	625.00	mm
Wear Plate Bearing Angle	delta2 or theta1	142.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1575.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		39465.82	Kgf
Pressure used in Analysis	P	2023.915	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	48.86	234.00
Long. Stress at Bottom of Midspan	92.75	234.00
Long. Stress at Top of Saddles	77.17	234.00
Long. Stress at Bottom of Saddles	67.23	234.00
Tangential Shear in Shell	27.25	138.58
Circ. Stress at Horn of Saddle	36.62	350.99
Circ. Compressive Stress in Shell	1.63	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (755/2 + 0) • 1575/1689
 = 1055.2 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(189, 0) • 1575/7000
 = 42.5 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 38411 + max(42.5, 1055, 0, 0)



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= 39465.8 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress (Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (1.97 • 1.2)²)
 = 3706.263 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	40617.23	Kgf
Transverse Shear Load	377.28	Kgf
Longitudinal Shear Load	188.97	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1800/2 + 16 + 16)sin(130/2), 2•910)
 = 1689.358 mm

The Computed K values from Table 4.15.1:

K1 = 0.1231	K2 = 1.0222	K3 = 0.7217	K4 = 0.3619
K5 = 0.7258	K6 = 0.0449	K7 = 0.0449	K8 = 0.3296
K9 = 0.2551	K10 = 0.0508	K1* = 0.2196	K6p = 0.0366
K7p = 0.0366			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -39466•1110 [1 - (1 - 1110/11964 + (910²-0²)/(2•1110•11964))/(1+(4•0)/(3•11964))]
 = -2700.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 39466•11964/4 ((1+2(910²-0²)/(11964²))/(1+(4•0)/(3•11964)) - 4•1110/11964)
 = 75601.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 2024 • 910/(2•13) - 75601/(π •910²•13)
 = 48.86 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 2024 • 910/(2 • 13) + 75601/(π • 910² • 13)
 = 92.75 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 2024•910/(2•13)--2700/(0.12• π •910²•13)
 = 77.17 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 2024•910/(2•13)+2700/(0.22• π •910²•13)
 = 67.23 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 39466 (11964 - 2 • 1110)/(11964 + (4 • 0/3))
 = 32142.7 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.02 \cdot 32143 / (910 \cdot 13)$$

$$= 27.25 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(910 \cdot 13)}$$

$$= 84.814 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(910 \cdot 13)}, 2 \cdot 1110)$$

$$= 594.63 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.73 \cdot 39466 \cdot 0.1 / (595(13 + 1 \cdot 16))$$

$$= -1.63 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -39466 / (4(13 + 1 \cdot 16)595) -$$

$$3 \cdot 0.045 \cdot 39466 / (2(13 + 1 \cdot 16)^2)$$

$$= -36.62 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1803.0000	mm
Baseplate Thickness	Bpthk	36.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	675.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(903 \cdot 13)} + 625)13^3) / 12$$

$$= 14.537 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 14.5 + 103(308 - 6.5)^2$$

$$= 93602.125 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	794.0	13.0	6.5	103.2	67094.8	0.936E+05
Wearplate	625.0	16.0	21.0	100.0	210000.0	0.822E+05
Web	24.0	607.0	332.5	145.7	4843859.5	0.456E+05
BasePlate	450.0	36.0	654.0	162.0	10594800.0	0.195E+06
Totals	...	...	...	510.9	15715755.0	0.416E+06

Distance to Centroid [C1]:



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$$= AY / A$$

$$= 15715755/511$$

$$= 307.608 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 130/2$$

$$= 115.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(115) - 0.5 \cdot \sin(115)^2) / (\pi - 2.01 + \sin(115) \cos(115))$$

$$= 0.2218$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.22 \cdot 39466$$

$$= 8754.3096 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 2.1059 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1575 - 903 \cdot \sin(130/2) / 1.13$$

$$= 853.606 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 7472.8755 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 5.4195 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(39466 + 1151) \cdot 450 / (4 \cdot 1803 \cdot 173)}$$

$$= 20.741 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 130/2) (900 + 16 + 16)$$

$$= 1689.358 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1689 / (5 - 1)$$

$$= 422.339 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 422 \cdot 450/2$$

$$= 950.264 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 40617 / (1803 \cdot 450)$$

$$= 5.006 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 950 \cdot 5.01$$

$$= 4757.143 \text{ Kgf}$$

Area of the Rib and Web [Ar]:



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= Rib Area + Web Area
 = 96.2 + 50.7
 = 146.921 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 4757/147
 = 3.175 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	425.0	...	102.0	...	0.154E+05

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 425 - 24
 = 401.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/147
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 425/2
 = 212.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{15353/147}$
 = 102.225 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1575 - cos(130/2)(900 + 16 + 16) - 36
 = 1145.120 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$
 = 123.211

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1145/102
 = 11.202

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 189 / (2 • 1803) • 422 • 1145/2
 = 12.672 Kg-m.

Compressive Allowable, KL/r < Cc (11.2 < 123) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (11.2)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.2) / (8 \cdot 123) - (11.2^3) / (8 \cdot 123^3))$
 = 152.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba



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$$= 3.18/152 + (12.7 \cdot 213/153531280)/173$$

$$= 0.022$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	401.0	0.0	96.2	0.0	0.154E+05
Web	422.3	24.0	0.0	101.4	0.0	48.7
Totals	...	...	...	197.6	...	0.154E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/198$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1575 - \cos(932/2)(36 + 0 + 0) - 0$$

$$= 607.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(15399/198)}$$

$$= 88.278 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 607/88.3$$

$$= 6.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 189/(2 \cdot 1803)$$

$$= 0.052 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.052 \cdot 422 \cdot 607$$

$$= 13.435 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 13.4/724660$$

$$= 0.182 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($6.88 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (6.88)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (6.88) / (8 \cdot 123) - (6.88)^3 / (8 \cdot 123^3))$$

$$= 153.8 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 4.72/154 + (13.4 \cdot 213/15399)/173$$

$$= 0.032$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8674 cm ²
Saddle Load Q0 (Weight)	Q0	39562.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	42.5 Kgf
Maximum Transverse Force	Ft	377.3 Kgf
Maximum Longitudinal Force	F1	189.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:
 $= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 189 / (4.87 \cdot 4)$
 $= 1.0 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 377 / (4.87 \cdot 4)$
 $= 1.9 \text{ N./mm}^2$. Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 189 / (29 \cdot 4)$
 $= 0.1599 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1575 \cdot 377 + 0$
 $= 594.23 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 594 / 39562$
 $= 15.02 \text{ mm} < Bplen / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.16, 0]$
 $= 0.1599 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.

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 Nozzle Calcs.: Noz N6 Nozl: 33 9:30am Sep 9,2026

Input, Nozzle Desc: Noz N6

From: 30

Pressure for Reinforcement Calculations	P	4550.000	KPa.
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Design Length of Section	L	1499.9696	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1068.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

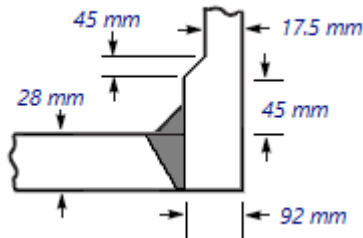
Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4800	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	45.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		92.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Noz N6

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.688 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4550 \cdot 603 / (138 \cdot 1 - 0.6 \cdot 4550)$
 $= 20.2843 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4550 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 4550)$
 $= 9.9195 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6373 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 74.5 / 0.5773$
 $= 129.0725 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.2800 mm
Parallel to Vessel Wall, opening length	d	580.6400 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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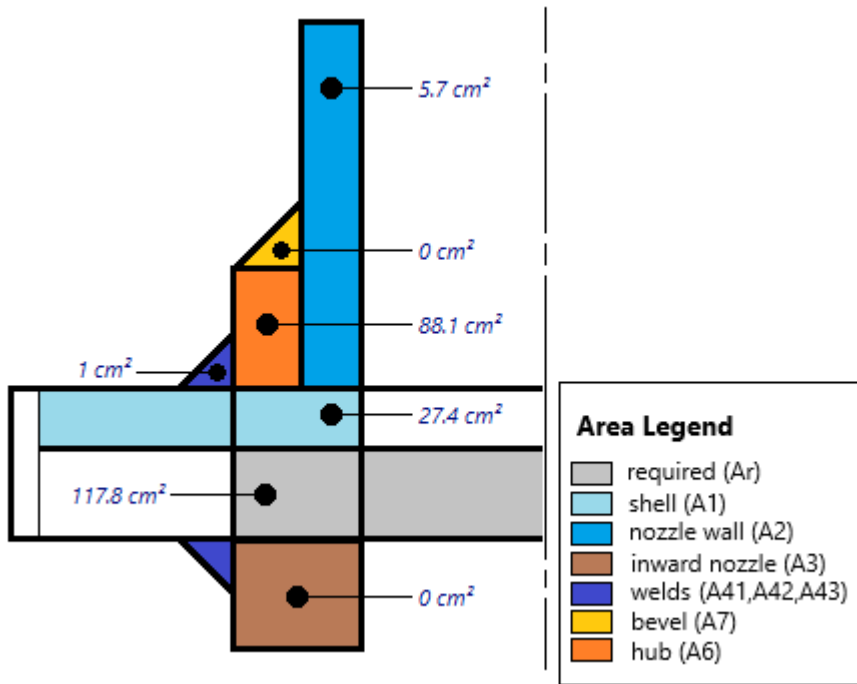
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Nozzle Calcs.: Noz N6

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	117.779	12.602	NA
Area in Shell	A1	27.381	119.955	NA
Area in Nozzle Wall	A2	5.701	16.053	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	88.079	88.079	NA
TOTAL AREA AVAILABLE	Atot	122.160	225.087	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 20.3 \cdot 1 + 2 \cdot 89 \cdot 20.3 \cdot 1 \cdot (1 - 1))$$

$$= 117.779 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 25 - 1 \cdot 20.3) - 2 \cdot 89$$

$$(1 \cdot 25 - 1 \cdot 20.3) \cdot (1 - 1)$$

$$= 27.381 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(14.5 - 9.92)1$$

$$= 5.701 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - c_n / 0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(62.5, 400, 45)) \cdot (92 - 17.5) \cdot 1$
 $= 88.079 \text{ cm}^2$ Includes Bevel Area of 21.011 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.9195 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 23.2843 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 23.2843 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4521 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.3, \max(23.3, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(12.9, 11.3)$
 $= 12.9195 \text{ mm}$

Available Nozzle Neck Thickness = 17.4800 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.6 N./mm ²	Passed
Occasional	: 44.5,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 15.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17.5, tr = 9.92, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.69$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-9 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 28, tr = 20.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.81$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-8 °C

Gov. MDMT of the nozzle to shell joint welded assembly :	-8 °C
--	-------

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
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Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4550.00/5100.00 = 0.892

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 74.5/0.5773
 = 129.0725 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 870.9600 mm
 Parallel to Vessel Wall, opening length d 435.4800 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 62.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		78.523	78.523
Area Available in Shell A1		13.690	13.690
Area Available in Nozzle Wall A2		5.701	5.701
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.000	1.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		88.079	88.079
TOTAL AREA AVAILABLE Atot		108.470	108.470

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Noz N6

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (118 - 27.4 + 2 • 89 • 1 •
 (1 • 25 - 20.3))138)
 = 139015.72 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A4-(Wi-Can/.707)²•fr2)•Sv
 = (93.8 + 1 - 0 • 1) • 138
 = 133369.14 Kgf



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Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (93.8 + 0 + 1 + (44.5)) \cdot 138$$

$$= 195987.62 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (93.8 + 0 + 1 + 0 + (44.5)) \cdot 138$$

$$= 195987.62 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 66024. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 298) \cdot (92 - 3) \cdot 0.7 \cdot 138$$

$$= 819512. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 610 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 139594. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (66024 + 819512) = 885536 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (66024 + 0 + 139594 + 0) = 205618 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (66024 + 139594 + 0) = 205618 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 885535 Kgf, must exceed W = 139015 Kgf or W1 = 133369 Kgf
 Path 2-2 = 205618 Kgf, must exceed W = 139015 Kgf or W2 = 195987 Kgf
 Path 3-3 = 205618 Kgf, must exceed W = 139015 Kgf or W3 = 195987 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4629 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 83.1858 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 511.1858 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 580.64 mm

*Warning:
 Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).*

Input Echo, WRC297 Item 1, Description: Noz N6 :



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1200.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	758.640	mm
Nozzle Thickness	92.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-765 II	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	4550.00	KPa.
Radial Load	P	-5506.84	Kgf
Circumferential Shear	Vc	3977.16	Kgf
Longitudinal Shear	Vl	3977.16	Kgf
Circumferential Moment	Mc	7339.62	Kg-m.
Longitudinal Moment	Ml	7339.62	Kg-m.
Torsional Moment	Mt	8440.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.604) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (8.246) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 758.64 + 2 \cdot 1.65 \cdot \sqrt{(615.5(28.0 - 3.0))}$$

$$= 1167.993 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1231.000	mm
Nozzle Outside Diameter	(d) =	758.640	mm
Vessel Thickness used	(T) =	25.000	mm
Nozzle Thickness used	(t) =	89.000	mm
T / t	=	0.281	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.325$$

$$Nr/P = 0.050$$



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Mr/P = 0.075
MØ/P = 0.026
NØ/P = 0.060
MrD/Mc = 0.254
NrDL/Mc = 0.091
MØd/Mc = 0.076
NØDL/Mc = 0.084
MrD/Ml = 0.069
NrDL/Ml = 0.044
MØD/Ml = 0.021
NØDL/Ml = 0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	107	112	107	112
Outplane Stress Summary	97	113	154	94

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	27	75	165	-36

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	97	114	166	130

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-69	69	69	-69
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53

Outplane Stress Summary	-9	102	154	-10
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-231	231	231	-231
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	112	107	112

Inplane Stress Summary	-94	295	395	-139
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	94	295	395	139
------------------------	----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20

Hoop Stress Summary	11	16	29	35
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2



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Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	4	6	15	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	16	29	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	7	13	33	38

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-7	18	27	-6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	C1 Inside	Du Outside	D1 Inside
Two * Max Shear Stress	15	18	33	45

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	C1	Du	D1
Circ. Pm (SUS)	107	112	107	112	107	112	107	112
Circ. Pl (SUS)	-4	-4	14	14	-9	-9	17	17
Circ. Q (SUS)	-6	6	32	-32	-193	193	269	-269
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-2	-2	10	10	-7	-7	17	17
Long. Q (SUS)	-24	24	100	-100	-56	56	82	-82
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	107.0	112.0	107.0	112.0	107.0	112.0	107.0	112.0
Pm+Pl (SUS)	103.0	108.0	121.0	126.0	98.0	103.0	124.0	129.0
Pm+Pl+Q (Total)	97.2	114.4	163.4	131.1	95.0	296.0	393.1	140.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	112.00	138.00	Passed
Pm+Pl (SUS)	129.02	207.00	Passed
Pm+Pl+Q (TOTAL)	393.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	C1	Du	D1
Circ. Pm (SUS)	15	20	15	20	15	20	15	20
Circ. Pl (SUS)	-4	-4	14	14	-7	-7	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-12	12	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0
Pm+Pl (SUS)	11.0	16.0	29.0	34.0	8.0	13.0	32.0	37.0



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Pm+Pl+Q (Total)	11.1	16.1	29.1	34.0	15.1	17.2	32.1	44.0
-----------------	------	------	------	------	------	------	------	------

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	37.00	207.00	Passed
Pm+Pl+Q (TOTAL)	44.05	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: Npz N2 250

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Input, Nozzle Desc: Npz N2 250

From: 60

Pressure for Reinforcement Calculations	P	1550.000	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5326.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

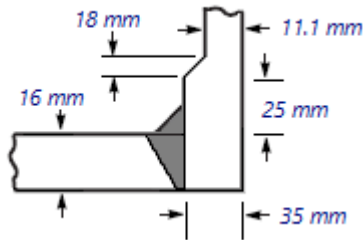
Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	25.0000	mm
Height of Beveled Transition	L'	18.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Npz N2 250

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Reinforcement CALCULATION, Description: Npz N2 250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.438 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1550 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1550)$
 $= 10.2120 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1550 \cdot 137 / (138 \cdot 1 + 0.4 \cdot 1550)$
 $= 1.5267 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 23.9 / 0.5773$
 $= 41.3744 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	513.6500	mm
Parallel to Vessel Wall, opening length	d	256.8250	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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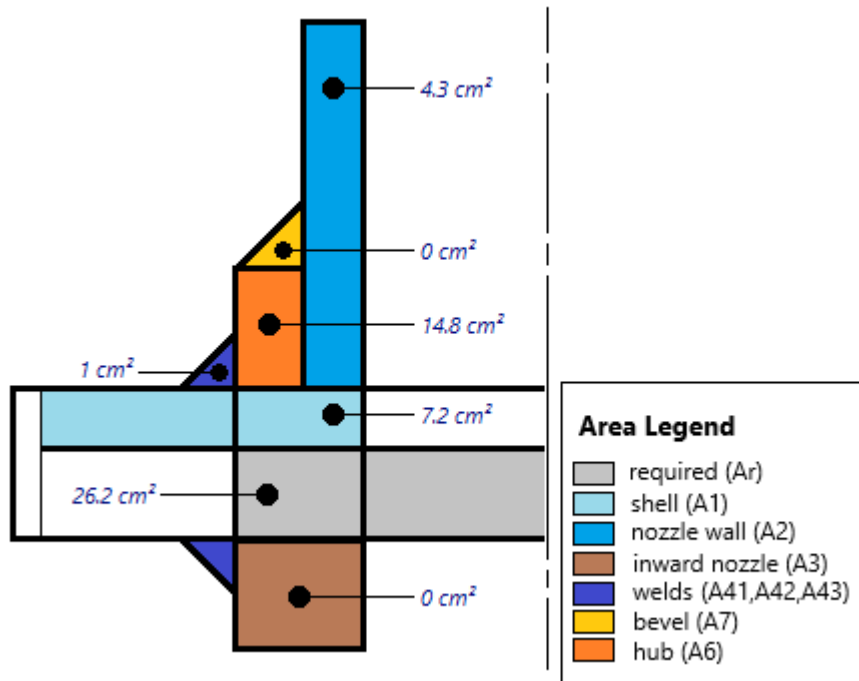
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Nozzle Calcs.: Npz N2 250

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	26.227	NA	NA
Area in Shell	A1	7.160	NA	NA
Area in Nozzle Wall	A2	4.281	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	14.780	NA	NA
TOTAL AREA AVAILABLE	Atot	27.221	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (257 \cdot 10.2 \cdot 1 + 2 \cdot 32 \cdot 10.2 \cdot 1 \cdot (1 - 1))$$

$$= 26.227 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 257(1 \cdot 13 - 1 \cdot 10.2) - 2 \cdot 32$$

$$(1 \cdot 13 - 1 \cdot 10.2) \cdot (1 - 1)$$

$$= 7.160 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(8.11 - 1.53)1$$

$$= 4.281 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(32.5, 250, 25)) \cdot (35 - 11.1) \cdot 1$
 $= 14.780 \text{ cm}^2$ Includes Bevel Area of 2.837 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5267 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2120 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2120 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.1, \max(13.2, 4.5)]$
 $= 11.1026 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.53, 11.1)$
 $= 11.1026 \text{ mm}$

Available Nozzle Neck Thickness = 11.1125 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 43.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 301.9 N./mm ²	Passed
Occasional	: 11.9,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 20.3,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.1, tr = 1.53, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -29 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1550.00/1960.01 = 0.791

Weld Size Calculations, Description: Npz N2 250

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (26.2 - 7.16 + 2 \cdot 32 \cdot 1 \cdot (1 \cdot 13 - 10.2)) \cdot 138)$
 $= 29340.57 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (19.1 + 1 - 0 \cdot 1) \cdot 138$
 $= 28229.22 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (19.1 + 0 + 1 + (8.32)) \cdot 138$
 $= 39936.76 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (19.1 + 0 + 1 + 0 + (8.32)) \cdot 138$
 $= 39936.76 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 138$
 $= 29573. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 132) \cdot (35 - 3) \cdot 0.7 \cdot 138$
 $= 131176. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 273 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 58060. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (29573 + 131176) = 160749 \text{ Kgf}$



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Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (29573 + 0 + 58060 + 0) = 87634 Kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (29573 + 58060 + 0) = 87634 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 160749 Kgf, must exceed W = 29340 Kgf or W1 = 28229 Kgf
 Path 2-2 = 87633 Kgf, must exceed W = 29340 Kgf or W2 = 39936 Kgf
 Path 3-3 = 87633 Kgf, must exceed W = 29340 Kgf or W3 = 39936 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1579 KPa.

The Drop for this Nozzle is : 10.4153 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.4153 mm

Input Echo, WRC297 Item 1, Description: Npz N2 250 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1800.000	mm	
Shell Thickness	16.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	320.825	mm	
Nozzle Thickness	35.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-765 II		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1550.00	KPa.
Radial Load	P	1907.00	Kgf
Circumferential Shear	Vc	1274.73	Kgf
Longitudinal Shear	Vl	1274.73	Kgf
Circumferential Moment	Mc	891.97	Kg-m.
Longitudinal Moment	Ml	891.97	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (9.166) is < 10
 check the limitations of WRC bulletin 297.



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Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 320.825 + 2 \cdot 1.65 \cdot \sqrt{909.5(16.0 - 3.0)}$
 $= 679.653 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1819.000 mm
 Nozzle Outside Diameter (d) = 320.825 mm
 Vessel Thickness used (T) = 13.000 mm
 Nozzle Thickness used (t) = 32.000 mm
 T / t = 0.406
 d / t = 10.026

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.086$$

Nr/P = 0.068
 Mr/P = 0.138
 M0/P = 0.041
 N0/P = 0.090
 MrD/Mc = 0.283
 NrDL/Mc = 0.097
 M0d/Mc = 0.087
 N0DL/Mc = 0.074
 MrD/Ml = 0.109
 NrDL/Ml = 0.070
 M0D/Ml = 0.034
 N0DL/Ml = 0.114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-9	-9	-9	-9
Outplane Bending (P)	-26	26	-26	26
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-33	33	33	-33
Normal Pressure Stress	106	108	106	108
Outplane Stress Summary	18	140	121	110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-7	-7	-7	-7
Inplane Bending (P)	-91	91	-91	91
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	-162	232	70	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			5	5	5	5
Shear Stress Summary			7	7	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Stress Intensities			Top	Top	Bottom	Bottom
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			181	232	122	110

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Membrane (P)			-9	-9	-9	-9
Outplane Bending (P)			-26	26	-26	26
Outplane Membrane (Mc)			-11	-11	11	11
Outplane Bending (Mc)			-84	84	84	-84
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			53	53	53	53
Outplane Stress Summary			-79	143	112	-2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Inplane Membrane (P)			-7	-7	-7	-7
Inplane Bending (P)			-91	91	-91	91
Inplane Membrane (Mc)			-15	-15	15	15
Inplane Bending (Mc)			-274	274	274	-274
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			106	108	106	108
Inplane Stress Summary			-282	451	297	-65

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			5	5	5	5
Shear Stress Summary			4	4	7	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	Dl
(Stress Intensities			Left	Left	Right	Right
			Outside	Inside	Outside	Inside
Two * Max Shear Stress			282	451	297	66

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
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(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-9	-9	-9	-9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	-21	-19	15	16

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	-11	11	-11	11
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-11	11	11	-11
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-25	21	7	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	27	41	15	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	-9	-9	-9	-9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	8	6	8
Hoop Stress Summary	-15	-13	8	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0



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Nozzle Calcs.: Npz N2 250

Nozl: 34

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Axial Bending (P)	-11	11	-11	11
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-37	37	37	-37
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-50	47	33	-18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	50	60	33	29

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	108	106	108	106	108	106	108
Circ. Pl (SUS)	-27	-27	9	9	-22	-22	8	8
Circ. Q (SUS)	-59	59	7	-7	-365	365	183	-183
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-18	-18	4	4	-20	-20	2	2
Long. Q (SUS)	-196	196	14	-14	-110	110	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	106.0	108.0	106.0	108.0	106.0	108.0	106.0	108.0
Pm+Pl (SUS)	79.0	81.0	115.0	117.0	84.0	86.0	114.0	116.0
Pm+Pl+Q (Total)	181.4	231.4	122.3	110.2	281.1	451.1	297.2	67.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	138.00	Passed
Pm+Pl (SUS)	117.02	207.00	Passed
Pm+Pl+Q (TOTAL)	451.05	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	8	6	8	6	8	6	8
Circ. Pl (SUS)	-27	-27	9	9	-20	-20	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	-22	22	0	0	-48	48	26	-26
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	6.0	8.0	6.0	8.0	6.0	8.0	6.0	8.0
Pm+Pl (SUS)	21.0	19.0	15.0	17.0	14.0	12.0	8.0	10.0
Pm+Pl+Q (Total)	24.2	40.2	15.5	17.4	49.1	59.1	33.2	29.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	21.00	207.00	Passed
Pm+Pl+Q (TOTAL)	59.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz N3 80

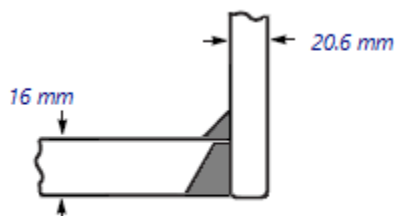
From: 60

Pressure for Reinforcement Calculations	P	1561.239	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7326.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm
Flange Material		SA-765 II	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





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ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 3.000 in.
Actual Thickness Used in Calculation 0.810 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1561 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1561)$
= 10.2865 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1561 \cdot 41.1 / (138 \cdot 1 - 0.6 \cdot 1561)$
= 0.4682 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	164.4000	mm
Parallel to Vessel Wall, opening length	d	82.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, 138/138)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



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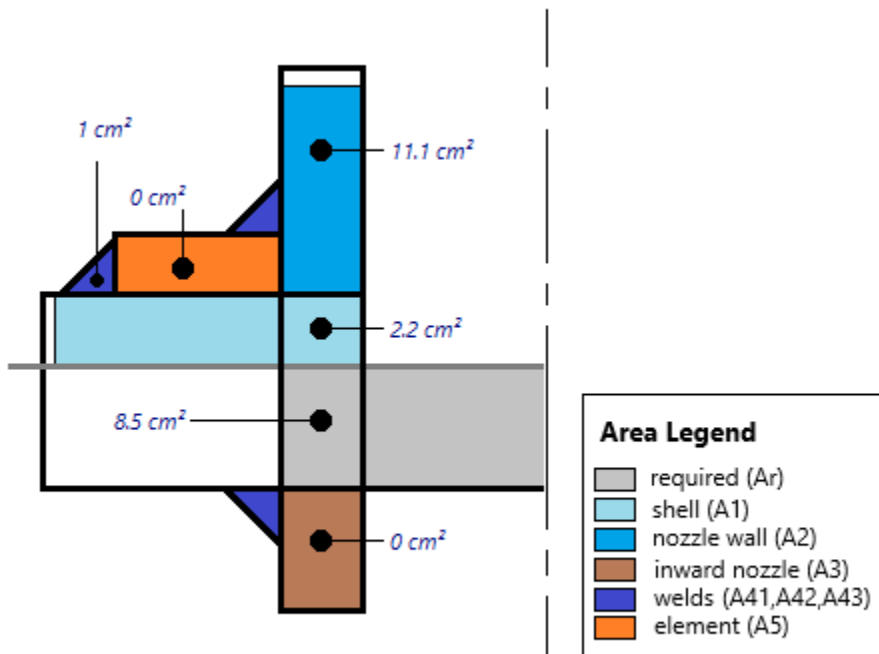


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.456	NA	NA
Area in Shell	A1	2.230	NA	NA
Area in Nozzle Wall	A2	11.119	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	14.349	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (82.2 \cdot 10.3 \cdot 1 + 2 \cdot 17.6 \cdot 10.3 \cdot 1 \cdot (1 - 1))$
 $= 8.456 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 82.2(1 \cdot 13 - 1 \cdot 10.3) - 2 \cdot 17.6$
 $(1 \cdot 13 - 1 \cdot 10.3) \cdot (1 - 1)$
 $= 2.230 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(17.6 - 0.47)1$
 $= 11.119 \text{ cm}^2$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4682 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2865 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2865 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 8.7300 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.73, \max(13.3, 4.5)]$$

$$= 8.7300 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.47, 8.73)$$

$$= 8.7300 \text{ mm}$$

Available Nozzle Neck Thickness = 20.5740 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	11.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	334.0 N./mm ²	Passed
Occasional	:	1.5,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	9.4,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.79, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1561.24/5100.00 = 0.306

Weld Size Calculations, Description: Noz N3 80

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm



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Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (8.46 - 2.23 + 2 \cdot 17.6 \cdot 1 \cdot (1 \cdot 13 - 10.3)) \cdot 138)$$

$$= 10101.70 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (11.1 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 17053.02 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (11.1 + 0 + 1 + (4.57)) \cdot 138$$

$$= 23482.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (11.1 + 0 + 1 + 0 + (4.57)) \cdot 138$$

$$= 23482.66 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 117 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 12710. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 49.9) \cdot (20.6 - 3) \cdot 0.7 \cdot 138$$

$$= 27130. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 117 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 24952. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (12710 + 27130) = 39840 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (12710 + 0 + 24952 + 0) = 37662 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (12710 + 24952 + 0) = 37662 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 39839	Kgf, must exceed W = 10101 Kgf	or W1 = 17053 Kgf
Path 2-2 = 37662	Kgf, must exceed W = 10101 Kgf	or W2 = 23482 Kgf
Path 3-3 = 37662	Kgf, must exceed W = 10101 Kgf	or W3 = 23482 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1958 KPa.



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Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.9146 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9396 mm

Input Echo, WRC107/537 Item 1, Description: Noz N3 80 :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Cylindrical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	1800.000	mm
Vessel Thickness	Tv	16.000	mm
Design Temperature	T1	65.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	76.200	mm
Nozzle Thickness	Tn	20.574	mm
Nozzle Material		SA-765 II	
Nozzle UNS Number		K03047	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	1561.239	KPa.
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-611.9	Kgf
Longitudinal Shear (SUS)	Vl	407.9	Kgf
Circumferential Shear (SUS)	Vc	407.9	Kgf
Circumferential Moment (SUS)	Mc	91.7	Kg-m.
Longitudinal Moment (SUS)	ML	91.7	Kg-m.
Torsional Moment (SUS)	Mt	130.5	Kg-m.

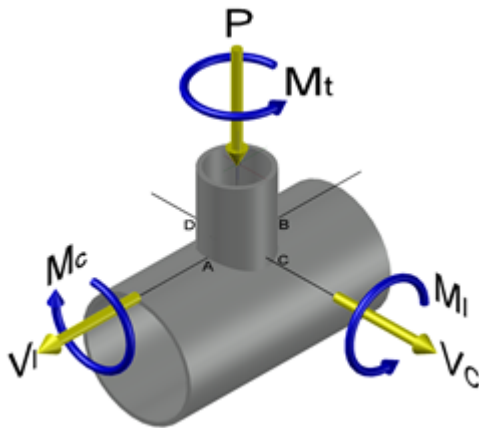
Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



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WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 117.348 + 2 \cdot 1.65 \cdot \sqrt{909.5(16.0 - 3.0)}$$

$$= 476.177 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-611.9	Kgf
Circumferential Shear	VC	407.9	Kgf
Longitudinal Shear	VL	407.9	Kgf
Circumferential Moment	MC	91.7	Kg-m.
Longitudinal Moment	ML	91.7	Kg-m.
Torsional Moment	MT	130.5	Kg-m.

Dimensionless Parameters used : Gamma = 69.96

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
$N(\text{PHI}) / (P/R_m)$	0.056	4C	12.477	(A,B)
$N(\text{PHI}) / (P/R_m)$	0.056	3C	12.150	(C,D)
$M(\text{PHI}) / (P)$	0.056	2C1	0.115	(A,B)
$M(\text{PHI}) / (P)$	0.056	1C	0.155	(C,D)
$N(\text{PHI}) / (MC/(R_m^{**2} * \text{Beta}))$	0.056	3A	1.482	(A,B,C,D)
$M(\text{PHI}) / (MC/(R_m * \text{Beta}))$	0.056	1A	0.100	(A,B,C,D)
$N(\text{PHI}) / (ML/(R_m^{**2} * \text{Beta}))$	0.056	3B	5.631	(A,B,C,D)
$M(\text{PHI}) / (ML/(R_m * \text{Beta}))$	0.056	1B	0.053	(A,B,C,D)
$N(x) / (P/R_m)$	0.056	3C	12.150	(A,B)
$N(x) / (P/R_m)$	0.056	4C	12.477	(C,D)
$M(x) / (P)$	0.056	1C1	0.158	(A,B)
$M(x) / (P)$	0.056	2C	0.115	(C,D)
$N(x) / (MC/(R_m^{**2} * \text{Beta}))$	0.056	4A	2.008	(A,B,C,D)
$M(x) / (MC/(R_m * \text{Beta}))$	0.056	2A	0.059	(A,B,C,D)
$N(x) / (ML/(R_m^{**2} * \text{Beta}))$	0.056	4B	1.540	(A,B,C,D)
$M(x) / (ML/(R_m * \text{Beta}))$	0.056	2B	0.086	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of	Stress Intensity Values at
---------	----------------------------



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Stress	Load	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		6.3	6.3	6.3	6.3	6.2	6.2	6.2	6.2
Circ. Bend. P		24.5	-24.5	24.5	-24.5	32.9	-32.9	32.9	-32.9
Circ. Memb. MC		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Circ. Bend. MC		0.0	0.0	0.0	0.0	-62.0	62.0	62.0	-62.0
Circ. Memb. ML		-8.3	-8.3	8.3	8.3	0.0	0.0	0.0	0.0
Circ. Bend. ML		-32.8	32.8	32.8	-32.8	0.0	0.0	0.0	0.0
Tot. Circ. Str.		-10.3	6.3	72.0	-42.6	-25.1	33.0	103.3	-86.6
Long. Memb. P		6.2	6.2	6.2	6.2	6.3	6.3	6.3	6.3
Long. Bend. P		33.7	-33.7	33.7	-33.7	24.5	-24.5	24.5	-24.5
Long. Memb. MC		0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-36.6	36.6	36.6	-36.6
Long. Memb. ML		-2.3	-2.3	2.3	2.3	0.0	0.0	0.0	0.0
Long. Bend. ML		-53.4	53.4	53.4	-53.4	0.0	0.0	0.0	0.0
Tot. Long. Str.		-15.9	23.6	95.6	-78.7	-8.7	15.4	70.3	-51.7
Shear VC		1.7	1.7	-1.7	-1.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.7	-1.7	1.7	1.7
Shear MT		4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Tot. Shear		6.2	6.2	2.9	2.9	2.9	2.9	6.2	6.2
Str. Int.		19.9	25.6	95.9	78.9	25.6	33.5	104.4	87.6

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N/mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		107.7	109.2	107.7	109.2	107.7	109.2	107.7	109.2
Circ. Pl (SUS)		-2.0	-2.0	14.7	14.7	4.0	4.0	8.4	8.4
Circ. Q (SUS)		-8.3	8.3	57.3	-57.3	-29.1	29.1	94.9	-94.9
Long. Pm (SUS)		53.8	53.8	53.8	53.8	53.8	53.8	53.8	53.8
Long. Pl (SUS)		3.9	3.9	8.4	8.4	3.4	3.4	9.3	9.3
Long. Q (SUS)		-19.7	19.7	87.1	-87.1	-12.1	12.1	61.0	-61.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.7	1.7	-1.7	-1.7	-1.7	-1.7	1.7	1.7
Shear Q (SUS)		4.6	4.6	4.6	4.6	4.6	4.6	4.6	4.6
Pm (SUS)		107.7	109.2	107.7	109.2	107.7	109.2	107.7	109.2
Pm+Pl (SUS)		105.7	107.3	122.4	124.0	111.7	113.3	116.1	117.7
Pm+Pl+Q (Total)		98.0	116.5	179.9	91.6	82.8	142.4	211.4	24.4

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	109.24	138.00	Passed
Pm+Pl (SUS)	123.96	207.00	Passed
Pm+Pl+Q (TOTAL)	211.41	413.99	Passed



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Nozzle Calcs.: Noz N3 80

Nozl: 35

9:30am

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*Because only sustained loads were specified, the $P_m + P_l + Q$ allowable was $3 * S_m h$.*

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 Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026

Input, Nozzle Desc: Noz N1 150

From: 60

Pressure for Reinforcement Calculations	P	1552.552	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		7326.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

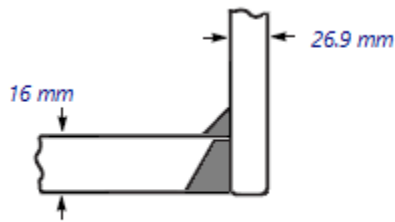
Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		56.92	deg
Diameter		6.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.9240	mm
Flange Material		SA-765 II	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	268.2880	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz N1 150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1553 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1553)$
= 10.2289 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1553 \cdot 79.2 / (138 \cdot 1 - 0.6 \cdot 1553)$
= 0.8972 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	381.1213	mm
Parallel to Vessel Wall, opening length	d	190.5607	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

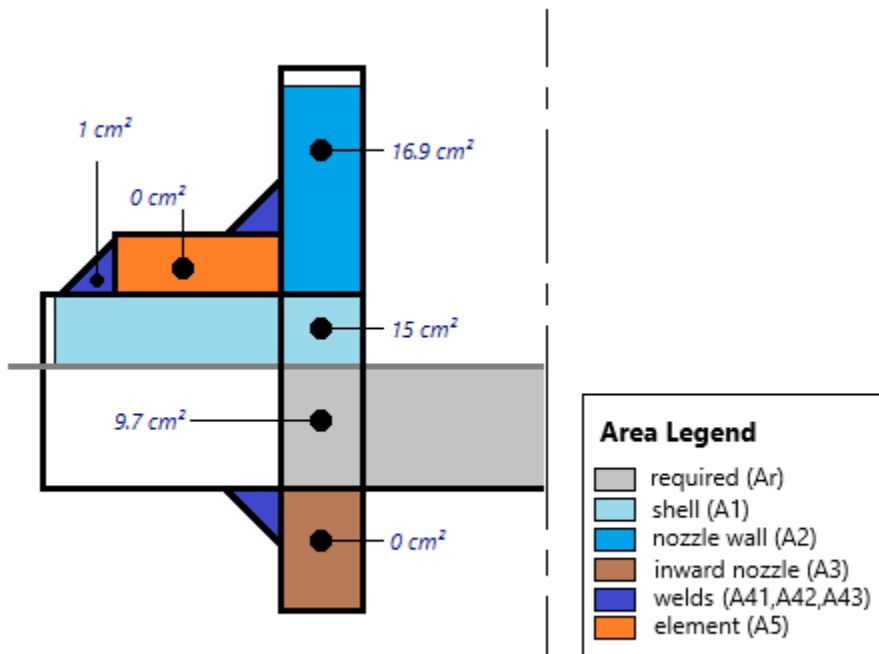


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.746	NA	NA
Area in Shell	A1	15.027	NA	NA
Area in Nozzle Wall	A2	16.851	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	32.878	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.23 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (191 \cdot 10.2 \cdot 0.5 + 2 \cdot 23.9 \cdot 10.2 \cdot 0.5 \cdot (1 - 1))$
 $= 9.746 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 191(1 \cdot 13 - 0.5 \cdot 10.2) - 2 \cdot 23.9$
 $(1 \cdot 13 - 0.5 \cdot 10.2) \cdot (1 - 1)$
 $= 15.027 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 32.5)(23.9 - 0.9)/\sin(62.7)$
 $= 16.851 \text{ cm}^2$



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 Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16$, $t_r = 10.2$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1552.55 / 5100.00 = 0.304$

Weld Size Calculations, Description: Noz N1 150

Intermediate Calc. for nozzle/shell Welds T_{min} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.75 - 15 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 13 - 5.11)) \cdot 138)$
 $= \max(0, -2121.32) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (16.9 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 25118.83 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (16.9 + 0 + 1 + (6.22)) \cdot 138$
 $= 33871.69 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (16.9 + 0 + 1 + 0 + (6.22)) \cdot 138$
 $= 33871.69 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



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$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 248 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 26874. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 110) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$$

$$= 81193. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 248 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 52760. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (26874 + 81193) = 108066 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (26874 + 0 + 52760 + 0) = 79634 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (26874 + 52760 + 0) = 79634 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 108066	Kgf, must exceed W = 0 Kg	or W1 = 25118 Kg
Path 2-2 = 79633	Kgf, must exceed W = 0 Kg	or W2 = 33871 Kg
Path 3-3 = 79633	Kgf, must exceed W = 0 Kg	or W3 = 33871 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1958 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz N1 150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	6.000 in.
Actual Thickness Used in Calculation	1.060 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1553 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1553)$$

$$= 10.2289 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1553 \cdot 79.2 / (138 \cdot 1 - 0.6 \cdot 1553)$$

$$= 0.8972 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	316.8000	mm
Parallel to Vessel Wall, opening length	d	158.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm



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Nozzle Calcs.: Noz N1 150

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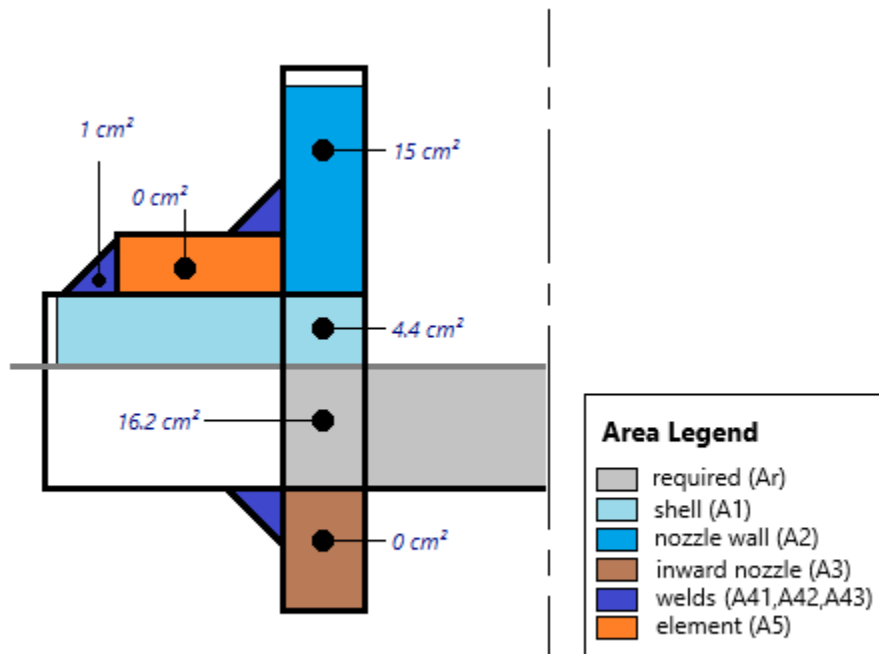


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.203	NA	NA
Area in Shell	A1	4.389	NA	NA
Area in Nozzle Wall	A2	14.967	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	20.357	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (158 \cdot 10.2 \cdot 1 + 2 \cdot 23.9 \cdot 10.2 \cdot 1 \cdot (1 - 1)) \\
 &= 16.203 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 158(1 \cdot 13 - 1 \cdot 10.2) - 2 \cdot 23.9 \\
 &\quad (1 \cdot 13 - 1 \cdot 10.2) \cdot (1 - 1) \\
 &= 4.389 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 32.5)(23.9 - 0.9)1 \\
 &= 14.967 \text{ cm}^2
 \end{aligned}$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8972 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2289 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2289 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.2, \max(13.2, 4.5)]$$

$$= 10.1600 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.9, 10.2)$$

$$= 10.1600 \text{ mm}$$

Available Nozzle Neck Thickness = 26.9240 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	9.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	335.7 N./mm ²	Passed
Occasional	:	2.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	6.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: Noz N1 150

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (16.2 - 4.39 + 2 \cdot 23.9 \cdot 1 \cdot (1 \cdot 13 - 10.2)) \cdot 138)$$

$$= 18488.80 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (15 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 22468.71 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (15 + 0 + 1 + (6.22)) \cdot 138$$

$$= 31221.56 \text{ Kg}$$



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Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (15 + 0 + 1 + 0 + (6.22)) \cdot 138$$

$$= 31221.56 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 206 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 22338. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 91.2) \cdot (26.9 - 3) \cdot 0.7 \cdot 138$$

$$= 67490. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 206 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 43856. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (22338 + 67490) = 89828 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (22338 + 0 + 43856 + 0) = 66194 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (22338 + 43856 + 0) = 66194 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 89828	Kgf, must exceed W = 18488 Kgf	or W1 = 22468 Kgf
Path 2-2 = 66194	Kgf, must exceed W = 18488 Kgf	or W2 = 31221 Kgf
Path 3-3 = 66194	Kgf, must exceed W = 18488 Kgf	or W3 = 31221 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1744 KPa.

The Drop for this Nozzle is : 80.3185 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 367.7748 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N1 150 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²



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See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	152.400	mm
Nozzle Thickness	26.9240	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-765 II	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1552.55	KPa.
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.660) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 206.248 + 2 \cdot 1.65 \cdot \sqrt{909.5(16.0 - 3.0)}$
 $= 565.076 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1819.000 mm
Nozzle Outside Diameter	(d) =	206.248 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	23.924 mm
T / t	=	0.543
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.341$$

Nr/P	=	0.090
Mr/P	=	0.183
M0/P	=	0.054
N0/P	=	0.117
MrD/Mc	=	0.299
NrDL/Mc	=	0.100
M0d/Mc	=	0.094
N0DL/Mc	=	0.053
MrD/Ml	=	0.159
NrDL/Ml	=	0.082
M0D/Ml	=	0.052
N0DL/Ml	=	0.131



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-28	28	28	-28
Normal Pressure Stress	107	108	107	108
Outplane Stress Summary	96	111	176	78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	73	-73	73	-73
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-86	86	86	-86
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	39	65	225	-92

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	97	112	226	170

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-50	50	50	-50
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53
Outplane Stress Summary	27	85	138	-6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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 Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026

(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	73	-73	73	-73
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-162	162	162	-162
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	108	107	108
Inplane Stress Summary	14	194	357	-111

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	194	357	111

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	7	5	7
Hoop Stress Summary	1	3	25	26

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-18	18	18	-18
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-4	0	44	-26

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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 Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	9	8	44	53
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	7	5	7

Hoop Stress Summary	8	10	17	19
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	-5	-5	5	5
Axial Bending (Mc)	-39	39	39	-39
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-25	21	65	-47
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	34	21	65	67
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



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 Nozzle Calcs.: Noz N1 150 Nozl: 36 9:30am Sep 9,2026

Circ. Pm (SUS)	107	108	107	108	107	108	107	108
Circ. Pl (SUS)	-4	-4	18	18	-4	-4	14	14
Circ. Q (SUS)	-7	7	49	-49	-89	89	235	-235
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-2	-2	12	12	3	3	11	11
Long. Q (SUS)	-13	13	159	-159	-29	29	71	-71
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	107.0	108.0	107.0	108.0	107.0	108.0	107.0	108.0
Pm+Pl (SUS)	103.0	104.0	125.0	126.0	103.0	104.0	121.0	122.0
Pm+Pl+Q (Total)	96.6	111.8	224.3	171.2	28.1	193.1	356.2	113.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	138.00	Passed
Pm+Pl (SUS)	126.02	207.00	Passed
Pm+Pl+Q (TOTAL)	356.16	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	7	5	7	5	7	5	7
Circ. Pl (SUS)	-4	-4	18	18	3	3	11	11
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-5	-5	5	5	-5	-5	5	5
Long. Q (SUS)	-2	2	34	-34	-23	23	55	-55
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	5.0	7.0	5.0	7.0	5.0	7.0	5.0	7.0
Pm+Pl (SUS)	4.0	6.0	23.0	25.0	11.0	13.0	16.0	18.0
Pm+Pl+Q (Total)	7.2	5.7	41.2	52.2	34.2	20.4	62.1	66.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	66.12	413.99	Passed



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Nozzle Calcs.: Noz N1 150

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*Because only sustained loads were specified, the $P_m + P_l + Q$ allowable was $3 * S_m h$.*

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 Nozzle Calcs.: Noz N4 100 Nozl: 37 9:30am Sep 9,2026

Input, Nozzle Desc: Noz N4 100

From: 60

Pressure for Reinforcement Calculations	P	1552.552	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	500.0000	mm
Distance from Bottom/Left Tangent		9326.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

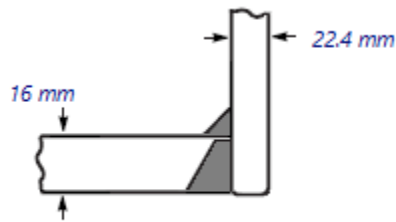
Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		56.92	deg
Diameter		4.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.3520	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	273.0500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Noz N4 100 Nozl: 37 9:30am Sep 9,2026



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz N4 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.880 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1553 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1553)$
 $= 10.2289 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1553 \cdot 53.8 / (138 \cdot 1 - 0.6 \cdot 1553)$
 $= 0.6094 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	258.2966	mm
Parallel to Vessel Wall, opening length	d	129.1483	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

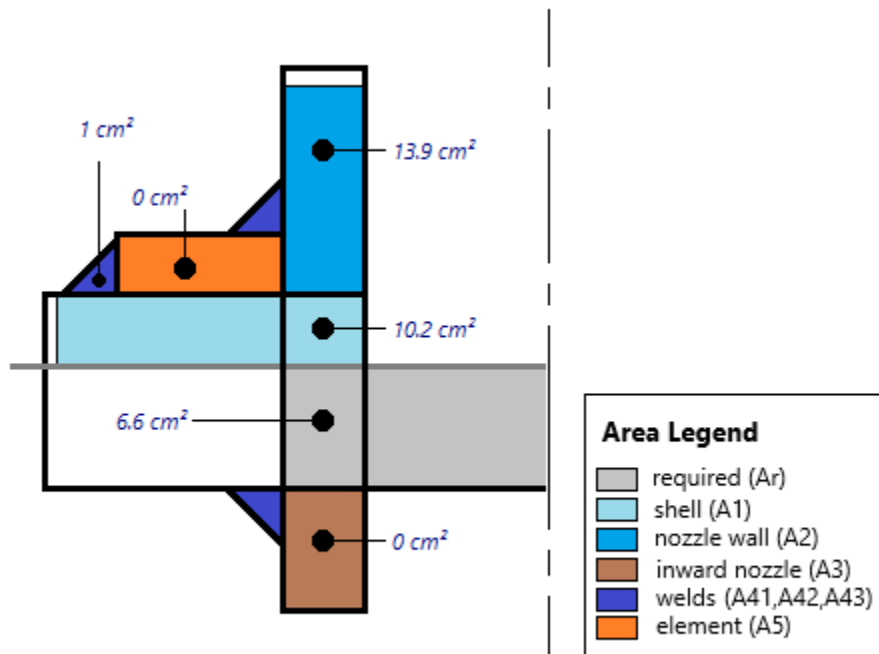


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	6.605	NA	NA
Area in Shell	A1	10.184	NA	NA
Area in Nozzle Wall	A2	13.950	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	25.134	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 56.42 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (129 \cdot 10.2 \cdot 0.5 + 2 \cdot 19.4 \cdot 10.2 \cdot 0.5 \cdot (1 - 1))$
 $= 6.605 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 129(1 \cdot 13 - 0.5 \cdot 10.2) - 2 \cdot 19.4$
 $(1 \cdot 13 - 0.5 \cdot 10.2) \cdot (1 - 1)$
 $= 10.184 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 32.5)(19.4 - 0.61)1/\sin(60.8)$
 $= 13.950 \text{ cm}^2$



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 Nozzle Calcs.: Noz N4 100 Nozl: 37 9:30am Sep 9,2026

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16$, $t_r = 10.2$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1552.55 / 5100.00 = 0.304$

Weld Size Calculations, Description: Noz N4 100

Intermediate Calc. for nozzle/shell Welds T_{min} 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) \cdot Sv)$
 $= \max(0, (6.61 - 10.2 + 2 \cdot 19.4 \cdot 1 \cdot (1 \cdot 13 - 5.11)) \cdot 138)$
 $= \max(0, -741.27) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / 0.707)^2 \cdot fr2) \cdot Sv$
 $= (13.9 + 0 + 1 - 0 \cdot 1) \cdot 138$
 $= 21036.40 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (13.9 + 0 + 1 + (5.03)) \cdot 138$
 $= 28116.54 \text{ Kgf}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (13.9 + 0 + 1 + 0 + (5.03)) \cdot 138$
 $= 28116.54 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:



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 Nozzle Calcs.: Noz N4 100 Nozl: 37 9:30am Sep 9,2026

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 176 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 19019. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 76.2) \cdot (22.4 - 3) \cdot 0.7 \cdot 138$$

$$= 45625. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 176 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 37340. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (19019 + 45625) = 64644 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (19019 + 0 + 37340 + 0) = 56359 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (19019 + 37340 + 0) = 56359 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 64644	Kgf, must exceed W = 0 Kg	or W1 = 21036 Kg
Path 2-2 = 56358	Kgf, must exceed W = 0 Kg	or W2 = 28116 Kg
Path 3-3 = 56358	Kgf, must exceed W = 0 Kg	or W3 = 28116 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1958 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz N4 100

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.880 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (Sv \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1553 \cdot 903 / (138 \cdot 1 \cdot 0.6 \cdot 1553)$$

$$= 10.2289 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1553 \cdot 53.8 / (138 \cdot 1 \cdot 0.6 \cdot 1553)$$

$$= 0.6094 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	215.2000	mm
Parallel to Vessel Wall, opening length	d	107.6000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

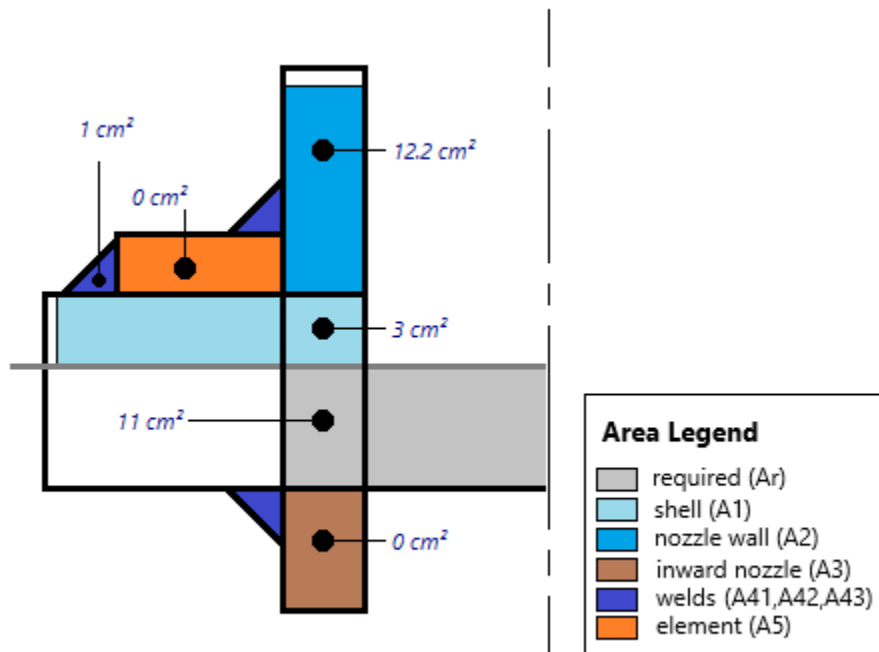


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	11.006	NA	NA
Area in Shell	A1	2.982	NA	NA
Area in Nozzle Wall	A2	12.183	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	16.164	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (108 \cdot 10.2 \cdot 1 + 2 \cdot 19.4 \cdot 10.2 \cdot 1 \cdot (1 - 1))$
 $= 11.006 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 108(1 \cdot 13 - 1 \cdot 10.2) - 2 \cdot 19.4$
 $(1 \cdot 13 - 1 \cdot 10.2) \cdot (1 - 1)$
 $= 2.982 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 32.5)(19.4 - 0.61)1$
 $= 12.183 \text{ cm}^2$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - can / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6094 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2289 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2289 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.22, \max(13.2, 4.5)]$$

$$= 9.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.61, 9.22)$$

$$= 9.2200 \text{ mm}$$

Available Nozzle Neck Thickness = 22.3520 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	10.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	335.0 N./mm ²	Passed
Occasional	:	1.8,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	7.8,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: Noz N4 100

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (11 - 2.98 + 2 \cdot 19.4 \cdot 1 \cdot (1 \cdot 13 - 10.2)) \cdot 138)$$

$$= 12801.12 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (Wi - Can / 0.707)^2 \cdot fr2) \cdot Sv$$

$$= (12.2 + 0 + 1 - 0 \cdot 1) \cdot 138$$

$$= 18550.09 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (12.2 + 0 + 1 + (5.03)) \cdot 138$$

$$= 25630.23 \text{ Kgf}$$



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Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (12.2 + 0 + 1 + 0 + (5.03)) \cdot 138$$

$$= 25630.23 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 146 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 15846. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 63.5) \cdot (22.4 - 3) \cdot 0.7 \cdot 138$$

$$= 38012. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 146 \cdot (16 - 3) \cdot 0.74 \cdot 138$$

$$= 31110. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (15846 + 38012) = 53858 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15846 + 0 + 31110 + 0) = 46955 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (15846 + 31110 + 0) = 46955 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 53858	Kgf, must exceed W = 12801 Kg	or W1 = 18550 Kg
Path 2-2 = 46955	Kgf, must exceed W = 12801 Kg	or W2 = 25630 Kg
Path 3-3 = 46955	Kgf, must exceed W = 12801 Kg	or W3 = 25630 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1904 KPa.

The Drop for this Nozzle is : 54.4318 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.6501 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N4 100 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²



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See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	101.600	mm
Nozzle Thickness	22.3520	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-765 II	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1552.55	KPa.
Radial Load	P	-764.84	Kgf
Circumferential Shear	Vc	509.89	Kgf
Longitudinal Shear	Vl	509.89	Kgf
Circumferential Moment	Mc	142.71	Kg-m.
Longitudinal Moment	Ml	142.71	Kg-m.
Torsional Moment	Mt	203.88	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.545) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 146.304 + 2 \cdot 1.65 \cdot \sqrt{909.5(16.0 - 3.0)}$
 $= 505.133 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1819.000 mm
Nozzle Outside Diameter	(d) =	146.304 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	19.352 mm
T / t	=	0.672
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.951$$

Nr/P	=	0.115
Mr/P	=	0.218
M0/P	=	0.067
N0/P	=	0.132
MrD/Mc	=	0.298
NrDL/Mc	=	0.109
M0d/Mc	=	0.101
N0DL/Mc	=	0.007
MrD/Ml	=	0.203
NrDL/Ml	=	0.098
M0D/Ml	=	0.070
N0DL/Ml	=	0.113



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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-6	-6	6	6
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	107	108	107	108
Outplane Stress Summary	100	113	160	79

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-68	68	68	-68
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	42	63	191	-62

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	101	114	191	142

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53
Outplane Stress Summary	42	75	111	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-101	101	101	-101
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	108	107	108

Inplane Stress Summary	62	150	277	-39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	63	150	278	48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-6	-6	6	6
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	3	5	16	18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-23	23	23	-23
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-7	1	52	-33
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	------------	-----------	------------	-----------



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 Nozzle Calcs.: Noz N4 100 Nozl: 37 9:30am Sep 9,2026

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	3	3	3	3

Shear Stress Summary	4	4	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	14	9	52	51
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	9	11	10	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-37	37	37	-37
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-21	14	66	-47
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	3	3	3	3

Shear Stress Summary	1	1	4	4
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	31	15	66	60
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



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Circ. Pm (SUS)	107	108	107	108	107	108	107	108
Circ. Pl (SUS)	-1	-1	11	11	-1	-1	11	11
Circ. Q (SUS)	-6	6	40	-40	-43	43	159	-159
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	0	0	10	10	5	5	5	5
Long. Q (SUS)	-10	10	126	-126	-17	17	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	107.0	108.0	107.0	108.0	107.0	108.0	107.0	108.0
Pm+Pl (SUS)	106.0	107.0	118.0	119.0	106.0	107.0	118.0	119.0
Pm+Pl+Q (Total)	100.4	113.5	189.3	142.1	63.4	150.1	277.1	48.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	138.00	Passed
Pm+Pl (SUS)	119.02	207.00	Passed
Pm+Pl+Q (TOTAL)	277.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	6	4	6	4	6	4	6
Circ. Pl (SUS)	-1	-1	11	11	5	5	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-4	4	42	-42	-18	18	56	-56
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	4.0	6.0	4.0	6.0	4.0	6.0	4.0	6.0
Pm+Pl (SUS)	7.3	9.2	15.1	17.1	13.2	15.1	9.6	11.3
Pm+Pl+Q (Total)	13.6	9.4	50.1	51.2	31.3	15.0	64.3	59.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	17.11	207.00	Passed
Pm+Pl+Q (TOTAL)	64.29	413.99	Passed



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Nozzle Calcs.: Noz N4 100

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*Because only sustained loads were specified, the $P_m + P_l + Q$ allowable was $3 * S_m h$.*

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 Nozzle Calcs.: Noz N5 200 Nozl: 38 9:30am Sep 9,2026

Input, Nozzle Desc: Noz N5 200

From: 60

Pressure for Reinforcement Calculations	P	1550.000	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9326.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

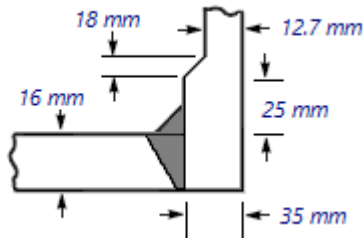
Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	25.0000	mm
Height of Beveled Transition	L'	18.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		35.0000	mm
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	263.5250	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Noz N5 200

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Reinforcement CALCULATION, Description: Noz N5 200

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 7.625 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1550 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1550)$
 $= 10.2120 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1550 \cdot 99.8 / (138 \cdot 1 - 0.6 \cdot 1550)$
 $= 1.1291 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22.3 / 0.5773$
 $= 38.6247 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	399.3500	mm
Parallel to Vessel Wall, opening length	d	199.6750	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	32.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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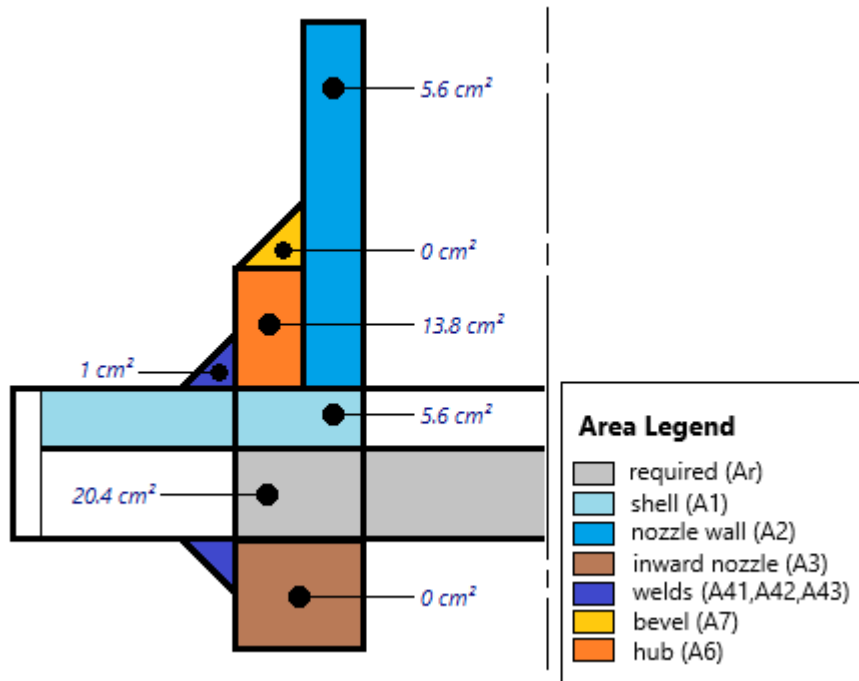
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	20.391	NA	NA
Area in Shell	A1	5.567	NA	NA
Area in Nozzle Wall	A2	5.571	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	13.798	NA	NA
TOTAL AREA AVAILABLE	Atot	25.936	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (200 \cdot 10.2 \cdot 1 + 2 \cdot 32 \cdot 10.2 \cdot 1 \cdot (1 - 1))$$

$$= 20.391 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 200(1 \cdot 13 - 1 \cdot 10.2) - 2 \cdot 32$$

$$(1 \cdot 13 - 1 \cdot 10.2) \cdot (1 - 1)$$

$$= 5.567 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 32.5)(9.7 - 1.13)1$$

$$= 5.571 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$
 $= (2 \cdot \min(32.5, 264, 25)) \cdot (35 - 12.7) \cdot 1$
 $= 13.798 \text{ cm}^2$ Includes Bevel Area of 2.648 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1291 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2120 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2120 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 10.1600 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[10.2, \max(13.2, 4.5)]$
 $= 10.1600 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(4.13, 10.2)$
 $= 10.1600 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	40.1,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	304.8 N./mm ²	Passed
Occasional :	7.6,	Allowable :	183.5 N./mm ²	Passed
Shear :	21.2,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1550.00 / 5100.00 = 0.304$

Weld Size Calculations, Description: Noz N5 200

Intermediate Calc. for nozzle/shell welds	Tmin	13.0000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (20.4 - 5.57 + 2 \cdot 32 \cdot 1 \cdot (1 \cdot 13 - 10.2)) \cdot 138)$
 $= 23370.29 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (19.4 + 1 - 0 \cdot 1) \cdot 138$
 $= 28662.71 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (19.4 + 0 + 1 + (8.32)) \cdot 138$
 $= 40370.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (19.4 + 0 + 1 + 0 + (8.32)) \cdot 138$
 $= 40370.26 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 264 \cdot 10 \cdot 0.49 \cdot 138$
 $= 28558. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 116) \cdot (35 - 3) \cdot 0.7 \cdot 138$
 $= 114707. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 264 \cdot (16 - 3) \cdot 0.74 \cdot 138$
 $= 56067. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (28558 + 114707) = 143265 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (28558 + 0 + 56067 + 0) = 84625 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (28558 + 56067 + 0) = 84625 \text{ Kgf}$

Summary of Failure Path Calculations:



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Path 1-1 = 143264 Kgf, must exceed W = 23370 Kgf or W1 = 28662 Kgf
 Path 2-2 = 84624 Kgf, must exceed W = 23370 Kgf or W2 = 40370 Kgf
 Path 3-3 = 84624 Kgf, must exceed W = 23370 Kgf or W3 = 40370 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1755 KPa.

The Drop for this Nozzle is : 9.7085 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 289.2335 mm

Input Echo, WRC297 Item 1, Description: Noz N5 200 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1800.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	35.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-765 II	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1550.00	KPa.
Radial Load	P	-1835.61	Kgf
Circumferential Shear	Vc	1325.72	Kgf
Longitudinal Shear	Vl	1325.72	Kgf
Circumferential Moment	Mc	815.51	Kg-m.
Longitudinal Moment	Ml	815.51	Kg-m.
Torsional Moment	Mt	937.84	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (7.534) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 263.675 + 2 \cdot 1.65 \cdot \sqrt{909.5(16.0 - 3.0)}$$

$$= 622.503 \text{ mm}$$

WRC 297 Curve Access Parameters:



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Vessel Mean Diameter (D) = 1819.000 mm
 Nozzle Outside Diameter (d) = 263.675 mm
 Vessel Thickness used (T) = 13.000 mm
 Nozzle Thickness used (t) = 32.000 mm
 T / t = 0.406
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.715$$

Nr/P = 0.076
 Mr/P = 0.159
 MØ/P = 0.046
 NØ/P = 0.099
 MrD/Mc = 0.292
 NrDL/Mc = 0.095
 MØd/Mc = 0.091
 NØDL/Mc = 0.066
 MrD/Ml = 0.128
 NrDL/Ml = 0.078
 MØD/Ml = 0.041
 NØDL/Ml = 0.122

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21
Outplane Bending (ML)	-44	44	44	-44
Normal Pressure Stress	106	108	106	108
Outplane Stress Summary	81	111	212	67

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-137	137	137	-137
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	11	83	314	-163

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6
Shear Stress Summary	8	8	4	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	82	114	314	231
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-97	97	97	-97
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53

Outplane Stress Summary	-15	120	203	-51
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-314	314	314	-314
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	106	108	106	108

Inplane Stress Summary	-114	312	547	-282
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-2	-2	2	2
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	114	312	548	282
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6



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Hoop Stress Summary	-6	-4	37	39
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-6	0	38	-18
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	9	8	39	58

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	10	10	10	10
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	6	4	6

Hoop Stress Summary	3	5	27	29
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	12	-12	12	-12
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-43	43	43	-43
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-33	27	66	-46
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Nozzle Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	37	27	66	75

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	108	106	108	106	108	106	108
Circ. Pl (SUS)	-11	-11	31	31	-9	-9	25	25
Circ. Q (SUS)	-15	15	73	-73	-213	213	415	-415
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-6	-6	22	22	-1	-1	21	21
Long. Q (SUS)	-36	36	238	-238	-68	68	126	-126
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	106.0	108.0	106.0	108.0	106.0	108.0	106.0	108.0
Pm+Pl (SUS)	95.1	97.1	137.1	139.1	97.1	99.1	131.1	133.1
Pm+Pl+Q (Total)	80.9	114.1	313.2	229.1	116.2	312.1	546.2	282.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	108.00	138.00	Passed
Pm+Pl (SUS)	139.06	207.00	Passed
Pm+Pl+Q (TOTAL)	546.18	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	6	4	6	4	6	4	6
Circ. Pl (SUS)	-11	-11	31	31	-1	-1	21	21
Circ. Q (SUS)	0	0	0	0	0	0	0	0



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Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-3	3	27	-27	-31	31	55	-55
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	4.0	6.0	4.0	6.0	4.0	6.0	4.0	6.0
Pm+Pl (SUS)	7.0	5.0	35.0	37.0	7.0	9.0	25.0	27.0
Pm+Pl+Q (Total)	9.0	5.8	37.0	56.1	38.2	27.2	63.1	74.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	37.00	207.00	Passed
Pm+Pl+Q (TOTAL)	74.11	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 213 of 317
 Nozzle Calcs.: D1 nozzle 50 Nozl: 39 9:30am Sep 9,2026

Input, Nozzle Desc: D1 nozzle 50

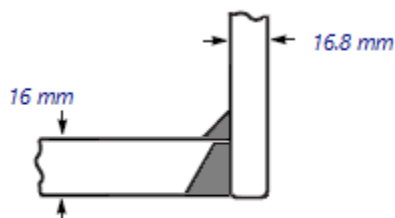
From: 60

Pressure for Reinforcement Calculations	P	1561.239	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6826.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.





PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 214 of 317
 Nozzle Calcs.: D1 nozzle 50 Nozl: 39 9:30am Sep 9,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1561 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1561)$
 $= 10.2865 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1561 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1561)$
 $= 0.3235 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm
Parallel to Vessel Wall, opening length	d	56.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1 nozzle 50.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3235	mm
Wall Thickness per UG16(b),	tr16b = 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2865	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2865	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000	mm
Wall Thickness per table UG-45	tb3 = 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(13.3, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User
FileName : 130-C1& C2-Final 23rd March Page 215 of 317
Nozzle Calcs.: D1 nozzle 50 Nozl: 39 9:30am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $1561.24/5100.00 = 0.306$

Weld Size Calculations, Description: D1 nozzle 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.9882 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.3632 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 216 of 317
 Nozzle Calcs.: K1 50 Nozl: 40 9:30am Sep 9,2026

Input, Nozzle Desc: K1 50

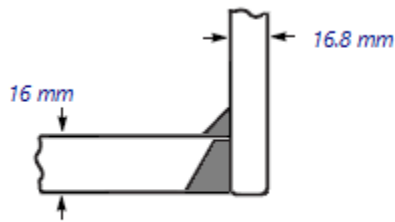
From: 60

Pressure for Reinforcement Calculations	P	1557.456	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		6826.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		109.07	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K1 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3872	mm
Parallel to Vessel Wall, opening length	d	60.1936	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1557.46 / 5100.00 = 0.305$



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 FileName : 130-C1& C2-Final 23rd March Page 218 of 317
 Nozzle Calcs.: K1 50 Nozl: 40 9:30am Sep 9,2026

Weld Size Calculations, Description: K1 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K1 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3227 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2614 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(13.3, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK



PV Elite 28 Licensee: SL User
FileName : 130-C1& C2-Final 23rd March Page 219 of 317
Nozzle Calcs.: K1 50 Nozl: 40 9:30am Sep 9,2026

Weld Size Calculations, Description: K1 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 220 of 317
 Nozzle Calcs.: K2 50 Nozl: 41 9:30am Sep 9,2026

Input, Nozzle Desc: K2 50

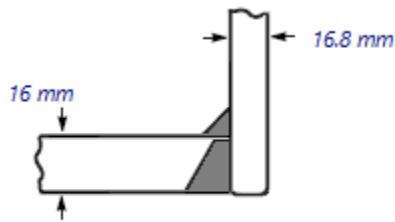
From: 60

Pressure for Reinforcement Calculations	P	1553.783	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		6826.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		70.93	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K2 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3874	mm
Parallel to Vessel Wall, opening length	d	60.1937	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1553.78 / 5100.00 = 0.305$



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 FileName : 130-C1& C2-Final 23rd March Page 222 of 317
 Nozzle Calcs.: K2 50 Nozl: 41 9:30am Sep 9,2026

Weld Size Calculations, Description: K2 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K2 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3220 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2371 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2371 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(13.2, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK



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FileName : 130-C1& C2-Final 23rd March Page 223 of 317
Nozzle Calcs.: K2 50 Nozl: 41 9:30am Sep 9,2026

Weld Size Calculations, Description: K2 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 224 of 317
 Nozzle Calcs.: K3 50 Nozl: 42 9:30am Sep 9,2026

Input, Nozzle Desc: K3 50

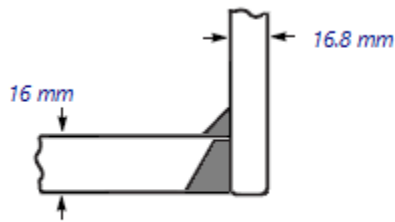
From: 60

Pressure for Reinforcement Calculations	P	1553.783	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		5826.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		70.93	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K3 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3874	mm
Parallel to Vessel Wall, opening length	d	60.1937	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1553.78 / 5100.00 = 0.305$



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 FileName : 130-C1& C2-Final 23rd March Page 226 of 317
 Nozzle Calcs.: K3 50 Nozl: 42 9:30am Sep 9,2026

Weld Size Calculations, Description: K3 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K3 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 = 10.2371 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 = 0.3220 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3220 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2371 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2371 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.8, max(13.2, 4.5)]
 = 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.32, 7.8)
 = 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK



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FileName : 130-C1& C2-Final 23rd March Page 227 of 317
Nozzle Calcs.: K3 50 Nozl: 42 9:30am Sep 9,2026

Weld Size Calculations, Description: K3 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 228 of 317
 Nozzle Calcs.: K4 50 Nozl: 43 9:30am Sep 9,2026

Input, Nozzle Desc: K4 50

From: 60

Pressure for Reinforcement Calculations	P	1557.456	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		5826.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

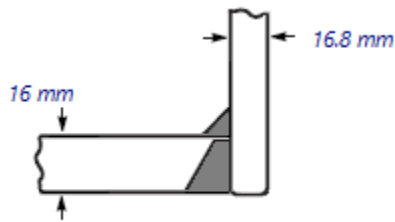
Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		109.07	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 FileName : 130-C1& C2-Final 23rd March Page 229 of 317
 Nozzle Calcs.: K4 50 Nozl: 43 9:30am Sep 9,2026



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K4 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3872	mm
Parallel to Vessel Wall, opening length	d	60.1936	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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 FileName : 130-C1& C2-Final 23rd March Page 230 of 317
 Nozzle Calcs.: K4 50 Nozl: 43 9:30am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1557.46/5100.00 = 0.305

Weld Size Calculations, Description: K4 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K4 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 = 10.2614 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 = 0.3227 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3227 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2614 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm



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Nozzle Calcs.: K4 50 Nozl: 43 9:30am Sep 9,2026

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.3, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.32, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Weld Size Calculations, Description: K4 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 232 of 317
 Nozzle Calcs.: K5 50 Nozl: 44 9:30am Sep 9,2026

Input, Nozzle Desc: K5 50

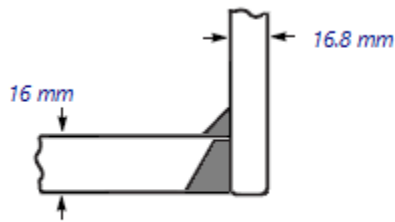
From: 60

Pressure for Reinforcement Calculations	P	1557.456	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		8626.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		109.07	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K5 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3872	mm
Parallel to Vessel Wall, opening length	d	60.1936	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1557.46 / 5100.00 = 0.305$



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 FileName : 130-C1& C2-Final 23rd March Page 234 of 317
 Nozzle Calcs.: K5 50 Nozl: 44 9:30am Sep 9,2026

Weld Size Calculations, Description: K5 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K5 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3227 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2614 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(13.3, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK



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FileName : 130-C1& C2-Final 23rd March Page 235 of 317
Nozzle Calcs.: K5 50 Nozl: 44 9:30am Sep 9,2026

Weld Size Calculations, Description: K5 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 236 of 317
 Nozzle Calcs.: K6 50 Nozl: 45 9:30am Sep 9,2026

Input, Nozzle Desc: K6 50

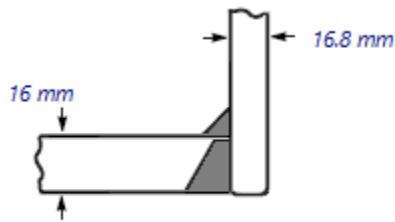
From: 60

Pressure for Reinforcement Calculations	P	1553.783	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		8626.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		70.93	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K6 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3874	mm
Parallel to Vessel Wall, opening length	d	60.1937	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1553.78 / 5100.00 = 0.305$



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 FileName : 130-C1& C2-Final 23rd March Page 238 of 317
 Nozzle Calcs.: K6 50 Nozl: 45 9:30am Sep 9,2026

Weld Size Calculations, Description: K6 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K6 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3220 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2371 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2371 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(13.2, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.32, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK



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FileName : 130-C1& C2-Final 23rd March Page 239 of 317
Nozzle Calcs.: K6 50 Nozl: 45 9:30am Sep 9,2026

Weld Size Calculations, Description: K6 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 240 of 317
 Nozzle Calcs.: K7 50 Nozl: 46 9:30am Sep 9,2026

Input, Nozzle Desc: K7 50

From: 60

Pressure for Reinforcement Calculations	P	1553.783	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		9726.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

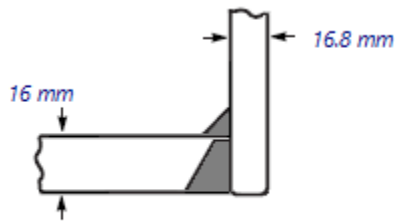
Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		70.93	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 FileName : 130-C1& C2-Final 23rd March Page 241 of 317
 Nozzle Calcs.: K7 50 Nozl: 46 9:30am Sep 9,2026



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K7 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 10.2371 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 $= 0.3220 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3874	mm
Parallel to Vessel Wall, opening length	d	60.1937	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.2, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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 FileName : 130-C1& C2-Final 23rd March Page 242 of 317
 Nozzle Calcs.: K7 50 Nozl: 46 9:30am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1553.78/5100.00 = 0.305

Weld Size Calculations, Description: K7 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K7 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1554 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1554)$
 = 10.2371 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1554 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1554)$
 = 0.3220 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3220 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.2371 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.2371 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm



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FileName : 130-C1& C2-Final 23rd March Page 243 of 317
Nozzle Calcs.: K7 50 Nozl: 46 9:30am Sep 9,2026

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.2, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.32, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Weld Size Calculations, Description: K7 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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PV Elite 28 Licensee: SL User
 FileName : 130-C1& C2-Final 23rd March Page 244 of 317
 Nozzle Calcs.: K8 50 Nozl: 47 9:30am Sep 9,2026

Input, Nozzle Desc: K8 50

From: 60

Pressure for Reinforcement Calculations	P	1557.456	KPa.
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1800.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	300.0000	mm
Distance from Bottom/Left Tangent		9626.17	mm
User Entered Minimum Design Metal Temperature		-33.00	°C

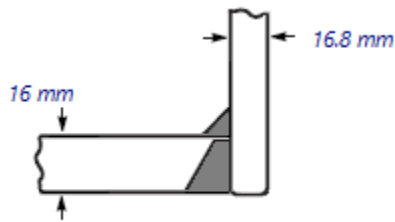
Type of Element Connected to the Parent : Nozzle

Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		109.07	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 FileName : 130-C1& C2-Final 23rd March Page 245 of 317
 Nozzle Calcs.: K8 50 Nozl: 47 9:30am Sep 9,2026



Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: K8 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 10.2614 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 $= 0.3227 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3872	mm
Parallel to Vessel Wall, opening length	d	60.1936	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 10.3, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -42 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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 FileName : 130-C1& C2-Final 23rd March Page 246 of 317
 Nozzle Calcs.: K8 50 Nozl: 47 9:30am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1557.46/5100.00 = 0.305

Weld Size Calculations, Description: K8 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: K8 50

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1557 \cdot 903 / (138 \cdot 1 - 0.6 \cdot 1557)$
 = 10.2614 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1557 \cdot 28.4 / (138 \cdot 1 - 0.6 \cdot 1557)$
 = 0.3227 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000 mm
Parallel to Vessel Wall, opening length	d	56.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8 50.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3227 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 13.2614 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.2614 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm



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Nozzle Calcs.: K8 50 Nozl: 47 9:30am Sep 9,2026

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.3, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.32, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Weld Size Calculations, Description: K8 50

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 16.1077 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.0598 mm

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 FileName : 130-C1& C2-Final 23rd March Page 248 of 317
 Nozzle Calcs.: Noz N7 Nozl: 48 9:30am Sep 9,2026

Input, Nozzle Desc: Noz N7

From: 90

Pressure for Reinforcement Calculations	P	4561.911	KPa.
Temperature for Internal Pressure	Temp	100	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	100	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Design Length of Section	L	1500.0299	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		14968.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

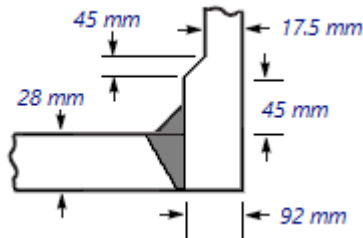
Material		SA-765 II	
Material UNS Number		K03047	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4800	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	45.0000	mm
Height of Beveled Transition	L'	45.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		92.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Noz N7

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.688 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4562 \cdot 603 / (138 \cdot 1 - 0.6 \cdot 4562)$
 $= 20.3385 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 4562 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 4562)$
 $= 9.9451 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6373 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 74.5 / 0.5773$
 $= 129.0725 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.2800 mm
Parallel to Vessel Wall, opening length	d	580.6400 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	62.5000 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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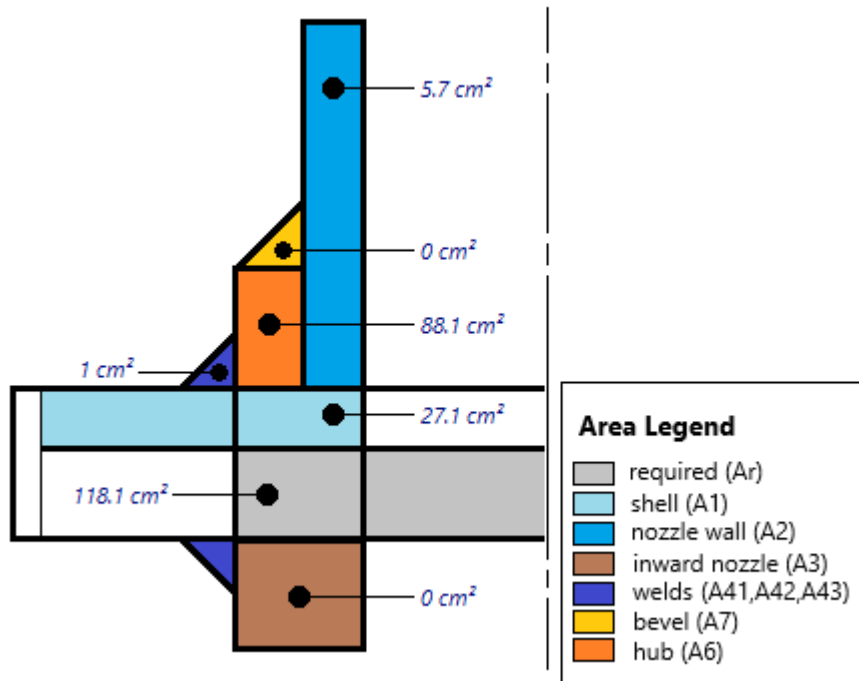
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Nozzle Calcs.: Noz N7

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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	118.094	12.603	NA
Area in Shell	A1	27.066	119.955	NA
Area in Nozzle Wall	A2	5.669	16.053	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub + Bevel	A6	88.079	88.079	NA
TOTAL AREA AVAILABLE	Atot	121.814	225.087	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 20.3 \cdot 1 + 2 \cdot 89 \cdot 20.3 \cdot 1 \cdot (1 - 1))$$

$$= 118.094 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 25 - 1 \cdot 20.3) - 2 \cdot 89$$

$$(1 \cdot 25 - 1 \cdot 20.3) \cdot (1 - 1)$$

$$= 27.066 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 62.5)(14.5 - 9.95)1$$

$$= 5.669 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_{an}/0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(62.5, 400, 45)) \cdot (92 - 17.5) \cdot 1$$

$$= 88.079 \text{ cm}^2 \text{ Includes Bevel Area of } 21.011 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 12.9451 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 23.3385 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 23.3385 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.4521 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(23.3, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(12.9, 11.3)$$

$$= 12.9451 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4800 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.5,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.5 N./mm ²	Passed
Occasional	: 44.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 15.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17.5, tr = 9.95, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.69, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 28, tr = 20.3, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.81, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	2 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -25 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -24 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 4561.91/5100.00 = 0.894

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 74.5/0.5773
 = 129.0725 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 870.9600 mm
 Parallel to Vessel Wall, opening length d 435.4800 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 62.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	78.733	78.733
Area Available in Shell	A1	13.533	13.533
Area Available in Nozzle Wall	A2	5.669	5.669
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.000	1.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	88.079	88.079
TOTAL AREA AVAILABLE	Atot	108.280	108.280

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: Noz N7

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
 = max(0, (118 - 27.1 + 2 * 89 * 1 *
 (1 * 25 - 20.3))138)
 = 139765.27 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A4-(Wi-Can/.707)²*fr2)*Sv
 = (93.7 + 1 - 0 * 1) * 138



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= 133324.06 Kgf

Weld Load [W2]:

= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
 = (93.7 + 0 + 1 + (44.5)) • 138
 = 195942.56 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2•tn•t•fr1))•S
 = (93.7 + 0 + 1 + 0 + (44.5)) • 138
 = 195942.56 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw
 = (3.14/2.0) • 610 • 10 • 0.49 • 138
 = 66024. Kgf

Shear, Nozzle Wall [Snw]:

= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
 = (3.14 • 298) • (92 - 3) • 0.7 • 138
 = 819512. Kgf

Tension, Shell Groove Weld [Tngw]:

= (π/2) • Dlo • Wgnvi • 0.74 • Sng
 = (3.14/2.0) • 610 • (14 - 0) • 0.74 • 138
 = 139594. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SONW + SNW) = (66024 + 819512) = 885536 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)
 = (66024 + 0 + 139594 + 0) = 205618 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)
 = (66024 + 139594 + 0) = 205618 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 885535 Kgf, must exceed W = 139765 Kgf or W1 = 133324 Kgf
 Path 2-2 = 205618 Kgf, must exceed W = 139765 Kgf or W2 = 195942 Kgf
 Path 3-3 = 205618 Kgf, must exceed W = 139765 Kgf or W3 = 195942 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4617 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 83.1858 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 511.1858 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 580.64 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).



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Input Echo, WRC297 Item 1, Description: Noz N7 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1200.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	758.640	mm	
Nozzle Thickness	92.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-765 II		
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	4561.91	KPa.
Radial Load	P	-5506.84	Kgf
Circumferential Shear	Vc	3977.16	Kgf
Longitudinal Shear	Vl	3977.16	Kgf
Circumferential Moment	Mc	7339.62	Kg-m.
Longitudinal Moment	Ml	7339.62	Kg-m.
Torsional Moment	Mt	8440.55	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.604) is > 0.5
 check the limitations of WRC bulletin 297.
Warning - The ratio, Dn/Tn (8.246) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 758.64 + 2 \cdot 1.65 \cdot \sqrt{615.5(28.0 - 3.0)}$
 $= 1167.993 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1231.000 mm
Nozzle Outside Diameter	(d) =	758.640 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	89.000 mm
T / t	=	0.281
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.325$$



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Nr/P = 0.050
 Mr/P = 0.075
 MØ/P = 0.026
 NØ/P = 0.060
 MrD/Mc = 0.254
 NrDL/Mc = 0.091
 MØd/Mc = 0.076
 NØDL/Mc = 0.084
 MrD/Ml = 0.069
 NrDL/Ml = 0.044
 MØD/Ml = 0.021
 NØDL/Ml = 0.062

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	107	112	107	112
Outplane Stress Summary	97	114	154	94

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	27	75	166	-36

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	98	114	166	130

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	D1 Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-69	69	69	-69
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53
Outplane Stress Summary	-9	102	154	-10

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	38	-38	38	-38
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-231	231	231	-231
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	112	107	112
Inplane Stress Summary	-94	295	395	-139

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	94	295	395	139

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	11	16	30	35

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	4	6	15	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)				
	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	11	16	30	35

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	20	15	20
Hoop Stress Summary	7	13	33	38

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)				
	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7
Axial Stress Summary	-7	18	27	-6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)				
	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	15	18	33	45

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	107	112	107	112	107	112	107	112
Circ. Pl (SUS)	-4	-4	14	14	-9	-9	17	17
Circ. Q (SUS)	-6	6	32	-32	-193	193	269	-269
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-2	-2	10	10	-7	-7	17	17
Long. Q (SUS)	-24	24	100	-100	-56	56	82	-82
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	107.0	112.0	107.0	112.0	107.0	112.0	107.0	112.0
Pm+Pl (SUS)	103.0	108.0	121.0	126.0	98.0	103.0	124.0	129.0
Pm+Pl+Q (Total)	97.2	114.4	163.4	131.1	95.0	296.0	393.1	140.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	112.00	138.00	Passed
Pm+Pl (SUS)	129.02	207.00	Passed
Pm+Pl+Q (TOTAL)	393.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	20	15	20	15	20	15	20
Circ. Pl (SUS)	-4	-4	14	14	-7	-7	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	-1	1	5	-5	-12	12	16	-16
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	15.0	20.0	15.0	20.0	15.0	20.0	15.0	20.0



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Pm+Pl (SUS)	11.0	16.0	29.0	34.0	8.0	13.0	32.0	37.0
Pm+Pl+Q (Total)	11.1	16.1	29.1	34.0	15.1	17.2	32.1	44.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	20.00	138.00	Passed
Pm+Pl (SUS)	37.00	207.00	Passed
Pm+Pl+Q (TOTAL)	44.05	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	771.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	65.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3995.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	17.20 °C
Channel Operating Temperature	Tcop	42.60 °C
Tube Operating Temperature	Ttop	42.60 °C
Tubesheet Operating Temperature	Tshop	42.60 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm



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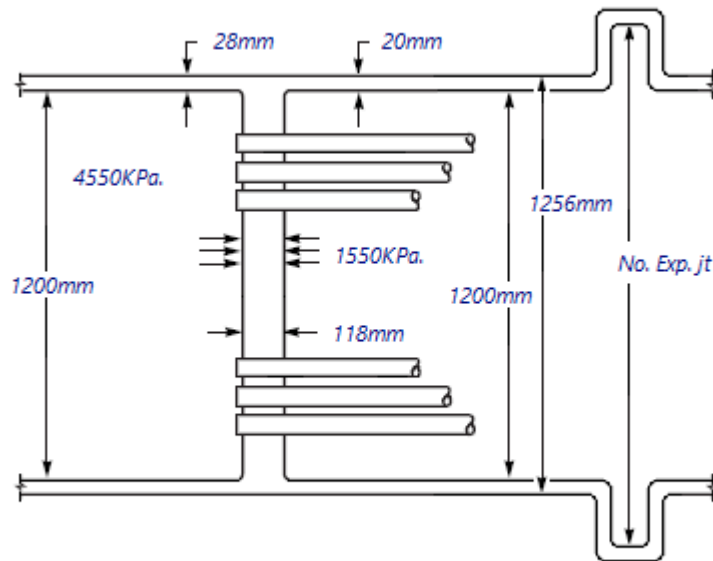
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Circumferential Joint Efficiency Eecc,w 1.00
 Length Lecc 1090.000 mm

Large Diameter Main Shell Cylinder Properties:

Material SA-516 70
 Material UNS Number K02700
 Allowable Stress at Temperature 138.00 N./mm²
 Allowable Stress at Operating Temperature 138.00 N./mm²
 Allowable Stress at Ambient 138.00 N./mm²
 Thickness ts,L 16.0000 mm
 Internal Corrosion Allowance 3.0000 mm
 External Corrosion Allowance 0.0000 mm
 Inside Diameter Ds,L 1800.000 mm
 Circumferential Joint Efficiency Es,L,w 1.00
 Length Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	42.6 °C	0.20147E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	42.6 °C	0.20147E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	42.6 °C	0.20147E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	42.6 °C	0.20147E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4550 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 4550)$
 $= 0.5262 + 0.0000 = 0.5262 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 100.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (E_{maxp}):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
E _{MAWP} = (4*B)/(3*(D/T)) = (4 *123)/(3 *12) = 13593 KPa.						

Results for Req'd Thickness for Ext. Pressure (T_{ca}):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5575	25.40	11964.00	45.56	50.0000	0.0005298	52.97
E _{MAWP} = (4*B)/(3*(D/T)) = (4 *53)/(3 *45.6) = 1550 KPa.						

Summary of Tube Required Thickness Results:



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Total Required Thickness including Corrosion all.	0.5575 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	1550.00 KPa.
Required Thickness due to Shell Side pressure	0.5575 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/118$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (198151376/198151376) \cdot (108/138) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 564 + 25.4$$

$$= 1154.200 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1154/2$$

$$= 577.1000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1154 \cdot 31.8) / (\pi \cdot 1154^2)) }$$

$$= 31.7500 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.2000$$

Ratio [ps]:

$$= as / ao$$

$$= 600/577$$

$$= 1.0397$$

Ratio [pc]:

$$= ac / ao$$

$$= 600/577$$

$$= 1.0397$$

Compute parameter [xt]:



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$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 1151 ((25.4 - 2 \cdot 2.11) / (2 \cdot 577))^2$$

$$= 0.6124$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 1151 (25.4 / (2 \cdot 577))^2$$

$$= 0.4426$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Stiffness of the Conical Section [Kecc]:

$$= 0.8 \cdot \pi \cdot tecc(Decc, S + tecc) Eecc / Lecc$$

$$= 0.8 \cdot \pi \cdot 20 (1200 + 20) 200187728 / 1090$$

$$= 1148516.3750 \text{ Kgf/mm}$$

Axial Stiffness of the Large Diameter Section [Ks,L]:

$$= \pi \cdot ts, L (Ds, L + ts, L) Es, L / Ls, L$$

$$= \pi \cdot 16 (1800 + 16) 200187728 / 9384$$

$$= 198578.2344 \text{ Kgf/mm}$$

Axial Shell Stiffness of the Small Diameter Section [Ks]:

$$= \pi \cdot ts(Ds + ts) Es / Ls$$

$$= 3.14 \cdot 20 (1200 + 20) 200187728 / 200$$

$$= 7824268.0000 \text{ Kgf/mm}$$

Overall Effective Shell Stiffness for Kettle Exchangers [Ks*]:

$$= Ks \cdot Ks, L \cdot Kecc / (2Kecc \cdot Ks, L + 2Ks, L \cdot Ks + Ks \cdot Kecc)$$

$$= 7824268 \cdot 198578 \cdot 1148516 /$$

$$(2 \cdot 1148516 \cdot 198578 + 2 \cdot 198578 \cdot 7824268 + 7824268 \cdot 1148516)$$

$$= 142191.0625 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(Dt - tt) Et / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 198151376 / 11964$$

$$= 260.7468 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 142191 / (1151 \cdot 261)$$

$$= 0.47378$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 142191 / 0)$$

$$= 1.0000 \text{ (= 1 if No Exp. Jt.)}$$

Shell coefficient [βs]:

$$= ((12(1 - vs^2))^{1/4}) / ((Ds + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1200 + 20) 20)$$

$$= 0.0116 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta s \cdot Es \cdot ts^3 / (6(1 - vs^2))$$

$$= 0.012 \cdot 200187728 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 3413465600.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta s + (h \cdot \beta s)^2 / 2)$$

$$= 6(1200 + 20) \cdot 3.41E+09 / 118^3 \cdot (1 + 118 \cdot 0.012 + (118 \cdot 0.012)^2 / 2)$$



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= 50430012.0000 KPa.

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 1200^2 / (4 \cdot 200187728 \cdot 20) \cdot (1 - 1200 \cdot 0.3 / (2(1200 + 20)))$$

$$= 0.76645E-01 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / ((1200 + 28)28))$$

$$= 0.0098 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.0098 \cdot 198151376 \cdot 28^3 / (6(1 - 0.3^2))$$

$$= 7810089984.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1200 + 28 / \cos(0)) \cdot 7.81E+09 / 118^3 \cdot (1 + 118 \cdot 0.0098 \cdot (1 + 1) / 2 + (118 \cdot 0.0098)^2 / 2 \cdot 1)$$

$$= 98972912.0 \text{ KPa.}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1200^2 / (4 \cdot 198151376 \cdot 28) \cdot (1 - 1200 \cdot 0.3 / (2(1200 + 28)))$$

$$= 0.55371E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 3.716536 ; \mu^* = 0.200000 ; E^* = 30782644 \text{ KPa.}$$

$$E^*/E = 0.155349 ; v^* = 0.458572$$

Note: As h/p (3.717) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - v^{*2})Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2)1151 \cdot 198151376 \cdot 2.11(25.4 - 2.11)^{577^2}) / (30782644 \cdot 11964 \cdot 118^3))^{1/4}$$

$$= 3.2883$$

Values from Table 4.18.3

$$Z_d = 0.041677 ; Z_v = 0.087307 ; Z_m = 0.436628$$

$$Z_a = 0.310011E+01 ; Z_w = 0.087307$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1256 / 1154$$

$$= 1.0882$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 30782644 \cdot (50430012 + 98972912 + 198151376 \cdot \ln(1.09))$$

$$= 2.9224$$



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Compute parameter [ϕ]:

$$= (1 + v) F$$

$$= (1 + 0.46) 2.92$$

$$= 4.2625$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.04 - 1 - 4.26 \cdot 0.087) / (1 + 4.26 \cdot 0.44)$$

$$= -0.116201$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.042 + -0.12 \cdot 0.087) / 2 \cdot 3.29^4$$

$$= 1.8433$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.087 + -0.12 \cdot 0.44) / 2 \cdot 3.29^4$$

$$= 2.1378$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.087 + (1.04 - 1) 0.44) 3.29^4 / (1 + 4.26 \cdot 0.44)$$

$$= 4.2757$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.04 \cdot 3.41E+09 \cdot 0.012 \cdot 0.07664492 (1 + 118 \cdot 0.012)$$

$$= 7512.8472 \text{ mm}^2$$

Calculate parameter [ω_s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 577^2 (1.04^2 - 1) \cdot (1.04 - 1) / 4 - 7513$$

$$= -7245.4409 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (C_m \theta + \beta_c + C_q \Delta \cdot h \cdot \beta_c^2) - \zeta_c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta_c / 2)]$$

$$= 1.04 \cdot 7.81E+09 [0.05537134 (1 + 0.0098 \cdot 1 \cdot 118) - 0.000 (1/118 + 1 \cdot 0.0098/2)]$$

$$= 9507.2373 \text{ mm}^2$$

Parameter [ω_c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 577^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.04 - 1) / 2] - 9507$$

$$= -9239.8311 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Small End Coefficient [As]:

$$= D_s (D_s + t_s)$$



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$$= 1200 (1200 + 20)$$

$$= 14640.0010 \text{ cm}^2$$

Large End Coefficient [As,L]:

$$= D_s, L (D_s, L + t_s, L)$$

$$= 1800 (1800 + 16)$$

$$= 32688.0020 \text{ cm}^2$$

Diameter Difference [Δecc]:

$$= \text{Decc}, L - \text{Decc}, S$$

$$= 1800 - 1200$$

$$= 600.0000 \text{ mm}$$

Equivalent Poisson's ratio for Kettle Exchangers [vs*]:

$$= 2 \cdot K_s \cdot A_s [A_s \cdot v_s / K_s + (v_{ecc} - (Lecc^2 - 2 \cdot v_{ecc} \cdot Lecc^2 - 3 \cdot \Delta ecc^2) / (8 \cdot Lecc^3) \cdot (\Delta ecc^2 + Lecc^2) \cdot 0.5) (Decc, L + Decc, S) (Decc, S + tecc) / (5 \cdot Kecc) + A_s, L \cdot v_s, L / (2 \cdot K_s, L)]$$

$$= 2 \cdot 142191 / 14640 [14640 \cdot 0.3 / 7824268 + [0.3 \cdot (1090^2 - 2 \cdot 0.3 \cdot 1090^2 - 3 \cdot 0.3^2) / (8 \cdot 1090^3) \cdot \sqrt{(600^2 + 1090^2)} (1800 + 1200) (1200 + 20) / (5 \cdot 1148516) + 32688 \cdot 0.3 / (2 \cdot 198578)]]$$

$$= 0.5367$$

Note: Use vs* in place of vs when determining P's

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2 / K_{st} \cdot (D_s / D_o)^2] v_s - [(p_s^2 - 1) / (J \cdot K_{st})] - [(1 - J) / (2J \cdot K_{st})] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 1550 \{ 0.44 + 2(1 - 0.44) 0.3 + [2 / 0.47 \cdot (1200 / 1154)^2] 0.54 - [(1.04^2 - 1) / (1 \cdot 0.47)] - [(1 - 1) / (2 \cdot 1 \cdot 0.47)] [(0^2 - (1200)^2) / 1154^2] \}$$

$$= 4735.3086 \text{ KPa.}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1 / (J \cdot K_{st})] P_t$$

$$= [0.61 + 2(1 - 0.61) 0.3 + 1 / (1 \cdot 0.47)] 4550$$

$$= 13448.1846 \text{ KPa.}$$

Pressure [Pγ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 1151 \cdot 261 \cdot 0 / (3.14 \cdot 577^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma_b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 4.28 \cdot 0 / (2 \cdot \pi \cdot 577^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U / a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_t)$$

$$= -(4.28 / 577^2) \cdot (-7245 \cdot 1550 - 9240 \cdot 4550)$$

$$= -395.5551 \text{ KPa.}$$

Calculate the pressure [Pω]:

$$= (U / a_o^2) \cdot (\omega_s \cdot P_s - \omega_c \cdot P_c)$$

$$= (4.28 / 577^2) \cdot (7513 \cdot 0 - 9507 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P_s - P_t + P_\gamma + P_w + P_{prim})$$

$$= 0.1000E+01 \cdot 0.47 / (1 + 1 \cdot 0.47 (1.84 + (1.04 - 1) 2.14)) \cdot (4735 - 13448 + 0 + 0 + -0 + -396)$$

$$= -2255.2095 \text{ KPa.}$$



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Determine factor [Q2]:

$$= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + W * \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-7245 \cdot 1550 - -9240 \cdot 4550) - (7513 \cdot 0 - 9507 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 4.26 \cdot 0.44)$$

$$= 1098.1470 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.12 + 2 \cdot 1098 / (-2255 \cdot 577^2)$$

$$= -0.14488$$

Fm Value from Table 4.18.3 = 0.072438

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.072 / 0.2) \cdot (2 \cdot 577 / (118 - 0))^2 \cdot -2255$$

$$= -117.2296 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.063 / 0.2) \cdot (2 \cdot 577 / (78.4 - 0))^2 \cdot -2017$$

$$= -206.9950 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 78.4 + 0 + 0$$

$$= 78.3570 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 118 / 577) \cdot 110$$

$$= 9.0293 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (9690 \cdot -1065) / (0.2 \cdot 13 \cdot 3629)$$

$$= -109.5070 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 13 + 0 + 0$$

$$= 12.9890 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(78.4 ; 13)$$

$$= 78.3570 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = 0.057, F_{tmax} = 1.7$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$



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$$= ((1550 \cdot 0.44 - 4550 \cdot 0.61) - (-2255) \cdot 0.057) / (0.61 - 0.44)$$

$$= -11.6163 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((1550 \cdot 0.44 - 4550 \cdot 0.61) - (-2255) \cdot 1.7) / (0.61 - 0.44)$$

$$= 10.1667 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(| \sigma_{t,1} |; | \sigma_{t,2} |) = 11.6 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= S_{ot} = 107.9980 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.042 + -0.14 \cdot 0.087) 3.29^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (\frac{d_t^2}{4} + (\frac{d_t - 2 \cdot t_t}{2})^2) }) / 4$$

$$= (\sqrt{ (\frac{25.4^2}{4} + (\frac{25.4 - 2 \cdot 2.11}{2})^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 198151376 / 189) }$$

$$= 143.8624$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 1500 / 8.27$$

$$= 181.4227$$

Buckling Allowable Stress [Stb]:

$$= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) (C_t \leq F_t)$$

$$= (1 / 2) \cdot (\pi^2 \cdot 198151376 / 181^2)$$

$$= 29.710 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 11.6 \cdot 1.54$$

$$= 182.87 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 108$$

$$= 1700.157 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 108$$



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= 1663.7775 Kgf

Weld Strength Factor [fw]:

= S / Sw where Sw = min(S and St)
 = 108/108
 = 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 108$
 = 1076.5620 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$
 = $2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$
 = 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1700 is >= tube strength 1700 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Small Shell Cylinder Axial Membrane Allowable Stress:

= $Ss \cdot Es, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Small Cylinder [σ_s, m]:

= $ao^2 / (ts (Ds + ts)) [Pe + (Ds^2 / (4 \cdot ao^2) - 1) (Ps - Pt)] + Ds^2 \cdot Pt / (4 \cdot ts (Ds + ts))$
 = $577^2 / (20 (1200 + 20)) [-2255 + (1200^2 / (4 \cdot 577^2) - 1) (1550 - 4550)] + 1200^2 \cdot 4550 / (4 \cdot 20 (1200 + 20))$
 = 33.0367 N./mm²

Large Shell Cylinder Axial Membrane Allowable Stress:

= $Ss, L \cdot Es, L, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Large Cylinder [σ_s, L, m]:

= $ao^2 / (ts, L (Ds, L + ts, L)) [Pe + (Ds, L^2 / (4 \cdot ao^2) - 1) (Ps - Pt)] + Ds, L^2 \cdot Pt / (4 \cdot ts, L (Ds, L + ts, L))$
 = $577^2 / (20 (1800 + 16)) [-2255 + (1800^2 / (4 \cdot 577^2) - 1) (1550 - 4550)] + 1800^2 \cdot 4550 / (4 \cdot 16 (1200 + 20))$
 = 51.7496 N./mm²

Cone Axial Membrane Allowable Stress:

= $Secc \cdot Eecc, w = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Cone Large End [$\sigma_{ecc, L, m}$]:

= $ao^2 / (tecc (Decc, L + tecc) \cos(\text{angle})) [Pe + (Decc, L^2 / (4 \cdot ao^2) - 1) (Ps - Pt)] + Decc, L^2 \cdot Pt / (4 \cdot tecc (Decc, L + tecc) \cos(\text{angle}))$
 = $577^2 / (20 (1800 + 20) \cos(28.8)) [-2255 + (1800^2 / (4 \cdot 577^2) - 1) (1550 - 4550)] + 1800^2 \cdot 4550 / (4 \cdot 20 (1800 + 20) \cos(28.8))$
 = 47.1536 N./mm²

Circumferential Load, Cone Large End (used in Appendix 1-5) [f1']:

= $\sigma_{ecc, L, m} \cdot tecc \cdot \cos(\text{theta ecc}) - Ps \cdot Decc, L / 4$
 = $47.2 \cdot 20 \cdot \cos(28.8) - 1550 \cdot 1800 / 4$
 = 13.1167 Kgf/mm



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Circumferential Load, Cone Large End, (used in Appendix 1-8) [f1']:
 $= -\sigma_{ecc,L,m} \cdot tecc \cdot \cos(\theta_{ecc}) + Ps \cdot Decc,L/4$
 $= -47.2 \cdot 20 \cdot \cos(28.8) + 1550 \cdot 1800/4$
 $= -13.1167 \text{ Kg/mm}$

Axial Membrane Stress in Cone Small End [$\sigma_{ecc,S,m}$]:
 $= ao^2 / (tecc(Decc,S + tecc) \cos(\angle)) [Pe + (Decc,S^2 / (4 \cdot ao^2) - 1)(Ps - Pt)] + Decc,S^2 \cdot Pt / (4 \cdot tecc(Decc,S + tecc) \cos(\angle))$
 $= 577^2 / (20(1200 + 20) \cos(28.8)) [-2255 + (1200^2 / (4 \cdot 577^2) - 1)(1550 - 4550)] + 1200^2 \cdot 4550 / (4 \cdot 20(1200 + 20) \cos(28.8))$
 $= 37.7111 \text{ N./mm}^2$

Circumferential Load, Cone Small End (used in Appendix 1-5) [f2']:
 $= \sigma_{ecc,S,m} \cdot tecc \cdot \cos(\theta_{ecc}) - Ps \cdot Decc,S/4$
 $= 37.7 \cdot 20 \cdot \cos(28.8) - 1550 \cdot 1200/4$
 $= 19.9562 \text{ Kg/mm}$

Circumferential Load, Cone Small End, (used in Appendix 1-8) [f2']:
 $= -\sigma_{ecc,S,m} \cdot tecc \cdot \cos(\theta_{ecc}) + Ps \cdot Decc,S/4$
 $= -37.7 \cdot 20 \cdot \cos(28.8) + 1550 \cdot 1200/4$
 $= -19.9562 \text{ Kg/mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 20 mm, the minimum Shell length adjacent to the tubesheet should be 278.855 mm

Shell Membrane Stress due to Joint Interaction [$\sigma_{s,m}$]:
 $= ao^2 / ((Ds + ts) ts) [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) ts)$
 $= 577^2 / ((1200 + 20) 20) [-2255 + (1.04^2 - 1)(1550 - 4550)] + 600^2 \cdot 4550 / ((1200 + 20) 20)$
 $= 33.0367 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [$\sigma_{s,b}$]:
 $= 6 \cdot ks / ts^2 \{ \beta_s [\delta_s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) (ao/h)^3 (1 + h \cdot \beta_s / 2) [Pe(Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$
 $= 6 \cdot 3.41E+09 / 20^2 \{ 0.012 [0.07664492 \cdot 1550 + 600^2 \cdot 0 / (4.00E+09)] + 6(1 - 0.46^2) / (30782644) (577/118)^3 (1 + 118 \cdot 0.012/2) [-2255(0.087 + 0.44 \cdot -0.12) + 2/577^2 \cdot 0.44 \cdot 1098] \}$
 $= -13.5464 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot Ss$
 $:: |33| + |-13.5| \leq 207 \text{ N./mm}^2$
 $:: 46.6, \text{ must be } \leq 207 \text{ N./mm}^2$

Note:
 For a given Channel thickness of 28 mm, the minimum Channel length adjacent to the tubesheet should be 329.945 mm

Channel Membrane Stress due to Joint Interaction [$\sigma_{c,m}$]:
 $= ac^2 \cdot Pt / ((Dc + tc) tc \cdot \cos(\theta))$
 $= 600^2 \cdot 4550 / ((1200 + 28) 28 \cdot \cos(0))$
 $= 47.641 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction [$\sigma_{c,b}$]:
 $= 6 \cdot kc / tc^2 \cdot \{ (\delta_c \cdot Cm \Delta + \beta_c - \zeta_c \cdot Cm \Theta) \cdot Pt - 6((1 - \nu^2) / E \cdot h) (ao/h)^3 (Cm \Theta + h \cdot \beta_c / 2 \cdot Cm \Delta) \cdot [Pe(Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$
 $= 6 \cdot 7.81E+09 / 28^2 \cdot \{ (0.05537134 \cdot 1 \cdot 0.0098 - 0.000 \cdot 1) \cdot 4550 - 6((1 - 0.46^2) / 30782644) \cdot$



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$$\begin{aligned} & (577/118)^3 * (1 + 118 * 0.0098/2 * 1) * [-2255 (0.087 + 0.44 \\ & \cdot -0.12) + 2/577^2 \cdot 0.44 \cdot 1098] \} \\ & = 239.7855 \text{ N./mm}^2 \end{aligned}$$

Channel Stress Summation vs. Allowable

$$\begin{aligned} & \therefore |\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c \\ & \therefore |47.6| + |240| \leq 207 \text{ N./mm}^2 \\ & \therefore 287 \text{ must be } \leq 207 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)} \end{aligned}$$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	117.2 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	25.5 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	11.6 <=	108.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-11.6 <=	-29.7 N./mm ²	Pass
Maximum Force on any one Tube	182.9 <=	1700.2 Kgf	Pass
Axial Membrane Stress in Shell	33.0 <=	138.0 N./mm ²	Pass
Large Cyl. Axial Membrane Stress	51.7 <=	138.0 N./mm ²	Pass
Large End Cone Axial Membrane Stress	47.2 <=	138.0 N./mm ²	Pass
Small End Cone Axial Membrane Stress	37.7 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	46.6 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	287.4 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(246, 492/2) = 246 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(239, 478/2) = 239 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4 * abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4 * abs(-13.5) / 246], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4 * abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4 * abs(240) / 239], 1) = 1

Compute Modified Modulus [Ec]:
 = Ec * factc
 = 198151376 * 1 = 197893808 KPa.

Tubesheet Bending Stress - final thickness [σf]:
 = (1.5 * Fm / μ*) * (2 * ao / (h - h'g))² * Pe
 = (1.5 * 0.063/0.2) * (2 * 577 / (78.3 - 0))² * -2016
 = -206.9949 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 = h + Cats + Catc
 = 78.3 + 0 + 0
 = 78.3457 mm



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For the Elastic-Plastic iteration, Shell Allowable = SPS,s
 ShellAllow = 492.0227 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Platic iteration, the channel allowable stress = SPS,c
 Challow = 478.0176 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	117.2 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	25.5 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	11.6 <=	108.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-11.6 <=	-29.7 N./mm ²	Pass
Maximum Force on any one Tube	182.9 <=	1700.2 Kgf	Pass
Axial Membrane Stress in Shell	33.0 <=	138.0 N./mm ²	Pass
Large Cyl. Axial Membrane Stress	51.7 <=	138.0 N./mm ²	Pass
Large End Cone Axial Membrane Stress	47.2 <=	138.0 N./mm ²	Pass
Small End Cone Axial Membrane Stress	37.7 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	46.6 <=	492.0 N./mm ²	Pass
Channel Stress (jt. inter.)	287.4 <=	478.0 N./mm ²	Pass

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend	Stresses Allwd	Case Type	Pass/ Fail
D1uc	105.768	...	173	207	39 110 Ps+Pt-Th	D1 Ok
D2uc	46.334	...	59	207	14 110 Ps+Pt-Th	D2 Ok
D3uc	78.357	...	117	207	26 110 Ps+Pt-Th	D3 Ok
O1uc	49.768	...	155	493	34 110 Ps+Pt+Th	O1 Ok
O2uc	8.153	...	29	493	7 110 Ps+Pt+Th	O2 Ok
O3uc	41.712	...	125	493	28 110 Ps+Pt+Th	O3 Ok
O4uc	3.810	...	...	493	... 110 Ps+Pt+Th	O4 Ok
D1c	105.646	...	166	207	41 110 Ps+Pt-Th-c	D1 Ok
D2c	51.450	...	58	207	14 110 Ps+Pt-Th-c	D2 Ok
D3c	80.514	...	114	207	27 110 Ps+Pt-Th-c	D3 Ok
O1c	54.490	...	153	493	36 110 Ps+Pt+Th-c	O1 Ok
O2c	15.357	...	29	493	7 110 Ps+Pt+Th-c	O2 Ok
O3c	47.062	...	124	493	29 110 Ps+Pt+Th-c	O3 Ok
O4c	9.810	...	...	493	... 110 Ps+Pt+Th-c	O4 Ok
Max:	105.7676	... mm	0.834	0.373	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+/-)Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:



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Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	15	137	...	...	:	...	...	...	...	:	Ok	
D2uc	18	137	...	...	:	...	...	...	...	:	Ok	
D3uc	33	137	...	...	:	...	...	...	...	:	Ok	
O1uc	13	524	...	...	:	...	...	...	...	:	Ok	
O2uc	9	524	...	...	:	...	...	...	...	:	Ok	
O3uc	22	524	...	...	:	...	...	...	...	:	Ok	
O4uc	...	524	...	...	:	...	...	...	...	:	Ok	
D1c	18	137	...	...	:	...	...	...	...	:	Ok	
D2c	22	137	...	...	:	...	...	...	...	:	Ok	
D3c	39	137	...	...	:	...	...	...	...	:	Ok	
O1c	15	524	...	...	:	...	...	...	...	:	Ok	
O2c	11	524	...	...	:	...	...	...	...	:	Ok	
O3c	26	524	...	...	:	...	...	...	...	:	Ok	
O4c	...	524	...	...	:	...	...	...	...	:	Ok	
Max RATIO	0.285		...	...	:	...	...	...	...	:	...	

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	12	137	...	...	:	Ok
D2uc	39	137	...	...	:	Ok
D3uc	52	137	...	...	:	Ok
O1uc	11	524	...	...	:	Ok
O2uc	20	524	...	...	:	Ok
O3uc	30	524	...	...	:	Ok
O4uc	...	524	...	...	:	Ok
D1c	15	137	...	...	:	Ok
D2c	49	137	...	...	:	Ok
D3c	64	137	...	...	:	Ok
O1c	14	524	...	...	:	Ok
O2c	24	524	...	...	:	Ok
O3c	38	524	...	...	:	Ok
O4c	...	524	...	...	:	Ok
Max RATIO	0.465		...	...	:	...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	17	137	...	...	:	747	9078	...	...	:	Ok	
D2uc	21	137	...	...	:	2355	9078	...	...	:	Ok	
D3uc	38	137	...	...	:	3102	9078	...	...	:	Ok	
O1uc	15	524	...	...	:	655	34473	...	...	:	Ok	
O2uc	10	524	...	...	:	1172	34473	...	...	:	Ok	
O3uc	25	524	...	...	:	1827	34473	...	...	:	Ok	



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O4uc	...	524	...	...	...	34473	...	...	Ok
D1c	20	137	...	...	887	9078	...	...	Ok
D2c	25	137	...	...	2794	9078	...	...	Ok
D3c	45	137	...	...	3681	9078	...	...	Ok
O1c	18	524	...	...	778	34473	...	...	Ok
O2c	12	524	...	...	1390	34473	...	...	Ok
O3c	30	524	...	...	2168	34473	...	...	Ok
O4c	...	524	...	...	...	34473	...	...	Ok

Max RATIO 0.326 ... 0.405 ...

Tube, Shell and Channel Stress Summary:

		---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd		Stress	Allwd	Stress	Allwd	Fail
D1uc	19	107	-15	-29	292	1700		152	492	345	478	Ok
D2uc	8	107	-8	-29	132	1700		142	206	58	206	Ok
D3uc	12	107	-12	-29	183	1700		47	492	287	478	Ok
O1uc	16	215	-14	-30	256	3400		133	524	303	520	Ok
O2uc	4	215	-4	-30	66	3400		70	524	29	520	Ok
O3uc	12	215	-12	-30	191	3400		81	524	274	520	Ok
O4uc	...	215	...	...	...	3400		...	524	...	520	Ok
D1c	21	107	-18	-29	331	1700		185	492	405	478	Ok
D2c	9	107	-9	-29	148	1700		170	206	72	206	Ok
D3c	13	107	-13	-29	206	1700		58	492	333	478	Ok
O1c	18	215	-16	-30	291	3400		162	524	356	520	Ok
O2c	5	215	-5	-30	74	3400		85	524	36	520	Ok
O3c	14	215	-13	-30	217	3400		99	524	320	520	Ok
O4c	...	215	...	...	...	3400		...	524	...	520	Ok
Max RATIO 0.195 0.598 0.195 0.823 0.848												

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	105.768	118.000	Ok
Tube thickness :	0.557	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Min Channel length of thk, (28.000) adj. to tubesheet: 329.945 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.172 mm
 Due to Pressure + Thermal Shell Extends by : +.172 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	5576.35	0.998	5672.30	0.999
Tubesheet Shear Stress	12205.25	1.000	11800.09	1.000
Tube Tensile Stress	23369.46	1.000	17756.80	1.000
Tube Compressive Stress	7611.38	1.000	4884.90	1.000
Tube-Tubesheet Joint load	23369.50	1.000	17756.78	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.987	4388.29	1.000
Tubesheet-Channel Junction	5366.65	1.000	4388.29	0.763



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Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAWP	5366.65		4388.29	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5505.42	0.998	5637.83	1.000
Tubesheet Shear Stress	12798.11	1.000	12493.61	1.000
Tube Tensile Stress	26417.57	1.000	19940.59	1.000
Tube Compressive Stress	8927.57	1.000	5611.65	1.000
Tube-Tubesheet Joint load	26417.56	1.000	19940.62	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.998	5439.98	1.000
Tubesheet-Channel Junction	5505.42	0.998	5439.98	0.968
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAPnc	5505.42		5439.98	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1721	0.1722	...	...	...	...
2	-0.1382	-0.1382	...	...	...	...
3	0.0339	0.0339	...	...	...	...
4	...	...	...	...	...	...
5	0.1486	0.1487	...	...	...	...
6	-0.0676	-0.0676	...	...	...	...
7	0.0810	0.0811	...	...	...	...
8	...	...	...	...	...	...

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Input Echo, Tubesheet Number 2, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	771.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	65.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3995.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	17.20 °C
Channel Operating Temperature	Tcop	21.00 °C
Tube Operating Temperature	Ttop	21.00 °C
Tubesheet Operating Temperature	Tshop	21.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm



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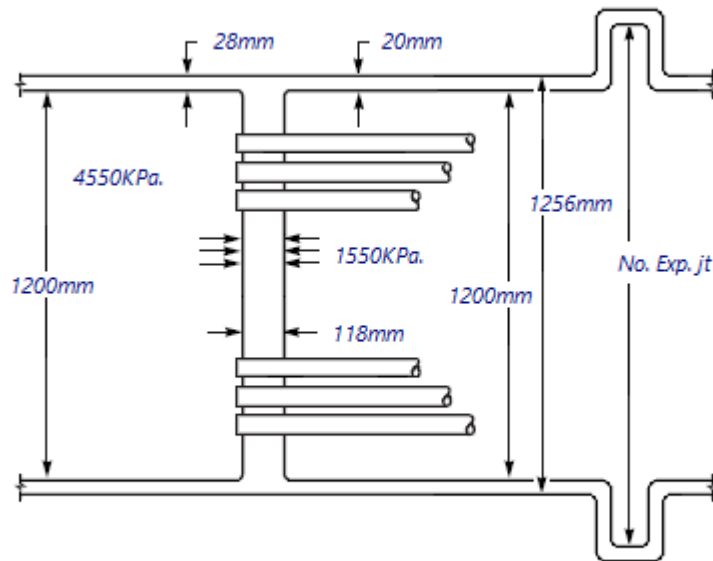
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Circumferential Joint Efficiency Eecc,w 1.00
 Length Lecc 1090.000 mm

Large Diameter Main Shell Cylinder Properties:

Material SA-516 70
 Material UNS Number K02700
 Allowable Stress at Temperature 138.00 N./mm²
 Allowable Stress at Operating Temperature 138.00 N./mm²
 Allowable Stress at Ambient 138.00 N./mm²
 Thickness ts,L 16.0000 mm
 Internal Corrosion Allowance 3.0000 mm
 External Corrosion Allowance 0.0000 mm
 Inside Diameter Ds,L 1800.000 mm
 Circumferential Joint Efficiency Es,L,w 1.00
 Length Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	21.0 °C	0.20271E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	17.2 °C	0.20296E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4550 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 4550)$
 $= 0.5262 + 0.0000 = 0.5262 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 100.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 123) / (3 \cdot 12) = 13593 \text{ KPa.}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5575	25.40	11964.00	45.56	50.0000	0.0005298	52.97
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 53) / (3 \cdot 45.6) = 1550 \text{ KPa.}$						

Summary of Tube Required Thickness Results:



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Total Required Thickness including Corrosion all.	0.5575 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	1550.00 KPa.
Required Thickness due to Shell Side pressure	0.5575 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	105.768 ...	173	207	39 110	Ps+Pt-Th	D1 Ok
D2uc	46.334 ...	59	207	14 110	Ps+Pt-Th	D2 Ok
D3uc	78.357 ...	117	207	26 110	Ps+Pt-Th	D3 Ok
O1uc	49.565 ...	155	496	34 110	Ps+Pt+Th	O1 Ok
O2uc	8.119 ...	29	496	7 110	Ps+Pt+Th	O2 Ok
O3uc	41.536 ...	125	496	28 110	Ps+Pt+Th	O3 Ok
O4uc	3.810 ...	...	496	... 110	Ps+Pt+Th	O4 Ok
D1c	105.646 ...	166	207	41 110	Ps+Pt-Th-c	D1 Ok
D2c	51.450 ...	58	207	14 110	Ps+Pt-Th-c	D2 Ok
D3c	80.514 ...	114	207	27 110	Ps+Pt-Th-c	D3 Ok
O1c	54.301 ...	153	496	36 110	Ps+Pt+Th-c	O1 Ok
O2c	15.320 ...	29	496	7 110	Ps+Pt+Th-c	O2 Ok
O3c	46.897 ...	124	496	29 110	Ps+Pt+Th-c	O3 Ok
O4c	9.810 ...	...	496	... 110	Ps+Pt+Th-c	O4 Ok
Max:	105.7676 ... mm	0.834	0.373	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

	----	Shell Stresses			----	:	----	Shell Band Stress			----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail		
D1uc	15	137	...	...	:	...	...	...	...	:		Ok	
D2uc	18	137	...	...	:	...	...	...	...	:		Ok	
D3uc	33	137	...	...	:	...	...	...	...	:		Ok	
O1uc	13	524	...	...	:	...	...	...	...	:		Ok	
O2uc	9	524	...	...	:	...	...	...	...	:		Ok	



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O3uc	22	524	...	...	...	...	...	...	Ok
O4uc	...	524	...	...	...	...	...	...	Ok
D1c	18	137	...	...	...	...	...	...	Ok
D2c	22	137	...	...	...	...	...	...	Ok
D3c	39	137	...	...	...	...	...	...	Ok
O1c	15	524	...	...	...	...	...	...	Ok
O2c	11	524	...	...	...	...	...	...	Ok
O3c	26	524	...	...	...	...	...	...	Ok
O4c	...	524	...	...	...	...	...	...	Ok

Max RATIO 0.285 ...

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					: Pass
Case#	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	12	137	...	...	Ok
D2uc	39	137	...	...	Ok
D3uc	52	137	...	...	Ok
O1uc	11	524	...	...	Ok
O2uc	20	524	...	...	Ok
O3uc	30	524	...	...	Ok
O4uc	...	524	...	...	Ok
D1c	15	137	...	...	Ok
D2c	49	137	...	...	Ok
D3c	64	137	...	...	Ok
O1c	14	524	...	...	Ok
O2c	24	524	...	...	Ok
O3c	38	524	...	...	Ok
O4c	...	524	...	...	Ok

Max RATIO 0.465 ...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---				: Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	17	137	...	...		747	9078	...	...	Ok
D2uc	21	137	...	...		2355	9078	...	...	Ok
D3uc	38	137	...	...		3102	9078	...	...	Ok
O1uc	15	524	...	...		653	34473	...	...	Ok
O2uc	10	524	...	...		1172	34473	...	...	Ok
O3uc	25	524	...	...		1825	34473	...	...	Ok
O4uc	...	524	...	...		...	34473	...	...	Ok
D1c	20	137	...	...		887	9078	...	...	Ok
D2c	25	137	...	...		2794	9078	...	...	Ok
D3c	45	137	...	...		3681	9078	...	...	Ok
O1c	18	524	...	...		775	34473	...	...	Ok
O2c	12	524	...	...		1390	34473	...	...	Ok
O3c	30	524	...	...		2166	34473	...	...	Ok
O4c	...	524	...	...		...	34473	...	...	Ok

Max RATIO 0.326 ... 0.405 ...

Tube, Shell and Channel Stress Summary:

Tube, Shell and Channel Stress Summary														
---- Tube Stresses ----							Tube Loads		Shell Stress		Channel Stress		Stress Pass	
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail			
D1uc	19	107	-15	-29	292	1700	152	492	345	478	Ok			
D2uc	8	107	-8	-29	132	1700	142	206	58	206	Ok			
D3uc	12	107	-12	-29	183	1700	47	492	287	478	Ok			
O1uc	16	215	-14	-30	257	3400	133	524	303	524	Ok			
O2uc	4	215	-4	-30	66	3400	70	524	29	524	Ok			



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O3uc	12	215	-12	-30	191	3400	80	524	275	524	Ok
O4uc	...	215	...	...	...	3400	...	524	...	524	Ok
D1c	21	107	-18	-29	331	1700	185	492	405	478	Ok
D2c	9	107	-9	-29	148	1700	170	206	72	206	Ok
D3c	13	107	-13	-29	206	1700	58	492	333	478	Ok
O1c	19	215	-16	-30	292	3400	161	524	356	524	Ok
O2c	5	215	-5	-30	74	3400	85	524	36	524	Ok
O3c	14	215	-13	-30	218	3400	98	524	321	524	Ok
O4c	...	215	...	...	...	3400	...	524	...	524	Ok

Max RATIO 0.195 0.598 0.195 0.823 0.848

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	105.768	118.000	Ok
Tube thickness :	0.557	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Min Channel length of thk, (28.000) adj. to tubesheet: 329.945 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.172 mm
 Due to Pressure + Thermal Shell Extends by : +.172 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5576.35	0.998	5672.30	0.999
Tubesheet Shear Stress	12193.48	1.000	11786.18	1.000
Tube Tensile Stress	23369.46	1.000	17756.80	1.000
Tube Compressive Stress	7611.38	1.000	4884.90	1.000
Tube-Tubesheet Joint load	23369.50	1.000	17756.78	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.987	4388.29	1.000
Tubesheet-Channel Junction	5366.65	1.000	4388.29	0.763
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAWP	5366.65		4388.29	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5505.42	0.998	5637.83	1.000
Tubesheet Shear Stress	12798.11	1.000	12493.61	1.000
Tube Tensile Stress	26417.57	1.000	19940.59	1.000
Tube Compressive Stress	8927.57	1.000	5611.65	1.000
Tube-Tubesheet Joint load	26417.56	1.000	19940.62	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.998	5439.98	1.000
Tubesheet-Channel Junction	5505.42	0.998	5439.98	0.968
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAPnc	5505.42		5439.98	



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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1721	0.1722	...	...	...	...
2	-0.1382	-0.1382	...	...	...	...
3	0.0339	0.0339	...	...	...	...
4	...	...	...	...	...	...
5	0.1477	0.1478	...	...	...	...
6	-0.0672	-0.0672	...	...	...	...
7	0.0805	0.0806	...	...	...	...
8	...	...	...	...	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(118/4, \min(118, 20))$$

$$= 29.500 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(1550/5440, 1)$$

$$= 0.285$$

Thickness Ratio = 0.28, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(118/4, \min(118, 28))$$

$$= 29.500 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(4550/5505, 1)$$

$$= 0.826$$

Thickness Ratio = 0.83, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

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Input Echo, Tubesheet Number 3, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	395.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	65.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3366.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	1.10 °C
Channel Operating Temperature	Tcop	21.00 °C
Tube Operating Temperature	Ttop	21.00 °C
Tubesheet Operating Temperature	Tshop	21.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm



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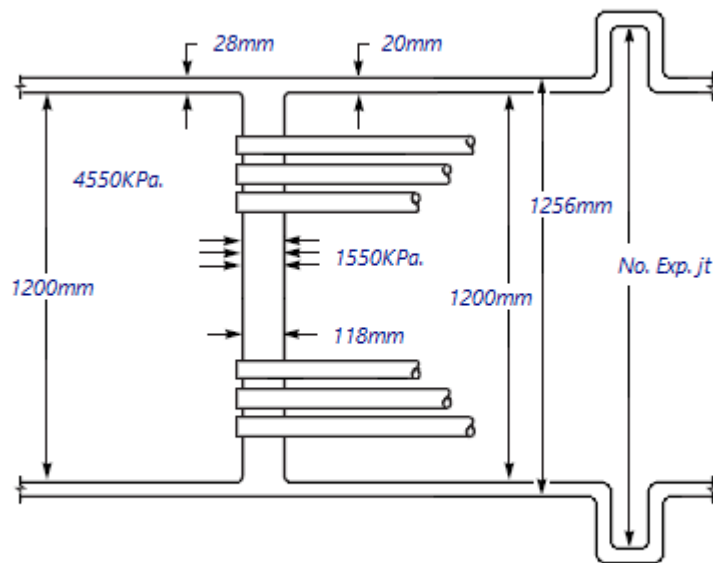
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Circumferential Joint Efficiency	Eecc,w	1.00
Length	Lecc	1090.000 mm

Large Diameter Main Shell Cylinder Properties:

Material	SA-516 70
Material UNS Number	K02700
Allowable Stress at Temperature	138.00 N./mm ²
Allowable Stress at Operating Temperature	138.00 N./mm ²
Allowable Stress at Ambient	138.00 N./mm ²
Thickness	ts,L 16.0000 mm
Internal Corrosion Allowance	3.0000 mm
External Corrosion Allowance	0.0000 mm
Inside Diameter	Ds,L 1800.000 mm
Circumferential Joint Efficiency	Es,L,w 1.00
Length	Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	21.0 °C	0.20271E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	21.0 °C	0.20271E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4550 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 4550)$$

$$= 0.5262 + 0.0000 = 0.5262 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 100.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
EMAWP = (4*B)/(3*(D/T)) = (4 *123)/(3 *12) = 13593 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5575	25.40	11964.00	45.56	50.0000	0.0005298	52.97
EMAWP = (4*B)/(3*(D/T)) = (4 *53)/(3 *45.6) = 1550 KPa.						

Summary of Tube Required Thickness Results:



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Total Required Thickness including Corrosion all.	0.5575 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	1550.00 KPa.
Required Thickness due to Shell Side pressure	0.5575 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	105.768 ...	173	207	39 110	Ps+Pt-Th	D1 Ok
D2uc	46.334 ...	59	207	14 110	Ps+Pt-Th	D2 Ok
D3uc	78.357 ...	117	207	26 110	Ps+Pt-Th	D3 Ok
O1uc	43.042 ...	130	496	29 110	Ps+Pt+Th	O1 Ok
O2uc	3.810 ...	15	496	3 110	Ps+Pt+Th	O2 Ok
O3uc	38.689 ...	115	496	26 110	Ps+Pt+Th	O3 Ok
O4uc	3.810 ...	...	496	... 110	Ps+Pt+Th	O4 Ok
D1c	105.646 ...	166	207	41 110	Ps+Pt-Th-c	D1 Ok
D2c	51.450 ...	58	207	14 110	Ps+Pt-Th-c	D2 Ok
D3c	80.514 ...	114	207	27 110	Ps+Pt-Th-c	D3 Ok
O1c	48.303 ...	129	496	30 110	Ps+Pt+Th-c	O1 Ok
O2c	9.810 ...	15	496	4 110	Ps+Pt+Th-c	O2 Ok
O3c	44.277 ...	114	496	27 110	Ps+Pt+Th-c	O3 Ok
O4c	9.810 ...	...	496	... 110	Ps+Pt+Th-c	O4 Ok
Max:	105.7676 ... mm	0.834	0.373	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

	----	Shell Stresses			----	:	----	Shell Band Stress			----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail		
D1uc	15	137	...	...		...	...	...	...			Ok	
D2uc	18	137	...	...		...	...	...	...			Ok	
D3uc	33	137	...	...		...	...	...	...			Ok	
O1uc	11	524	...	...		...	...	...	...			Ok	
O2uc	5	524	...	...		...	...	...	...			Ok	



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O3uc	16	524	...	...	...	...	...	...	Ok
O4uc	...	524	...	...	...	...	...	...	Ok
D1c	18	137	...	...	...	...	...	...	Ok
D2c	22	137	...	...	...	...	...	...	Ok
D3c	39	137	...	...	...	...	...	...	Ok
O1c	13	524	...	...	...	...	...	...	Ok
O2c	6	524	...	...	...	...	...	...	Ok
O3c	19	524	...	...	...	...	...	...	Ok
O4c	...	524	...	...	...	...	...	...	Ok

Max RATIO 0.285 ...

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					: Pass
Case#	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	12	137	...	...	Ok
D2uc	39	137	...	...	Ok
D3uc	52	137	...	...	Ok
O1uc	9	524	...	...	Ok
O2uc	10	524	...	...	Ok
O3uc	19	524	...	...	Ok
O4uc	...	524	...	...	Ok
D1c	15	137	...	...	Ok
D2c	49	137	...	...	Ok
D3c	64	137	...	...	Ok
O1c	11	524	...	...	Ok
O2c	12	524	...	...	Ok
O3c	24	524	...	...	Ok
O4c	...	524	...	...	Ok

Max RATIO 0.465 ...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---				:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	17	137	...	...		747	9078	...	...		Ok
D2uc	21	137	...	...		2355	9078	...	...		Ok
D3uc	38	137	...	...		3102	9078	...	...		Ok
O1uc	13	524	...	...		551	34473	...	...		Ok
O2uc	5	524	...	...		600	34473	...	...		Ok
O3uc	18	524	...	...		1152	34473	...	...		Ok
O4uc	...	524	...	...		...	34473	...	...		Ok
D1c	20	137	...	...		887	9078	...	...		Ok
D2c	25	137	...	...		2794	9078	...	...		Ok
D3c	45	137	...	...		3681	9078	...	...		Ok
O1c	15	524	...	...		655	34473	...	...		Ok
O2c	6	524	...	...		712	34473	...	...		Ok
O3c	21	524	...	...		1367	34473	...	...		Ok
O4c	...	524	...	...		...	34473	...	...		Ok

Max RATIO 0.326 ... 0.405 ...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses					Tube Loads		Shell Stress		Channel Stress		Stress Pass	
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail	
D1uc	19	107	-15	-29	292	1700	152	492	345	478	Ok	
D2uc	8	107	-8	-29	132	1700	142	206	58	206	Ok	
D3uc	12	107	-12	-29	183	1700	47	492	287	478	Ok	
O1uc	14	215	-11	-30	216	3400	112	524	255	524	Ok	
O2uc	2	215	-2	-30	34	3400	36	524	15	524	Ok	



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O3uc	12	215	-10	-30	183	3400	85	524	241	524	Ok
O4uc	...	215	...	...	...	3400	...	524	...	524	Ok
D1c	21	107	-18	-29	331	1700	185	492	405	478	Ok
D2c	9	107	-9	-29	148	1700	170	206	72	206	Ok
D3c	13	107	-13	-29	206	1700	58	492	333	478	Ok
O1c	16	215	-13	-30	245	3400	136	524	300	524	Ok
O2c	2	215	-2	-30	38	3400	43	524	18	524	Ok
O3c	13	215	-12	-30	207	3400	104	524	282	524	Ok
O4c	...	215	...	...	...	3400	...	524	...	524	Ok

Max RATIO 0.195 0.598 0.195 0.823 0.848

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	105.768	118.000	Ok
Tube thickness :	0.557	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Min Channel length of thk, (28.000) adj. to tubesheet: 329.945 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.172 mm
 Due to Pressure + Thermal Shell Extends by : +.172 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5576.35	0.998	5672.30	0.999
Tubesheet Shear Stress	12203.49	1.000	11797.98	1.000
Tube Tensile Stress	23369.46	1.000	17756.80	1.000
Tube Compressive Stress	7611.38	1.000	4884.90	1.000
Tube-Tubesheet Joint load	23369.50	1.000	17756.78	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.987	4388.29	1.000
Tubesheet-Channel Junction	5366.65	1.000	4388.29	0.763
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAWP	5366.65		4388.29	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5505.42	0.998	5637.83	1.000
Tubesheet Shear Stress	12798.11	1.000	12493.61	1.000
Tube Tensile Stress	26417.57	1.000	19940.59	1.000
Tube Compressive Stress	8927.57	1.000	5611.65	1.000
Tube-Tubesheet Joint load	26417.56	1.000	19940.62	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.998	5439.98	1.000
Tubesheet-Channel Junction	5505.42	0.998	5439.98	0.968
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAPnc	5505.42		5439.98	



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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1721	0.1722	...	...	...	...
2	-0.1382	-0.1382	...	...	...	...
3	0.0339	0.0339	...	...	...	...
4	...	...	...	...	...	...
5	0.1245	0.1245	...	...	...	...
6	-0.0344	-0.0344	...	...	...	...
7	0.0900	0.0901	...	...	...	...
8	...	...	...	...	...	...

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Input Echo, Tubesheet Number 4, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	395.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	65.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3366.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	1.10 °C
Channel Operating Temperature	Tcop	4.00 °C
Tube Operating Temperature	Ttop	4.00 °C
Tubesheet Operating Temperature	Tshop	4.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm



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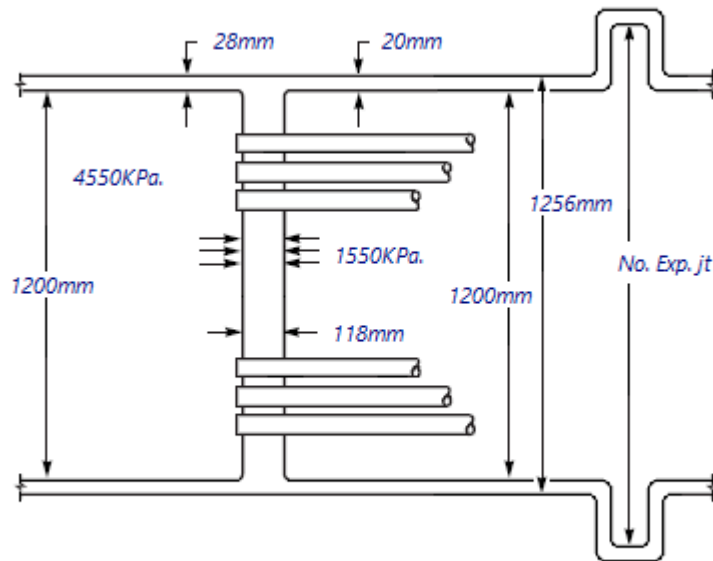
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Circumferential Joint Efficiency	Eecc,w	1.00
Length	Lecc	1090.000 mm

Large Diameter Main Shell Cylinder Properties:

Material	SA-516 70
Material UNS Number	K02700
Allowable Stress at Temperature	138.00 N./mm ²
Allowable Stress at Operating Temperature	138.00 N./mm ²
Allowable Stress at Ambient	138.00 N./mm ²
Thickness	ts,L 16.0000 mm
Internal Corrosion Allowance	3.0000 mm
External Corrosion Allowance	0.0000 mm
Inside Diameter	Ds,L 1800.000 mm
Circumferential Joint Efficiency	Es,L,w 1.00
Length	Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	4.0 °C	0.20383E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	4.0 °C	0.20383E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	4.0 °C	0.20383E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	4.0 °C	0.20383E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (4550 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 4550)$$

$$= 0.5262 + 0.0000 = 0.5262 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 100.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
EMAWP = (4*B)/(3*(D/T)) = (4 *123)/(3 *12) = 13593 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5575	25.40	11964.00	45.56	50.0000	0.0005298	52.97
EMAWP = (4*B)/(3*(D/T)) = (4 *53)/(3 *45.6) = 1550 KPa.						

Summary of Tube Required Thickness Results:



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Total Required Thickness including Corrosion all.	0.5575 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	1550.00 KPa.
Required Thickness due to Shell Side pressure	0.5575 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	105.768 ...	173	207	39 110	Ps+Pt-Th	D1 Ok
D2uc	46.334 ...	59	207	14 110	Ps+Pt-Th	D2 Ok
D3uc	78.357 ...	117	207	26 110	Ps+Pt-Th	D3 Ok
O1uc	43.087 ...	130	496	29 110	Ps+Pt+Th	O1 Ok
O2uc	3.810 ...	15	496	3 110	Ps+Pt+Th	O2 Ok
O3uc	38.734 ...	115	496	26 110	Ps+Pt+Th	O3 Ok
O4uc	3.810 ...	...	496	... 110	Ps+Pt+Th	O4 Ok
D1c	105.646 ...	166	207	41 110	Ps+Pt-Th-c	D1 Ok
D2c	51.450 ...	58	207	14 110	Ps+Pt-Th-c	D2 Ok
D3c	80.514 ...	114	207	27 110	Ps+Pt-Th-c	D3 Ok
O1c	48.342 ...	129	496	30 110	Ps+Pt+Th-c	O1 Ok
O2c	9.810 ...	15	496	4 110	Ps+Pt+Th-c	O2 Ok
O3c	44.317 ...	114	496	27 110	Ps+Pt+Th-c	O3 Ok
O4c	9.810 ...	...	496	... 110	Ps+Pt+Th-c	O4 Ok
Max:	105.7676 ... mm	0.834	0.373	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				:	Pass
	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	15	137	...	...	:	...	...	...	...	:	Ok
D2uc	18	137	...	...	:	...	...	...	...	:	Ok
D3uc	33	137	...	...	:	...	...	...	...	:	Ok
O1uc	11	524	...	...	:	...	...	...	...	:	Ok
O2uc	5	524	...	...	:	...	...	...	...	:	Ok



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O3uc	16	524	...	...	...	...	...	...	Ok
O4uc	...	524	...	...	...	...	...	...	Ok
D1c	18	137	...	...	...	...	...	...	Ok
D2c	22	137	...	...	...	...	...	...	Ok
D3c	39	137	...	...	...	...	...	...	Ok
O1c	13	524	...	...	...	...	...	...	Ok
O2c	6	524	...	...	...	...	...	...	Ok
O3c	19	524	...	...	...	...	...	...	Ok
O4c	...	524	...	...	...	...	...	...	Ok

Max RATIO 0.285 ...

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					: Pass
Case#	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	12	137	...	...	Ok
D2uc	39	137	...	...	Ok
D3uc	52	137	...	...	Ok
O1uc	9	524	...	...	Ok
O2uc	10	524	...	...	Ok
O3uc	19	524	...	...	Ok
O4uc	...	524	...	...	Ok
D1c	15	137	...	...	Ok
D2c	49	137	...	...	Ok
D3c	64	137	...	...	Ok
O1c	11	524	...	...	Ok
O2c	12	524	...	...	Ok
O3c	24	524	...	...	Ok
O4c	...	524	...	...	Ok

Max RATIO 0.465 ...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---				:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	17	137	...	...		747	9078	...	...		Ok
D2uc	21	137	...	...		2355	9078	...	...		Ok
D3uc	38	137	...	...		3102	9078	...	...		Ok
O1uc	12	524	...	...		550	34473	...	...		Ok
O2uc	5	524	...	...		601	34473	...	...		Ok
O3uc	18	524	...	...		1151	34473	...	...		Ok
O4uc	...	524	...	...		...	34473	...	...		Ok
D1c	20	137	...	...		887	9078	...	...		Ok
D2c	25	137	...	...		2794	9078	...	...		Ok
D3c	45	137	...	...		3681	9078	...	...		Ok
O1c	15	524	...	...		653	34473	...	...		Ok
O2c	6	524	...	...		712	34473	...	...		Ok
O3c	21	524	...	...		1366	34473	...	...		Ok
O4c	...	524	...	...		...	34473	...	...		Ok

Max RATIO 0.326 ... 0.405 ...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses					Tube Loads		Shell Stress		Channel Stress		Stress Pass	
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail	
D1uc	19	107	-15	-29	292	1700	152	492	345	478	Ok	
D2uc	8	107	-8	-29	132	1700	142	206	58	206	Ok	
D3uc	12	107	-12	-29	183	1700	47	492	287	478	Ok	
O1uc	14	215	-11	-30	217	3400	112	524	256	524	Ok	
O2uc	2	215	-2	-30	34	3400	36	524	15	524	Ok	



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O3uc	12	215	-10	-30	183	3400	85	524	241	524	Ok
O4uc	...	215	...	...	...	3400	...	524	...	524	Ok
D1c	21	107	-18	-29	331	1700	185	492	405	478	Ok
D2c	9	107	-9	-29	148	1700	170	206	72	206	Ok
D3c	13	107	-13	-29	206	1700	58	492	333	478	Ok
O1c	16	215	-13	-30	246	3400	136	524	300	524	Ok
O2c	2	215	-2	-30	38	3400	43	524	18	524	Ok
O3c	13	215	-12	-30	208	3400	103	524	282	524	Ok
O4c	...	215	...	...	...	3400	...	524	...	524	Ok

Max RATIO 0.195 0.598 0.195 0.823 0.848

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	105.768	118.000	Ok
Tube thickness :	0.557	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Min Channel length of thk, (28.000) adj. to tubesheet: 329.945 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.172 mm
 Due to Pressure + Thermal Shell Extends by : +.172 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5576.35	0.998	5672.30	0.999
Tubesheet Shear Stress	12192.92	1.000	11785.51	1.000
Tube Tensile Stress	23369.46	1.000	17756.80	1.000
Tube Compressive Stress	7611.38	1.000	4884.90	1.000
Tube-Tubesheet Joint load	23369.50	1.000	17756.78	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.987	4388.29	1.000
Tubesheet-Channel Junction	5366.65	1.000	4388.29	0.763
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAWP	5366.65		4388.29	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5505.42	0.998	5637.83	1.000
Tubesheet Shear Stress	12798.11	1.000	12493.61	1.000
Tube Tensile Stress	26417.57	1.000	19940.59	1.000
Tube Compressive Stress	8927.57	1.000	5611.65	1.000
Tube-Tubesheet Joint load	26417.56	1.000	19940.62	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.998	5439.98	1.000
Tubesheet-Channel Junction	5505.42	0.998	5439.98	0.968
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAPnc	5505.42		5439.98	



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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1721	0.1722	...	...	...	...
2	-0.1382	-0.1382	...	...	...	...
3	0.0339	0.0339	...	...	...	...
4	...	...	...	...	...	...
5	0.1238	0.1238	...	...	...	...
6	-0.0342	-0.0342	...	...	...	...
7	0.0895	0.0896	...	...	...	...
8	...	...	...	...	...	...

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Input Echo, Tubesheet Number 5, Description:

Shell Data:

Main Shell Description: LHS Cone Shell

Shell Maximum Design Pressure	Psd,max	1550.00 KPa.
Shell Maximum Operating Pressure	Psox,max	409.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	65.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4550.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	3366.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	28.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	100.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	1151
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12200.00 mm
Straight Tube Length (bet. inner tubsht faces) L		11964.00 mm
Design Temperature of the Tubes		100.00 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. & wld. tube	
Tube Allowable Stress at Temperature		108.00 N./mm ²
Tube Allowable Stress At Ambient		108.00 N./mm ²
Tube Allowable Stress at Operating Temperature		108.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	189.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



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Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	564.400 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	9690.4 cm ²
Perimeter of Tube Layout	Cp	3628.9 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	100.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	118.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1256.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1500.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	1.10 °C
Channel Operating Temperature	Tcop	41.90 °C
Tube Operating Temperature	Ttop	41.90 °C
Tubesheet Operating Temperature	Tshop	41.90 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Additional Data for Kettle Type Exchangers:

Eccentric Cone Properties:

Material		SA-516 70
Material UNS Number		K02700
Allowable Stress at Temperature		138.00 N./mm ²
Allowable Stress at Operating Temperature		138.00 N./mm ²
Allowable Stress at Ambient		138.00 N./mm ²
Thickness	tecc	20.0000 mm
Internal Corrosion Allowance		3.0000 mm
External Corrosion Allowance		0.0000 mm
Small End Inside Diameter	Decc,S	1200.000 mm
Large End Inside Diameter	Decc,L	1800.000 mm



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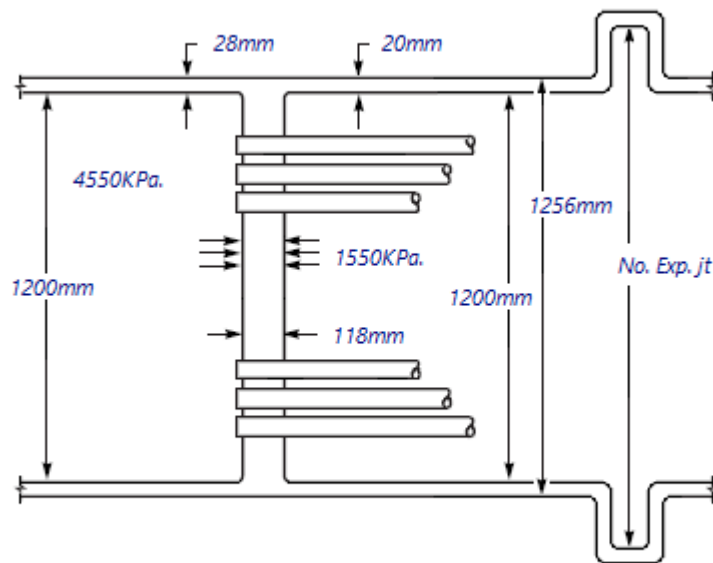
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Circumferential Joint Efficiency	Eecc,w	1.00
Length	Lecc	1090.000 mm

Large Diameter Main Shell Cylinder Properties:

Material	SA-516 70
Material UNS Number	K02700
Allowable Stress at Temperature	138.00 N./mm ²
Allowable Stress at Operating Temperature	138.00 N./mm ²
Allowable Stress at Ambient	138.00 N./mm ²
Thickness	ts,L 16.0000 mm
Internal Corrosion Allowance	3.0000 mm
External Corrosion Allowance	0.0000 mm
Inside Diameter	Ds,L 1800.000 mm
Circumferential Joint Efficiency	Es,L,w 1.00
Length	Ls,L 9384.001 mm

Tubesheet Integral w/Shell and Channel



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	41.9 °C	0.20151E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	41.9 °C	0.20151E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	41.9 °C	0.20151E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	100.0 °C	0.19815E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	41.9 °C	0.20151E+09 KPa.

Large Cyl. - TE-1 Carbon & Low Alloy Steels, Group 1
 Large Cyl. - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Cone - TE-1 Carbon & Low Alloy Steels, Group 1
 Cone - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	65.0 °C	0.20019E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	1.1 °C	0.20402E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4550 \cdot (25.4/2 - 0)) / (108 \cdot 1 + 0.4 \cdot 4550)$
 $= 0.5262 + 0.0000 = 0.5262 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 100.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	11964.00	12.04	50.0000	0.0075908	122.73
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 123) / (3 \cdot 12) = 13593 \text{ KPa.}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5575	25.40	11964.00	45.56	50.0000	0.0005298	52.97
EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 53) / (3 \cdot 45.6) = 1550 \text{ KPa.}$						

Summary of Tube Required Thickness Results:



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Total Required Thickness including Corrosion all.	0.5575 mm
Allowable Internal Pressure at Corroded thickness	19219.18 KPa.
Required Internal Design Pressure	4550.00 KPa.
Allowable External Pressure at Corroded thickness	13593.04 KPa.
Required External Design Pressure	1550.00 KPa.
Required Thickness due to Shell Side pressure	0.5575 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	105.768 ...	173	207	39 110	Ps+Pt-Th	D1 Ok
D2uc	46.334 ...	59	207	14 110	Ps+Pt-Th	D2 Ok
D3uc	78.357 ...	117	207	26 110	Ps+Pt-Th	D3 Ok
O1uc	43.160 ...	130	494	29 110	Ps+Pt+Th	O1 Ok
O2uc	3.810 ...	16	494	4 110	Ps+Pt+Th	O2 Ok
O3uc	38.637 ...	115	494	25 110	Ps+Pt+Th	O3 Ok
O4uc	3.810 ...	...	494	... 110	Ps+Pt+Th	O4 Ok
D1c	105.646 ...	166	207	41 110	Ps+Pt-Th-c	D1 Ok
D2c	51.450 ...	58	207	14 110	Ps+Pt-Th-c	D2 Ok
D3c	80.514 ...	114	207	27 110	Ps+Pt-Th-c	D3 Ok
O1c	48.414 ...	129	494	30 110	Ps+Pt+Th-c	O1 Ok
O2c	9.810 ...	15	494	4 110	Ps+Pt+Th-c	O2 Ok
O3c	44.231 ...	113	494	27 110	Ps+Pt+Th-c	O3 Ok
O4c	9.810 ...	...	494	... 110	Ps+Pt+Th-c	O4 Ok
Max:	105.7676 ... mm	0.834	0.373	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

	----	Shell Stresses			----	:	----	Shell Band Stress			----	:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail		
D1uc	15	137	...	...		...	...	...	...			Ok	
D2uc	18	137	...	...		...	...	...	...			Ok	
D3uc	33	137	...	...		...	...	...	...			Ok	
O1uc	11	524	...	...		...	...	...	...			Ok	
O2uc	5	524	...	...		...	...	...	...			Ok	



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O3uc	16	524	...	...	...	...	...	...	Ok
O4uc	...	524	...	...	...	...	...	...	Ok
D1c	18	137	...	...	...	...	...	...	Ok
D2c	22	137	...	...	...	...	...	...	Ok
D3c	39	137	...	...	...	...	...	...	Ok
O1c	13	524	...	...	...	...	...	...	Ok
O2c	6	524	...	...	...	...	...	...	Ok
O3c	19	524	...	...	...	...	...	...	Ok
O4c	...	524	...	...	...	...	...	...	Ok

Max RATIO 0.285 ...

Large Diameter Shell Cylinder Axial Membrane Stress Summary:

---- Shell Stresses ----					: Pass
Case#	Ten	Allwd	Cmp	Allwd	: Fail
D1uc	12	137	...	...	Ok
D2uc	39	137	...	...	Ok
D3uc	52	137	...	...	Ok
O1uc	9	524	...	...	Ok
O2uc	10	524	...	...	Ok
O3uc	20	524	...	...	Ok
O4uc	...	524	...	...	Ok
D1c	15	137	...	...	Ok
D2c	49	137	...	...	Ok
D3c	64	137	...	...	Ok
O1c	11	524	...	...	Ok
O2c	13	524	...	...	Ok
O3c	24	524	...	...	Ok
O4c	...	524	...	...	Ok

Max RATIO 0.465 ...

Cone Small End & Cone Large End Axial Membrane Stress Summary:

---- Small End Stresses ----					:	--- Large End Stresses ---				:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail
D1uc	17	137	...	...	:	747	9078	...	...	:	Ok
D2uc	21	137	...	...	:	2355	9078	...	...	:	Ok
D3uc	38	137	...	...	:	3102	9078	...	...	:	Ok
O1uc	13	524	...	...	:	553	34473	...	...	:	Ok
O2uc	5	524	...	...	:	621	34473	...	...	:	Ok
O3uc	18	524	...	...	:	1174	34473	...	...	:	Ok
O4uc	...	524	...	...	:	...	34473	...	...	:	Ok
D1c	20	137	...	...	:	887	9078	...	...	:	Ok
D2c	25	137	...	...	:	2794	9078	...	...	:	Ok
D3c	45	137	...	...	:	3681	9078	...	...	:	Ok
O1c	15	524	...	...	:	657	34473	...	...	:	Ok
O2c	7	524	...	...	:	737	34473	...	...	:	Ok
O3c	21	524	...	...	:	1394	34473	...	...	:	Ok
O4c	...	524	...	...	:	...	34473	...	...	:	Ok

Max RATIO 0.326 ... 0.405 ...

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Stress Pass	
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail	
D1uc	19	107	-15	-29	292	1700	152	492	345	478	Ok	
D2uc	8	107	-8	-29	132	1700	142	206	58	206	Ok	
D3uc	12	107	-12	-29	183	1700	47	492	287	478	Ok	
O1uc	14	215	-11	-30	216	3400	112	524	255	521	Ok	
O2uc	2	215	-2	-30	35	3400	37	524	15	521	Ok	



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O3uc	11	215	-10	-30	181	3400	85	524	240	521	Ok
O4uc	...	215	...	...	...	3400	...	524	...	521	Ok
D1c	21	107	-18	-29	331	1700	185	492	405	478	Ok
D2c	9	107	-9	-29	148	1700	170	206	72	206	Ok
D3c	13	107	-13	-29	206	1700	58	492	333	478	Ok
O1c	16	215	-13	-30	245	3400	137	524	300	521	Ok
O2c	2	215	-2	-30	39	3400	45	524	19	521	Ok
O3c	13	215	-12	-30	206	3400	103	524	281	521	Ok
O4c	...	215	...	...	...	3400	...	524	...	521	Ok

Max RATIO 0.195 0.598 0.195 0.823 0.848

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	105.768	118.000	Ok
Tube thickness :	0.557	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 278.855 mm

Min Channel length of thk, (28.000) adj. to tubesheet: 329.945 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.172 mm
 Due to Pressure + Thermal Shell Extends by : +.172 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5576.35	0.998	5672.30	0.999
Tubesheet Shear Stress	12210.79	1.000	11806.62	1.000
Tube Tensile Stress	23369.46	1.000	17756.80	1.000
Tube Compressive Stress	7611.38	1.000	4884.90	1.000
Tube-Tubesheet Joint load	23369.50	1.000	17756.78	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.987	4388.29	1.000
Tubesheet-Channel Junction	5366.65	1.000	4388.29	0.763
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAWP	5366.65		4388.29	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	5505.42	0.998	5637.83	1.000
Tubesheet Shear Stress	12798.11	1.000	12493.61	1.000
Tube Tensile Stress	26417.57	1.000	19940.59	1.000
Tube Compressive Stress	8927.57	1.000	5611.65	1.000
Tube-Tubesheet Joint load	26417.56	1.000	19940.62	1.000
Axial Shell, Cone, Large Cyl.	5505.42	0.998	5439.98	1.000
Tubesheet-Channel Junction	5505.42	0.998	5439.98	0.968
Tube Pressure Stress	19219.17	1.000	13593.03	1.000
Minimum MAPnc	5505.42		5439.98	



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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1721	0.1722	...	...	...	...
2	-0.1382	-0.1382	...	...	...	...
3	0.0339	0.0339	...	...	...	...
4	...	...	...	...	...	...
5	0.1252	0.1253	...	...	...	...
6	-0.0359	-0.0359	...	...	...	...
7	0.0893	0.0894	...	...	...	...
8	...	...	...	...	...	...

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 FileName : 130-C1& C2-Final 23rd March Page 310 of 317
 Conical Section: LHS Cone Shell Cone: 11 9:30am Sep 9,2026

Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

LHS Cone Shell

Design Internal Pressure, Small End	1567.673	KPa.
Design Internal Pressure, Large End	1567.673	KPa.
Temperature for Internal Pressure	65.0	°C
Design External Pressure	0.000	KPa.
Temperature for External Pressure	65.0	°C

CONE Properties:

Cone Material		SA-516 70	
Cone Allowable Stress at Temperature	Ss	138.00	N./mm ²
Cone Allowable Stress At Ambient		138.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	20.0000	mm
Corrosion Allowance for Cone	Cac	3.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1200.000	mm
Diameter of Large End of Cone	Dl	1800.000	mm
Half Apex Angle for Cone		28.83	degrees
Axial Length of Cone	Lc	1090.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	
Operating Allowable Stress	Ss	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	20.0000	mm
Corrosion Allowance	Cas	3.0000	mm
Axial Length of Attached Element	Ls	11964.001	mm

LARGE-END Properties:

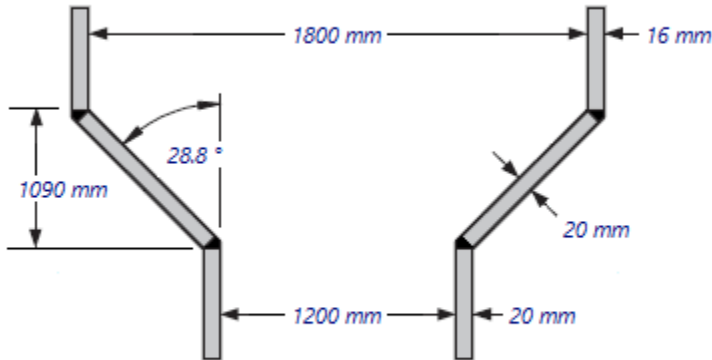
Attached Large End Material Specification		SA-516 70	
Operating Allowable Stress	Sl	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	16.0000	mm
Corrosion Allowance	Cal	3.0000	mm
Axial Length of Attached Element	Ll	11964.001	mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 65 °C

Elastic Modulus of Cone Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large End Reinforcement	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small End Reinforcement	0.200E+09 KPa. at 65 °C

Elastic Modulus Data from ASME Section II Part D at 65 °C

Elastic Modulus of Cone Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large End Reinforcement	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small End Reinforcement	0.200E+09 KPa. at 65 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Note: Neither end of Cone is a Line of Support

Maximum Centroid Reinforcement Distance Large End	27.2809 mm
Maximum Centroid Reinforcement Distance Small End	25.6661 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01136

Large End Maximum half apex angle w/o reinforcement [δ]:

$$= 326.6 \cdot \sqrt{\left(\frac{P}{(S_s \cdot E_1)} \right)}$$
 per 1-5 equation 1

$$= 326.6 \cdot \sqrt{\left(\frac{1568}{(138 \cdot 1)} \right)}$$



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 Conical Section: LHS Cone Shell Cone: 11 9:30am Sep 9, 2026

= 34.811 degrees

Large end actual half apex angle [alpha]: 28.831 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the large end under internal pressure.

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01136

Small End Maximum half apex angle w/o reinforcement [δ]:
 $= 89 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-5 equation 6
 $= 89 \cdot \sqrt{(1568 / (138 \cdot 1))}$
 = 9.486 degrees

Small end actual half apex angle [alpha]: 28.831 degrees

Intermediate Value [k]:
 $= \max(Y / (Sr \cdot Ers), 1)$
 $= \max(27.63E+09 / (138 \cdot 200187728), 1)$
 = 1.0000

where [Y] is:
 = Small End All. Stress • Small End Elastic Modulus (Int. temp.)
 = $138 \cdot 200187728$
 = $27625406464.0 \text{ (N./mm}^2\text{)}^2$

Decay Length, Cone Small End:
 $= 1.4 \cdot \sqrt{(Rs(t_s - ca))}$
 $= 1.4 \cdot \sqrt{(603(20 - 3))}$
 = 141.746 mm

Required Area of Reinforcement, Small End, Internal [Ars]:
 $= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha)$ per 1-5 equation 7
 $= 1 \cdot 78.7 \cdot 603 / (138 \cdot 1) \cdot (1 - 9.49/28.8) \cdot 0.55$
 = 12.46 cm²

Force per Length, Cone Small End, tensile side [QS+]:
 $= P(Rs/2) - Faxial / (\pi(Ds + Ts)) + Moment / (\pi(Rs + Ts/2)(Rs + Ts/2)) + f2'$
 $= 1568(603/2) - 0 / (\pi(1206 + 20)) +$
 $0 / (\pi(603 + 20/2)(603 + 20/2)) + 30.5$
 = 78.721 Kg/mm

Note:
 Note: f2' is the axial load due to the heat exchanger constraint and is taken as the maximum value considering all of the exchanger load cases and conditions.

Area of Reinforcement Available in Small End Shell [Aes]:
 $= 0.78[\sqrt{(Rs \cdot ts)} \cdot (ts - t) + \sqrt{(Rs \cdot tc / \cos(\alpha))} \cdot (tc - tr)]$ per 1-5 eqn. 8
 $= 0.78[\sqrt{(603 \cdot 17)} \cdot (17 - 6.9) +$
 $\sqrt{(603 \cdot 17 / 0.88)} \cdot (17 - 7.87)]$
 = 15.6786 cm²

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	12.4593	cm ²
Area of reinforcement in shell per App. 1-5(8)	15.6786	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.



The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	133.26	492.00	Small Cyl. Long.
Compres. Stress	-76.87	-492.00	Small Cyl. Long.
Membrane Stress	114.22	-207.00	Small End Tang.
Tensile Stress	137.25	492.00	Cone Longitudinal
Compres. Stress	-72.88	-492.00	Cone Longitudinal
Compres Stress	122.20	-207.00	Cone Tangential
Tensile Stress	348.77	492.00	Large Cyl. Long.
Compres. Stress	-239.08	-492.00	Large Cyl. Long.
Membrane Stress	-19.66	-207.00	Large End Tang.
Tensile Stress	219.75	492.00	Cone Longitudinal
Compres. Stress	-124.01	-492.00	Cone Longitudinal
Compres Stress	-33.60	-207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure KPa.	Reason for Failure at this Pressure
MAWP	2200.237	Small End Reinforcement
MAPnc	2620.779	Small End Reinforcement

These pressures were determined by iteration.



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 FileName : 130-C1& C2-Final 23rd March Page 314 of 317
 Conical Section: RHS Cone shell Cone: 12 9:30am Sep 9,2026

Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

RHS Cone shell

Design Internal Pressure, Small End	1561.258	KPa.
Design Internal Pressure, Large End	1561.258	KPa.
Temperature for Internal Pressure	65.0	°C
Design External Pressure	0.000	KPa.
Temperature for External Pressure	65.0	°C

CONE Properties:

Cone Material		SA-516 70	
Cone Allowable Stress at Temperature	Ss	138.00	N./mm ²
Cone Allowable Stress At Ambient		138.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	20.0000	mm
Corrosion Allowance for Cone	Cac	3.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1200.000	mm
Diameter of Large End of Cone	Dl	1800.000	mm
Half Apex Angle for Cone		28.83	degrees
Axial Length of Cone	Lc	1090.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	
Operating Allowable Stress	Ss	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	20.0000	mm
Corrosion Allowance	Cas	3.0000	mm
Axial Length of Attached Element	Ls	11964.001	mm

LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	
Operating Allowable Stress	Sl	138.00	N./mm ²
Ambient Allowable Stress		138.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	16.0000	mm
Corrosion Allowance	Cal	3.0000	mm
Axial Length of Attached Element	Ll	11964.001	mm

Type of Reinforcement at Large End of Cone: None

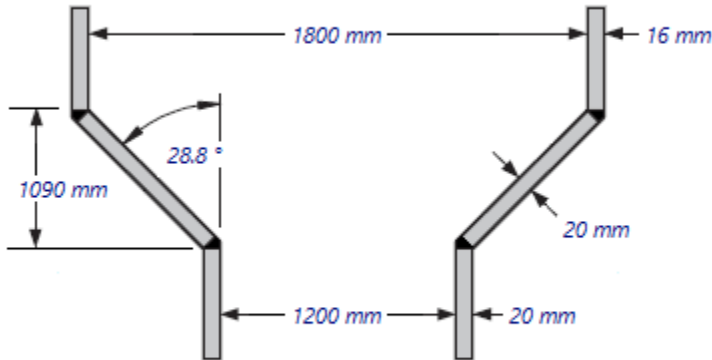
Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange No

Force Arl & Ars calculation, cone -> cylinder No



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 FileName : 130-C1& C2-Final 23rd March
 Conical Section: RHS Cone shell Cone: 12 9:30am Sep 9,2026



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 65 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 65 °C
Elastic Modulus of Small Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large End Reinforcement	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small End Reinforcement	0.200E+09 KPa. at 65 °C

Elastic Modulus Data from ASME Section II Part D at 65 °C

Elastic Modulus of Cone Material	0.199E+09 KPa. at 65 °C
Elastic Modulus of Small Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large Cylinder Material	0.200E+09 KPa. at 65 °C
Elastic Modulus of Large End Reinforcement	0.200E+09 KPa. at 65 °C
Elastic Modulus of Small End Reinforcement	0.200E+09 KPa. at 65 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Note: Neither end of Cone is a Line of Support

Maximum Centroid Reinforcement Distance Large End	27.2809 mm
Maximum Centroid Reinforcement Distance Small End	25.6661 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.01131

Large End Maximum half apex angle w/o reinforcement [δ]:

$$= 326.6 \cdot \sqrt{\left(\frac{P}{(S_s \cdot E_1)} \right)}$$
 per 1-5 equation 1

$$= 326.6 \cdot \sqrt{\left(\frac{1561}{(138 \cdot 1)} \right)}$$



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 Conical Section: RHS Cone shell Cone: 12 9:30am Sep 9, 2026

= 34.740 degrees

Large end actual half apex angle [alpha]: 28.831 degrees

Because delta is greater than alpha, reinforcement calculations are not required for the large end under internal pressure.

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.01131

Small End Maximum half apex angle w/o reinforcement [δ]:
 $= 89 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-5 equation 6
 $= 89 \cdot \sqrt{(1561 / (138 \cdot 1))}$
 = 9.467 degrees

Small end actual half apex angle [alpha]: 28.831 degrees

Intermediate Value [k]:
 $= \max(Y / (Sr \cdot Ers), 1)$
 $= \max(27.63E+09 / (138 \cdot 200187728), 1)$
 = 1.0000

where [Y] is:
 $= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$
 $= 138 \cdot 200187728$
 $= 27625406464.0 \text{ (N./mm}^2\text{)}^2$

Decay Length, Cone Small End:
 $= 1.4 \cdot \sqrt{(Rs (ts - ca))}$
 $= 1.4 \cdot \sqrt{(603 (20 - 3))}$
 = 141.746 mm

Required Area of Reinforcement, Small End, Internal [Ars]:
 $= k \cdot QS \cdot Rs / (Ss \cdot E1) \cdot (1 - \delta / \alpha) \cdot \tan(\alpha)$ per 1-5 equation 7
 $= 1 \cdot 78.5 \cdot 603 / (138 \cdot 1) \cdot (1 - 9.47 / 28.8) \cdot 0.55$
 = 12.44 cm²

Force per Length, Cone Small End, tensile side [QS+]:
 $= P (Rs/2) - \text{Faxial} / (\pi(Ds + Ts)) + \text{Moment} / (\pi(Rs + Ts/2)(Rs + Ts/2)) + f2'$
 $= 1561 (603/2) - 0 / (\pi(1206 + 20)) +$
 $0 / (\pi(603 + 20/2)(603 + 20/2)) + 30.5$
 = 78.524 Kg/mm

Note:
 Note: f2' is the axial load due to the heat exchanger constraint and is taken as the maximum value considering all of the exchanger load cases and conditions.

Area of Reinforcement Available in Small End Shell [Aes]:
 $= 0.78 [\sqrt{(Rs \cdot ts) \cdot (ts - t)} + \sqrt{(Rs \cdot tc / \cos(\alpha)) \cdot (tc - tr)}]$ per 1-5 eqn. 8
 $= 0.78 [\sqrt{(603 \cdot 17) \cdot (17 - 6.87)} +$
 $\sqrt{(603 \cdot 17 / 0.88) \cdot (17 - 7.84)}]$
 = 15.7284 cm²

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	12.4405	cm ²
Area of reinforcement in shell per App. 1-5(8)	15.7284	cm ²
Area of reinforcement in stiffening ring	0.0000	cm ²



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.



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 FileName : 130-C1& C2-Final 23rd March Page 317 of 317
 Conical Section: RHS Cone shell Cone: 12 9:30am Sep 9,2026

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	132.72	492.00	Small Cyl. Long.
Compres. Stress	-76.55	-492.00	Small Cyl. Long.
Membrane Stress	113.76	-207.00	Small End Tang.
Tensile Stress	136.69	492.00	Cone Longitudinal
Compres. Stress	-72.58	-492.00	Cone Longitudinal
Compres Stress	121.70	-207.00	Cone Tangential
Tensile Stress	347.34	492.00	Large Cyl. Long.
Compres. Stress	-238.10	-492.00	Large Cyl. Long.
Membrane Stress	-19.58	-207.00	Large End Tang.
Tensile Stress	218.85	492.00	Cone Longitudinal
Compres. Stress	-123.50	-492.00	Cone Longitudinal
Compres Stress	-33.46	-207.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure KPa.	Reason for Failure at this Pressure
MAWP	2206.652	Small End Reinforcement
MAPnc	2620.779	Small End Reinforcement

These pressures were determined by iteration.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: